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FIRM-SPECIFIC NEWS SENTIMENT AND SHORT- HORIZON RISK-ADJUSTED RETURN PREDICTABILITY: EVIDENCE FROM LARGE LANGUAGE MODELS

S.S.D. ECON-09/B

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ABSTRACT

Financial markets operate in a narrative-driven information environment in which news, disclosures, and digital communication shape how investors interpret firms' future prospects. Although asset-pricing theory emphasizes fundamentals, prices ultimately reflect heterogeneous expectations and evaluative beliefs, commonly referred to as investor sentiment. The economic relevance of sentiment depends critically on how it is defined, measured, and aligned with the asset-pricing unit of analysis.

This thesis develops a firm-level framework for analysing narrative sentiment in financial markets and examines whether sentiment embedded in financial news contains economically meaningful information at short horizons. It further investigates whether advances in large language models (LLMs) enhance the extraction of firm-specific sentiment and improve its predictive and economic value.

The central premise is that sentiment measurement should correspond to the economic object being evaluated. Financial news articles frequently discuss multiple firms within a single narrative and may convey heterogeneous or contrasting evaluations across entities. While document-level sentiment measures provide useful aggregate signals, they do not explicitly attribute evaluative tone to individual firms. The thesis, therefore, adopts a target-level approach in which sentiment is systematically linked to the specific firm under analysis, ensuring conceptual alignment between narrative evaluation and firm-level returns.

To operationalize this framework, the thesis constructs and publicly releases two expert-annotated Bloomberg-based datasets: one comprising 1476 firm-specific headlines and another containing 1162 full-length news articles. Both datasets are manually labelled at the firm level under structured annotation guidelines and achieve high intercoder reliability. Using these datasets, the thesis benchmarks lexicon-based approaches, discriminative transformer-based models, and state-of-the-art generative LLMs, with the latter evaluated under zero-shot and few-shot prompting strategies.

Lexicon-based tools exhibit limited sensitivity to contextual nuance and entity-specific attribution. Discriminative transformer models perform more strongly but remain dependent on training-domain alignment. In contrast, contemporary generative LLMs consistently achieve superior performance in target-level classification tasks without task-specific fine-tuning.

The thesis then evaluates whether firm-level sentiment signals improve short-horizon return forecasting and generate economically meaningful performance. Sentiment derived from headlines and full-length articles is integrated with lagged returns and technical indicators to predict one-, two-, and three-day-ahead return directions for large U.S. technology firms. Predictive frameworks include linear benchmarks and nonlinear ensemble methods. Performance is assessed using classification metrics and economic criteria, including Sharpe ratios and risk-adjusted alpha within the Fama-French five-factor framework. Several consistent findings emerge. Linear models relying solely on past returns exhibit limited predictive power. Nonlinear ensemble methods substantially outperform linear benchmarks, indicating that sentiment-return relationships are interaction-dependent. Sentiment extracted from full-length articles provides stronger predictive gains than headline-based sentiment, highlighting the importance of narrative depth. Predictive improvements strengthen at multi-day horizons, consistent with gradual information diffusion. Most importantly, models incorporating LLM-derived article-level sentiment generate positive and statistically significant risk-adjusted alpha under out-of-sample evaluation, whereas price-only strategies do not. Taken together, the evidence demonstrates that firm-specific narrative sentiment embedded in financial news contains economically relevant short-horizon information when measured at the appropriate level of granularity and evaluated under disciplined asset-pricing benchmarks. The thesis contributes to behavioural asset pricing, financial natural language processing, and quantitative finance by establishing a scalable and economically validated framework for linking narrative information to return predictability.

Keywords: Investor sentiment; financial news; firm-level sentiment analysis; stock return prediction; risk-adjusted performance; lexicon-based method; discriminative transformer-based models; generative large language models.

Publications and Collaborations

Parts of this thesis are based on joint research that has been published, accepted for publication, or is currently under review. The work has been conducted in close collaboration with my supervisor, Prof. Marco Minozzo, my co-supervisor, Prof. Marco Rospocher, and external collaborators, reflecting the interdisciplinary nature of the research at the intersection of financial economics and natural language processing.

Chapter III, titled “*On Assessing the Performance of Large Language Models for Target-Level Sentiment Analysis in Financial News Headlines*”, is based on work published in the journal *Algorithms*¹. This study was conducted in collaboration with Prof. Marco Rospocher and focuses on constructing an expert-annotated dataset of financial news headlines and evaluating LLMs for target-level sentiment analysis.

Chapter IV, entitled “*Benchmarking Large Language Models for Target-Based Sentiment Analysis and Stock Returns*”, was developed in collaboration with Prof. Marco Rospocher, Prof. Slavko Žitnik, and Dr. Timotej Knez. A preliminary version of this work was published in the *proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)*² under the title “*Benchmarking Large Language Models for Target-Based Sentiment Analysis*”. The extended version presented in this chapter is currently under review at the *International Journal of Data Science and Analytics*. The journal submission extends the conference publication by incorporating additional analyses, including sentiment-return correlation, directional accuracy evaluation, and expanded zero-shot and few-shot prompting strategies.

Chapter V, titled “*Forecasting Risk-Adjusted Stock Returns with Firm-Specific News Sentiment: Evidence from Large Language Models*”, is joint work with Prof. Marco Minozzo, Prof. Marco Rospocher, and Prof. Chris Biemann. This study investigates the economic relevance of firm-level sentiment measures derived from LLMs for short-horizon return prediction and trading performance. The manuscript is currently under review at the *Journal of Forecasting*.

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² <https://aclanthology.org/2025.clicit-1.74/>

Note: For consistency, the first-person plural (“we”, “our”) is used throughout the thesis.

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“The best way out is always through.”
— Robert Frost

The completion of this dissertation marks the end of a formative and demanding intellectual journey. Although a doctoral thesis bears a single author’s name, it is invariably shaped by the guidance, generosity, and support of many individuals and institutions. I am deeply grateful to all who have contributed to this work and to my development as a scholar.

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This dissertation reflects not only years of research but years of shared belief, patience, and perseverance. For that, I remain profoundly grateful.

Lovingly dedicated to my daughter, Aena Iftikhar.

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LIST OF ABBREVIATIONS

AAII	American Association of Individual Investors
ABSA	Aspect-Based Sentiment Analysis
AI	Artificial Intelligence
API	Application Programming Interface
ASTE	Aspect Sentiment Triplet Extraction
BERT	Bidirectional Encoder Representations from Transformers
CAPM	Capital Asset Pricing Model
CEFR	Common European Framework of Reference for Languages
CMA	Conservative Minus Aggressive
CNNs	Convolutional Neural Networks
DA	Directional Accuracy
EMH	Efficient Market Hypothesis
FEARS	Financial and Economic Attitudes Revealed by Search
FSA	Financial Sentiment Analysis
GPT	Generative Pre-trained Transformer
HML	High Minus Low
IPO	Initial Public Offering
LDA	Latent Dirichlet Allocation
LLMs	Large Language Models
LR	Logistic Regression
LSTM	Long Short-Term Memory
MDD	Maximum Drawdown
MKT	Market Risk Premium
ML	Machine Learning
NLP	Natural Language Processing
PCA	Principal Component Analysis
PLS	Partial Least Squares
RF	Random Forest
RMW	Robust Minus Weak
RNNs	Recurrent Neural Networks

RSI	Relative Strength Index
SMA	Simple Moving Average
SMB	Small Minus Big
TBFSA	Target-Based Financial Sentiment Analysis
TIs	Technical Indicators
TLFSA	Target-Level Financial Sentiment Analysis
TLSA	Target-Level Sentiment Analysis
VADER	Valence Aware Dictionary and Sentiment Reasoner
XGB	Extreme Gradient Boosting

CHAPTER I

GENERAL INTRODUCTION

Financial markets are driven not only by economic fundamentals, such as discount rates, expected cash flows, and risk premia, but also by beliefs about the future. While classical asset-pricing theory formalises how information should be incorporated into prices, prices are ultimately determined by investors who interpret information under uncertainty and form expectations about future outcomes (Sharpe, 1964; Fama, 1970; Ross, 1976). These expectations, commonly referred to as investor sentiment, shape trading behaviour and price dynamics (Barberis et al., 1998; Kumar and Lee, 2006; Muarya et al., 2026).

Sentiment /'sen.tɪ.mənt/ *noun*

“A thought, opinion, or idea based on a feeling about a situation, or a way of thinking about something.”

– *Cambridge English Dictionary*

In everyday language, sentiment refers to an opinion or attitude that is influenced by emotions. Within finance, however, defining investor sentiment is more complex. Scholarly definitions emphasise different aspects, including beliefs, expectations, and deviations from fundamentals (Baker and Wurgler, 2006; Shefrin, 2008; Glaser et al., 2009). Baker and Wurgler (2006), for example, define sentiment as the inclination of investors to speculate based on subjective valuations and the optimism or pessimism they express toward an asset or market. Glaser et al. (2009) describe sentiment as asset price expectations formed on subjective principles, while Shefrin (2008) characterises it as investors' subjective beliefs relative to fundamental values. Despite these nuances, a common thread emerges: sentiment captures subjective assessments of future asset performance that are not mechanically implied by observable fundamentals.

The intellectual foundations of sentiment can be traced to Keynes' notion of “animal spirits,” which emphasised the psychological forces underlying economic decision-

making (Keynes, 1937). Modern behavioural asset-pricing models formalise this intuition by treating sentiment as a latent state variable that influences prices when arbitrage is limited or risky (Shleifer and Vishny, 1997; Barberis et al., 1998; Baker and Wurgler, 2006; Ryu et al., 2025; Ali et al., 2025). Importantly, sentiment need not be purely irrational to be economically relevant. Investors operate under cognitive constraints, heterogeneous information sets, and limited attention (Hong and Stein, 1999; Shiller, 2019). Even when behaving prudently given available information, they may form beliefs that deviate from fundamentals. Moreover, rational investors may anticipate the behaviour of others and trade on expected sentiment-driven mispricing (De Long et al., 1990). The distinction between “rational” and “irrational” investors is therefore overly simplistic; market participants occupy a continuum of belief formation and information processing (Black, 1986; Barberis and Thaler, 2003).

In this thesis, investor sentiment is defined as an investor’s aggregate opinion about the future performance of a risky asset or assets. This deliberately inclusive definition avoids imposing a strict rational-irrational dichotomy and aligns with both the natural language definition and the broad conceptual framing in Baker and Wurgler (2006). It also aligns with behavioural finance and natural language processing (NLP), where sentiment is understood as the opinion, tone, or evaluative valence conveyed through language (Pang and Lee, 2008; Liu, 2015).

Over the past decades, investor sentiment has moved from the periphery to the centre of empirical asset pricing, particularly within behavioural finance (Barberis et al., 1998; Baker and Wurgler, 2006; Shiller, 2019; Sakariyahu et al., 2024). A substantial body of evidence suggests that sentiment influences asset prices, volatility, liquidity, and trading activity, particularly when arbitrage is limited and information diffusion is gradual (De Long et al., 1990; Shleifer and Vishny, 1997; Ling et al., 2014; Kumari and Mahakud, 2015). These effects tend to be most pronounced over short horizons, before mispricing is corrected through learning and rational trading (Tetlock, 2007; Heston and Sinha, 2017).

Yet sentiment is inherently unobservable. If investors’ private beliefs were directly observable, longstanding puzzles such as excess volatility, delayed reactions to

information, and horizon-dependent return predictability would be considerably easier to resolve (Shiller, 1988; Barberis et al., 1998; Gennaioli and Shleifer, 2018; Bordalo et al., 2024). In practice, beliefs are heterogeneous, dynamic, and latent. Empirical research must therefore rely on observable proxies.

Traditional sentiment proxies, including survey-based measures, closed-end fund discounts, option-implied indicators, and internet search activity, have provided valuable insight into aggregate market mood (Brown and Cliff, 2004; Tetlock, 2007; Da et al., 2015). However, these measures are typically broad and indirect. While useful for studying market-wide fluctuations, they are less suited to explaining cross-sectional variation in individual firm returns (Žitnik et al., 2022). Because asset prices are determined at the firm level, accurate empirical inference requires sentiment measures that are aligned with that level of aggregation.

At the same time, the informational environment of financial markets has undergone a profound transformation. Financial news articles, earnings calls, analyst commentary, and digital media generate continuous textual narratives through which beliefs are framed, interpreted, and transmitted in real time (Doey & de Jong, 2025; Liu et al., 2025; Bachate & Suchitra, 2025). These narratives do not merely convey objective facts; they shape investor attention and interpretation (Barber & Odean, 2008; Shiller, 2019). Advances in natural language processing (NLP) have made it possible to extract sentiment from large volumes of financial text in a systematic and scalable manner (Araci, 2019; Lopez-Lira & Tang, 2023). Textual sentiment, therefore, provides a high-frequency and content-rich window into belief formation.

However, the economic relevance of textual sentiment depends critically on how it is measured. Much of the finance literature relies on document-level sentiment measures that assign a single polarity to an entire document (Moraes et al., 2013; Dou, 2017; Behdenna et al., 2018; Irum and Tahir, 2025; Wang et al., 2026). Financial texts frequently reference multiple firms and may convey heterogeneous or even opposing evaluations within the same narrative (Muhammad et al., 2025). For example, a single article may report that one firm exceeded analysts' earnings expectations while another firm in the same industry experienced declining revenues or weaker future guidance.

In such cases, the same narrative may simultaneously convey positive sentiment toward one firm and negative sentiment toward another. From an asset-pricing perspective, this creates a potential alignment challenge: assets are priced at the firm level, whereas sentiment is often measured at the document level. While document-level measures provide useful aggregate indicators of tone, they do not explicitly attribute evaluative language to individual firms. This difference in granularity may influence how effectively narrative sentiment maps into firm-level return variation and motivates the development of target-level measurement approaches.

Recent advances in large language models (LLMs) offer a potential solution. Unlike traditional lexicon-based tools or models trained on homogeneous corpora, generative LLMs can perform context-sensitive target-level (entity-level) sentiment³ extraction in zero-shot and few-shot settings (Brown et al., 2020; Lopez-Lira & Tang, 2023; Kirtac & Germano, 2024; Qi, 2025). These models enable context-aware attribution of evaluative tone to specific firms within complex multi-entity narratives, thereby improving alignment between sentiment measurement and firm-level return evaluation.

However, improvements in classification accuracy do not automatically translate into economic relevance. In asset pricing, the value of a measurement technology must ultimately be assessed by whether it enhances return predictability and generates statistically and economically significant risk-adjusted performance. If sentiment represents a priced informational state variable, improvements in its measurement should strengthen statistical inference and, where economically relevant, improve out-of-sample risk-adjusted performance.

The central objective of this thesis is therefore to examine whether improved firm-level narrative sentiment measurement, particularly using large language models, enhances short-horizon asset-pricing outcomes.

³ In this thesis, the terms “firm-level sentiment,” “target-level sentiment,” “target-based sentiment,” and “entity-level sentiment” are used interchangeably to refer to sentiment attributed to a specific firm or entity mentioned within a financial text (e.g., Amazon, Nvidia, Netflix, or Alphabet), rather than assigning a single sentiment label to the document as a whole.

The research is structured around three interconnected themes:

1. Measurement: constructing high-quality, reproducible firm-level annotated sentiment datasets.
2. Methodological evaluation: benchmarking lexicon-based, discriminative transformer-based, and generative large language model approaches for target-based financial sentiment analysis.
3. Economic validation: assessing whether improved firm-level sentiment measurement translates into statistically robust and economically meaningful risk-adjusted returns.

By progressing from measurement construction to methodological evaluation and finally to economic validation, the thesis establishes a direct link between advances in NLP and economically interpretable pricing outcomes. In doing so, it bridges computational linguistics and empirical asset pricing, addressing a central challenge in behavioural finance: translating latent belief dynamics into tradable economic value.

The remainder of this chapter develops the research motivation, articulates the central problem and research questions, outlines the empirical strategy, and concludes with an overview of the structure of the thesis.

1.1. Research Motivation

The motivation for this dissertation arises from a practical and scholarly tension. On the one hand, financial narratives are increasingly central to how markets interpret events, form expectations, and allocate attention. On the other hand, much of the empirical evidence on sentiment and returns remains sensitive to how sentiment is measured and how predictive models are evaluated. In particular, improvements in sentiment classification quality do not necessarily imply improvements in economic outcomes such as risk-adjusted abnormal performance.

Recent generative LLMs have demonstrated strong capabilities in extracting context-sensitive evaluative signals from complex financial narratives. These tools may improve the measurement of firm-specific sentiment in ways that are difficult for traditional lexicons or even domain-specific transformers. Yet the economic value of such improvements, especially for short-horizon return predictability, remains insufficiently established in settings where rigorous evaluation includes out-of-sample testing and risk-adjusted benchmarks.

This dissertation is motivated by the need to assess whether advances in language modelling can be translated into disciplined empirical asset-pricing evidence. Specifically, it evaluates whether firm-specific sentiment extracted from news narratives using modern language models provides incremental information for predicting short-horizon stock return direction and whether any predictive gains translate into economically meaningful performance as measured by trading metrics and Fama-French alpha relative to standard factor controls.

1.2. Problem Statement and Research Questions

Although extensive research documents that investor sentiment influences asset prices, much of the empirical literature relies on sentiment measures that are highly aggregated or imperfectly aligned with firm-level return realizations. In multi-entity financial narratives, document-level sentiment may obscure heterogeneous evaluations across firms, potentially weakening the alignment between narrative tone and firm-specific price dynamics. At the same time, recent advances in LLMs provide powerful tools for contextual and entity-specific language understanding, enabling fine-grained extraction of firm-level sentiment. Despite their strong performance in NLP tasks, systematic evidence on the economic value of LLM-extracted firm-specific sentiment within asset-pricing frameworks remains limited.

The central problem addressed in this thesis is therefore:

Do firm-level narrative sentiment signals extracted using LLMs provide incremental and economically meaningful information for predicting short-horizon stock returns

and generating risk-adjusted abnormal returns beyond traditional sentiment models and established asset pricing factors?

To address this problem, the thesis investigates the following research questions:

RQ1: To what extent can LLMs reliably extract firm-level (target-specific) sentiment from financial news, and how does their performance compare with lexicon-based and fine-tuned transformer-based approaches?

RQ2: Does few-shot prompting improve the accuracy and robustness of LLM-based target-level financial sentiment analysis relative to zero-shot approaches?

RQ3: Does LLM-derived firm-specific sentiment exhibit a statistically significant association with short-horizon stock returns?

RQ4: How do methodological choices, such as textual information depth (headlines versus full-length articles) and predictive modelling frameworks (linear versus nonlinear methods), affect the accuracy and robustness of sentiment-enhanced return forecasts?

RQ5: Does LLM-derived firm-specific narrative sentiment provide incremental predictive information and generate statistically and economically significant risk-adjusted performance in realistic out-of-sample trading settings?

By answering these research questions, the thesis makes three primary contributions:

First, it provides measurement infrastructure by introducing reproducible, expert-annotated target-level sentiment datasets for financial news headlines and full-length articles.

Second, it provides methodological insights through the systematic benchmarking of lexicon-based approaches, discriminative transformer-based models, and generative large language models in a target-based financial setting.

Third, it delivers economic validation by demonstrating that improved firm-specific narrative sentiment measurement, particularly when derived from richer article narratives and incorporated into nonlinear forecasting frameworks, enhances short-horizon predictive performance and generates statistically and economically significant risk-adjusted abnormal returns under established asset-pricing benchmarks.

1.3. Research Methodology

The thesis adopts a measurement-driven empirical design integrating financial economics and natural language processing. The empirical strategy proceeds in three stages:

1. Construction of expert-annotated datasets of financial news headlines and full-length articles labelled at the firm (target) level.
2. Systematic benchmarking of lexicon-based methods, discriminative transformer-based models, and generative LLMs under zero-shot and few-shot paradigms.
3. Economic validation, incorporating validated firm-level sentiment measures into out-of-sample short-horizon forecasting and trading frameworks, with performance evaluated using multifactor asset-pricing models and Fama-French alpha.

By combining dataset construction, methodological benchmarking, and economic validation, the thesis directly links improvements in sentiment measurement to economically interpretable asset-pricing outcomes.

1.4. Structure of the Thesis

The thesis develops a coherent research programme that progresses from sentiment measurement to economic relevance. It is organised into six chapters, with Chapters 3-5 forming the empirical core.

Chapter I establishes the conceptual foundation of the thesis, develops the research motivation and central problem, formulates the guiding research questions, and provides an overview of the organisation of the dissertation.

Chapter II reviews the literature on investor sentiment and asset pricing; textual sentiment analysis in finance; target-level sentiment analysis; LLMs in financial NLP; and short-horizon return predictability and economic evaluation. It identifies gaps that motivate the empirical analyses in Chapters III-V.

Chapter III focuses on sentiment measurement in financial news headlines. It introduces a manually annotated dataset of Bloomberg news headlines labelled at the target (firm) level and benchmarks lexicon-based methods, discriminative transformer-based models, and zero-shot LLMs. The results indicate that traditional tools exhibit subpar performance in target-level tasks, whereas LLMs achieve strong performance without task-specific fine-tuning.

Chapter IV extends the analysis to full-length financial news articles, introducing a second manually annotated dataset and evaluating multiple LLMs under zero-shot and few-shot paradigms. The findings indicate that LLMs consistently outperform traditional sentiment tools in extracting firm-specific sentiment from complex multi-entity texts.

Chapter V assesses economic relevance by incorporating validated firm-specific sentiment into short-horizon forecasting and trading frameworks. It compares linear and nonlinear predictive models and demonstrates that sentiment-augmented nonlinear models generate statistically significant and economically meaningful risk-adjusted returns. Predictive gains peak at multi-day horizons, consistent with gradual information diffusion and attention constraints.

Chapter VI concludes the thesis by synthesising the empirical findings and systematically addressing the research questions introduced in Chapter I. Drawing on the evidence presented in the preceding chapters, it evaluates the theoretical, methodological, and policy implications of the results. The chapter also acknowledges the principal limitations of the study and outlines directions for future research that extend and refine the contributions of this work.

CHAPTER II

MEASURING INVESTOR SENTIMENT: METHODS, GRANULARITY, AND ECONOMIC ALIGNMENT

This chapter examines the principal approaches to measuring investor sentiment and establishes the conceptual and methodological foundations for the empirical analyses that follow. It first develops a taxonomy of sentiment proxies and discusses their theoretical underpinnings. The chapter then reviews market-implied, textual, survey-based, internet-query, and unconventional mood-based measures of sentiment. It subsequently examines methodological developments in textual sentiment analysis, from lexicon-based approaches to modern LLMs and discusses issues of sentiment granularity and entity-level attribution. The chapter concludes by connecting sentiment measurement to asset-pricing evaluation and identifying research gaps that motivate the empirical analyses in Chapters III-V.

2.1. Measuring Investor Sentiment

Investment decisions are influenced by investor sentiment, which is an inherently latent construct that reflects beliefs, attitudes, or affective states that are not entirely justified by economic fundamentals (Ph and Uchil, 2020). Sentiment is not a unique empirical proxy due to its ability to be interpreted in a variety of ways, including speculative optimism and noise-trader demand, as well as broader concepts such as attention, confidence, or mood (Aggarwal, 2022). Therefore, any sentiment measure that is observed is inherently imperfect, as it combines the latent behavioural component of interest with substantial non-sentiment variation arising from risk, liquidity, information shocks, institutional constraints, and measurement noise (Ung et al., 2024).

Consequently, the empirical literature has evolved in two complementary directions. First, researchers suggest proxies that are likely to plausibly represent investor sentiment, as determined by observed behaviour, stated beliefs, or expressed language

(Kumari and Mahakud, 2015; Zhou, 2018). Second, numerous studies aggregate multiple indicators to reduce measurement error and isolate a common sentiment component, acknowledging that individual proxies are noisy (Brown and Cliff, 2004; Baker and Wurgler, 2006). Although this logic has been advantageous for investigating aggregate market dynamics, it also underscores an inherent trade-off between scalability, temporal resolution, interpretability, and alignment with asset-level economic outcomes.

The channel through which sentiment becomes observable is a useful organizing framework for distinguishing sentiment measures. One category of metrics derives sentiment from market outcomes, including derivatives prices, trading activity, or issuance decisions (Yu, 2014). A second class of studies depends on the direct elicitation of beliefs through surveys or indirectly through revealed information demand, such as internet search behaviour (Burdea and Woon, 2022). A third category encompasses sentiment that is either conveyed or induced through language, such as corporate or policy communication, social platforms, and news media (Shahare, 2017; Yu et al., 2020). In terms of economic interpretation, aggregation, and frequency, each class presents unique advantages and constraints.

Crucially, most conventional sentiment measures, particularly market-implied and survey-based proxies, are developed at relatively low frequencies and high levels of aggregation. Although they are well-suited to the analysis of broad sentiment cycles and market-wide effects, they offer limited insight into the formation, transmission, and pricing of sentiment toward individual firms, particularly over short horizons (Orlowski and Sopoer, 2019). This limitation is essential for firm- or asset-level analysis, as a misalignment between the sentiment measure and the underlying economic object can introduce substantial measurement error and weaken empirical inference (Žitnik et al., 2022).

The following subsections review the principal approaches to measuring investor sentiment: market-implied sentiment, internet query-based sentiment, survey-based sentiment, unconventional mood-based measures, and textual sentiment, as illustrated in Figure 2.1.

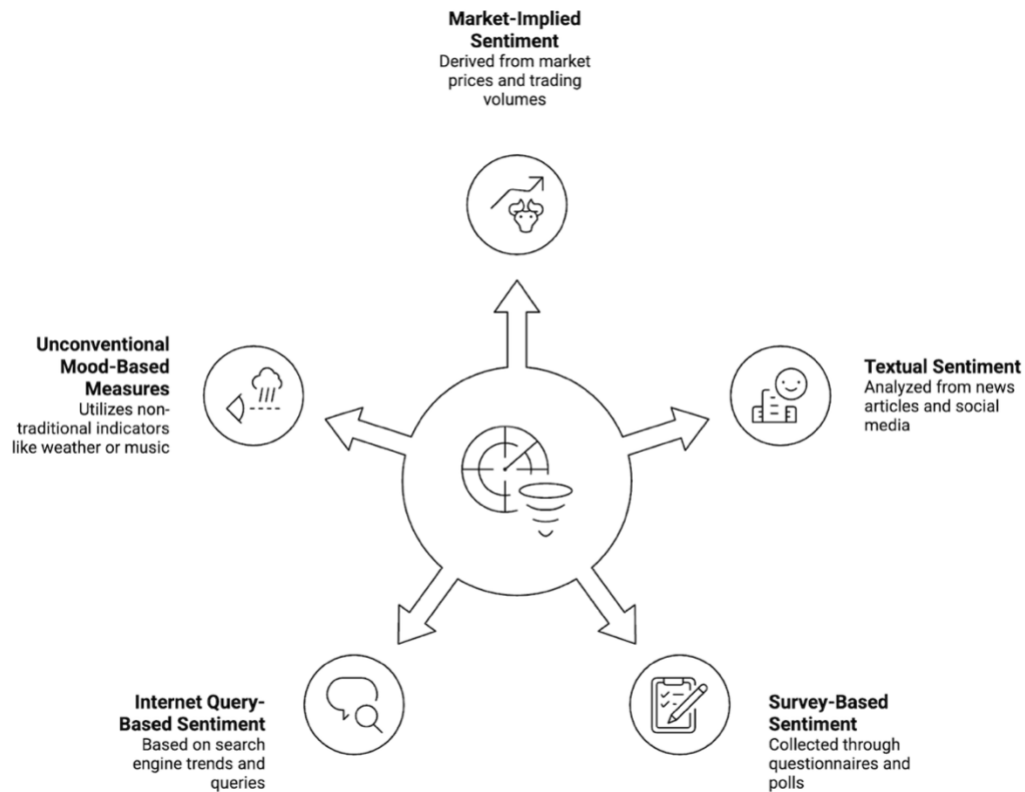


Figure 2.1: Investor sentiment measurement methods.

2.1.1. Market-Implied Sentiment

Market-implied sentiment refers to the component of investor optimism or pessimism that is inferred from observable market behaviour and corporate actions, rather than from explicitly elicited beliefs (Derman et al., 1996). The central intuition that underpins this approach is well-established in the field of behavioural finance. When investor sentiment is high, speculative demand for risky assets increases, and prices may rise above levels that are justified by fundamentals. Corporate managers may capitalize on favourable sentiment by timing equity issuance, particularly initial public offerings, during periods of high valuations (Brown and Cliff, 2004; Baker and Wurgler, 2006, 2007; Ljungqvist et al., 2006). On the other hand, periods of depressed sentiment are linked to decreased risk appetite, restricted financing activity, and lower asset valuations (Ljungqvist et al., 2006).

A diverse array of sentiment proxies that are market-based are driven by this behavioural logic. Common indicators include trading volume and turnover, equity issuance shares, closed-end fund discounts, IPO (Initial Public Offering) volume and first-day returns, and derivatives-based measures that capture speculative demand or beliefs about tail risk (Lee et al., 1991; Baker and Stein, 2004; Baker and Wurgler, 2006). However, a fundamental identification challenge is that each proxy is concurrently impacted by non-sentiment forces, including time-varying liquidity, investment opportunities, regulatory constraints, risk premia, hedging demand, and market structure. Consequently, any singular market-implied proxy can only offer a noisy and incomplete estimate of the underlying latent sentiment factor (Shleifer and Vishny, 1997; Baker and Wurgler, 2006; Stambaugh et al., 2012).

Recognizing this challenge, a significant portion of the empirical literature transitions from the use of individual proxies to the development of composite sentiment indices that compile data from numerous market indicators. The classical measurement-error argument is the foundation of the underlying rationale: if multiple indicators are reliant on a shared sentiment component but contain idiosyncratic noise that is imperfectly correlated across series, extracting their shared variation enhances signal-to-noise and results in a more dependable sentiment estimate (Brown and Cliff, 2004; Baker and Wurgler, 2006; Baker et al., 2012). Baker and Wurgler (2006) provide the canonical framework by synthesizing multiple market-based proxies into a single sentiment index and demonstrating that it elucidates cross-sectional return patterns, with particularly robust effects among equities that are challenging to value or expensive to arbitrage. These findings have been further supported and expanded by subsequent research, which has solidified composite indices as a fundamental component of empirical behavioural finance (Baker and Wurgler, 2007; Yu and Yuan, 2011; Stambaugh et al., 2012).

Composite market-implied sentiment indices have since been modified to accommodate a diverse array of institutional contexts, asset classes, and sampling frequencies. Baker et al. (2012) developed sentiment indices that are internationally comparable, allowing for the analysis of sentiment transmission and pricing effects across multiple countries. Conversely, other studies customize indicator sets to

specific markets or sectors. To mitigate microstructure disturbance and reflect data availability, most market-implied indices are constructed at a monthly frequency. Higher-frequency sentiment measures are the subject of more recent research. Gao and Liu (2020) developed an intraday composite sentiment index using high-frequency trading data and demonstrated that sentiment can impact prices even at extremely short time horizons. This expansion underscores an essential trade-off: while higher-frequency indices facilitate the study of intraday sentiment transmission and price pressure, they also intensify concerns regarding noise, endogeneity, and economic interpretation (Chordia et al., 2008; Gao and Liu, 2020).

Methodologically, the construction of composite sentiment indices is primarily influenced by two linear factor-extraction approaches: principal component analysis (PCA) and partial least squares (PLS). PCA extracts orthogonal linear combinations of candidate indicators that maximise explained variance, with the first principal component commonly interpreted as the shared sentiment factor (McCown et al., 1989). This approach, employed in early research by Brown and Cliff (2004) and subsequently formalized in the Baker-Wurgler framework, is appealing due to its lack of predetermined economic objective variables and its minimal structure. Nevertheless, this agnosticism is also a limitation: PCA is designed to maximize statistical co-movement rather than economic relevance, which may result in the capture of other systematic forces, such as liquidity regimes or issuance cycles, if these drive co-variation among the underlying indicators (Baker and Stein, 2004; Baker and Wurgler, 2006).

PLS provides an alternative extraction strategy that explicitly integrates economic relevance. Rather than maximising variance, PLS identifies the linear combination of sentiment proxies that maximizes covariance with a prespecified dependent variable, most commonly next-period aggregate market returns (Eichengreen and Mody, 1998; Gelper and Croux, 2010). Huang et al. (2015) propose a sentiment-aligned index, constructed using PLS, offering superior in-sample and out-of-sample return predictability compared to PCA-based indices. They contend that this index mitigates a prevalent noise component that is present in traditional proxies. The interpretive distinction between PCA and PLS is therefore substantive: PCA is consistent with a

latent-factor measurement perspective, whereas PLS implicitly defines sentiment in relation to predictive content for market outcomes. Consequently, PLS-based indices are frequently employed as complements or robustness checks rather than primary sentiment measures. Careful implementation is necessary to prevent the overly tight embedding of outcome variables into the proxy construction and mechanical look-ahead (Lowry and Gaskin, 2014; Yu et al., 2018).

Market-implied sentiment measurement has been further refined by recent methodological advancements. One strand acknowledges that the informativeness of specific proxies, market structure, and investor composition evolves over time, allowing for time-variation in indicator weights. Rolling-window estimation, state-space models, or dynamic factor models with time-varying loadings may be justified by the fact that fixed-weight indices estimated over lengthy samples may mischaracterize sentiment in specific regimes (Kim and Byun, 2010; Ung et al., 2024). Another strand utilizes derivatives-market information to infer sentiment related to skewness preference, tail-risk beliefs, or speculative demand by utilizing option-implied distributions and discrepancies between implied and realised moments (Han, 2008; Bali et al., 2011; Bollerslev et al., 2011; Büchner and Kelly, 2022). Although derivatives-based measures offer valuable complementary information, they necessitate meticulous interpretation because option prices also reflect rational risk compensation and hedging demand.

Overall, market-implied sentiment indices offer a potent set of instruments for the examination of aggregate short cycles and broad market dynamics. Nevertheless, they are less well-suited to settings that necessitate firm-level alignment and high-frequency variation, which are central to the analysis of short-horizon return dynamics using target-level narrative sentiment, because they are indirectly inferred from market outcomes and traditionally constructed at high levels of aggregation.

2.1.2. Internet Query Sentiment

Internet query-based measures infer investor sentiment from revealed information demand, based on the concept that individuals' search behaviour reflects underlying

concerns, expectations, and uncertainty that may not be explicitly observable in prices or surveys (Dennis et al., 2002). Ginsberg et al. (2009) provide the methodological foundation for this approach by demonstrating that aggregate Google search volumes can track influenza outbreaks in real time. This establishes search activity as a high-frequency and unobtrusive proxy for latent psychological states, such as concern and attention.

Da et al. (2011) introduce this insight to finance by employing firm-level Google search volume as a metric of investor attention. The results indicate that abnormal search intensity is predictive of short-term stock returns, trading volume, and liquidity. This is consistent with transient price pressure that is driven by heightened investor attention rather than changes in fundamentals. Based on this framework, Da et al. (2015) developed a sentiment index at the market level by conducting queries for terms that are economically pessimistic. The resulting FEARS (Financial and Economic Attitudes Revealed by Search) index is developed to capture negative investor sentiment and is shown to predict short-horizon market returns, volatility, and fund movements, with the strongest effects occurring during periods of elevated uncertainty.

Since its introduction, FEARS has established itself as a standard measure in the literature on search-based sentiment. Jiang et al. (2019) demonstrate that investor sentiment, including search-based components, contains incremental information that surpasses traditional macroeconomic predictors and contributes to explaining return predictability. The relevance of internet query sentiment is further substantiated by evidence from international and non-US markets. Ding and Hou (2015) demonstrate that Google search activity captivates the attention of retail investors in the Chinese stock market, whereas Naeem et al. (2021) demonstrate that search-based sentiment predicts volatility and returns in commodity markets. These results indicate that internet queries capture a behavioural dimension of sentiment that is distinct from, yet complementary to, market-implied measures.

The search-based sentiment framework has been expanded in recent research in a variety of ways. Gao et al. (2020) develop sentiment measures that differentiate

between positive and negative search intensity, thereby enabling asymmetric effects on asset prices. Hasan (2022) demonstrates that search-based attention is particularly informative during periods of market duress by applying Google search data to emerging markets. Do et al. (2023) propose that internet inquiries capture a distinctive dimension of investor psychology related to concern, attention, and uncertainty. This is supported by hybrid approaches that integrate search data with market-based sentiment indicators, which further enhance predictive performance.

Internet query-based sentiment measures provide numerous advantages. They are private, high-frequency, and expeditious, as search behaviour does not necessitate explicit belief reporting, making it less susceptible to social desirability or response bias (Diaz-Garcia et al., 2020). Additionally, search data are free from certain endogeneity concerns that influence market-implied sentiment measures, as they are not mechanically derived from asset prices (Zaefarian et al., 2017). At the same time, interpretation necessitates caution, as search intensity is indicative of both sentiment and rational information acquisition. Consequently, empirical studies frequently depend on short-horizon predictability, interaction with uncertainty measures, and robustness tests to differentiate between rational information demand and behavioural effects (Da et al., 2015; Jiang et al., 2019).

Overall, internet query-based sentiment measures offer a valuable addition to conventional sentiment proxies by capturing real-time fluctuations in investor concerns and attention. They enhance comprehension of the evolution and influence of investor psychology on financial markets, particularly over short time frames, when implemented in conjunction with survey-based, textual and market-implied sentiment measures.

2.1.3. Survey Sentiment

Survey-based measures of investor sentiment are designed to directly elicit beliefs, expectations, and attitudes by inquiring about the current and prospective economic or financial conditions of market participants, including households, individual investors, or professionals (Qiu and Welch, 2004). Surveys offer a conceptually transparent and

explicit mapping to sentiment as a psychological construct, in contrast to market-implied sentiment measures, which infer sentiment indirectly from observed behaviour (Shiller et al., 1984). This is the reason why survey sentiment is a central figure in the early empirical literature on behavioural finance and investor sentiment, and it has frequently been used as a benchmark against which alternative sentiment proxies are assessed (De Long et al., 1990; Baker and Wurgler, 2007).

Face validity is the primary allure of survey sentiment. Surveys are designed to measure the beliefs of respondents, rather than the inferences that researchers make based on prices, trading activity or textual expression (Mossholder et al., 1995). This feature was first utilized in early studies to investigate whether the expectations that were expressed are consistently correlated with the subsequent market returns, volatility, or predictability (Shiller et al., 1984; Solt and Statman, 1988). Survey-based sentiment frequently demonstrates contrarian characteristics at extremes, which is a robust empirical regularity. For instance, Fisher and Statman (2000) demonstrate that the self-reported optimism of individual investors tends to reach its maximum at market highs and its lowest at market lows. Brown and Cliff (2004) also demonstrate that aggregate survey sentiment is strongly correlated with contemporaneous returns and volatility, but its capacity to predict long-horizon returns is restricted once macroeconomic and financial conditioning variables are considered. Survey measures, according to Qiu and Welch (2004), are predominantly indicative of perceived risk or time-varying expected returns, rather than persistent mispricing.

Despite their conceptual clarity, survey-based sentiment measures are subject to significant methodological and practical constraints. The statistical power and feasibility of high-frequency analysis are both constrained by the limited size of historical survey samples, particularly prior to the digitisation of data collection (Clayton et al., 2009). Survey design has the potential to introduce systematic biases, such as anchoring, framing effects, and social desirability bias, which can distort reported beliefs in the opposite direction of underlying expectations (Suchman, 1968; Kahneman and Tversky, 2013). Additionally, surveys elicit expressed beliefs rather than disclosed preferences; respondents' responses do not necessarily correspond to their actual portfolio holdings or trading decisions (Dominitz and Manski, 2011).

Consequently, the correlation between survey sentiment and asset prices is frequently nonlinear, state-dependent, or contrarian, particularly during periods of extreme optimism or pessimism (Neal and Wheatley, 1998; Fisher and Statman, 2000).

These constraints contribute to the explanation of why survey sentiment occupies a lesser position in contemporary empirical asset-pricing research than sentiment measures derived from market data or large-scale textual sources. The construction of sentiment proxies that are scalable, high-frequency, and directly linked to observable behaviour has been made possible by advancements in data availability, automated text analysis, and machine learning (Tetlock, 2007; Loughran and McDonald, 2011; Da et al., 2015). Nevertheless, survey sentiment continues to be valuable for a variety of reasons. First, it serves as a natural benchmark for the evaluation of alternative sentiment measures (Baker and Wurgler, 2006). Second, it provides historical continuity, which allows for comparisons across long time horizons and market regimes (Schmeling, 2009). Third, surveys capture sentimental dimensions, including confidence, expectations, and perceived risk, that may not be entirely visible in prices or textual data (Shiller, 2015).

Several survey instruments have been particularly influential in the literature. The University of Michigan Consumer Sentiment Index is a widely used proxy for broad economic optimism and aggregate investor sentiment, and it assesses households' assessments of personal finances and general economic conditions (Carroll et al., 1994; Ludvigson, 2004; Kellstedt et al., 2015). Empirical evidence indicates that consumer confidence is a predictor of stock returns in both domestic and international markets, with stronger effects in economies with lower financial sophistication or stricter arbitrage limits (Lemmon and Portniaguina, 2006; Schmeling, 2009; Ferrer et al., 2016).

The American Association of Individual Investors (AAII) Sentiment Survey is another notable indicator. It provides weekly reports on the percentage of respondents who are bullish, bearish, or impartial about the stock market over a six-month period (Qiu and Welch, 2004). The AAI survey has been extensively utilized in empirical sentiment research due to its relatively high frequency and explicit emphasis on equity-market

expectations (Clarke and Statman, 1998; Brown and Cliff, 2004; Wang et al., 2006). A consistent finding is that extreme bullishness or bearishness among AAI respondents is associated with subsequent market reversals, thereby supporting a contrarian interpretation of retail investor sentiment that is consistent with noise-trader models (De Long et al., 1990; Baker and Wurgler, 2007).

The Investor Intelligence Survey, which assesses the sentiment of newsletter writers and market commentators, is frequently utilized as a proxy for professional or institutional sentiment (Solt and Statman, 1988; Chau et al., 2016). This measure also demonstrates contrarian properties; however, its dynamics are distinct from those of retail surveys, as it reflects the professional background of respondents and their potential influence on investor beliefs, rather than merely reflecting them (Brown and Cliff, 2005; Ameriks et al., 2009). The UBS/Gallup Investor Optimism Index also enables the analysis of expectation heterogeneity across wealth groups, thereby emphasizing systematic differences in sentiment formation among different investor types (Guzmán, 2009; Dominitz and Manski, 2011).

Furthermore, the Sentix Investor Confidence Index has been extensively employed to investigate sentiment effects in global and European equity markets (Anamika and Subramaniam, 2023; Bormann, 2013; Schmeling, 2007). The evidence suggests that Sentix sentiment accurately predicts international stock returns, with the strongest effects observed in markets with higher arbitrage limits. This supports the notion that survey sentiment captures a behavioural component that becomes more influential when mispricing is more challenging to correct (Hengelbrock et al., 2013).

In recent research, survey sentiment is increasingly being regarded as a complement to newer sentiment measures, rather than a substitute. Comparative studies demonstrate that survey sentiment is more closely aligned with slow-moving sentiment cycles, while textual and search-based measures capture higher-frequency fluctuations in beliefs and attention (Tetlock, 2007; Da et al., 2015; García, 2013). Other research integrates survey sentiment into conditional asset-pricing frameworks, where it serves as a proxy for time-varying expected returns or risk premia (Schmeling, 2009; Huang et al., 2023).

A further methodological consideration concerns the statistical treatment of survey responses as measures of latent sentiment. Many survey-based indicators rely on rating-scale or ordinal responses, such as bullish, neutral, and bearish expectations, or Likert-type assessments of confidence and economic outlook. These responses do not directly observe sentiment itself but rather provide discrete indicators of an underlying latent construct. Consequently, the construction of survey sentiment measures involves choices regarding ordinal scaling, response aggregation, and the transformation of categorical answers into quantitative indices. Methods such as composite index construction, latent-factor approaches, or ordered-response models can therefore influence both the interpretation and comparability of survey-based sentiment indicators across studies.

To summarize, survey-based sentiment measures offer a conceptually transparent and direct perspective on articulated investor beliefs; however, they are restricted by bias, scalability, frequency, and behavioural interpretation. While they are no longer the predominant instrument in empirical sentiment research, survey measures are still indispensable as historical benchmarks, validation devices, and indicators of broad investor confidence. In this regard, survey sentiment continues to inform and contextualize contemporary methods of measuring investor sentiment, even though contemporary research increasingly relies on market-implied and textual proxies to investigate firm-level and high-frequency sentiment dynamics.

2.1.4. Unconventional Sentiment Measures

A related strand of the investor sentiment literature quantifies sentiment by utilizing variables that appear to be unrelated to financial markets but have an impact on investor behaviour through mood. The underlying concept is that affective states can influence asset prices even in the absence of new fundamental information by subconsciously shaping risk perception and decision-making. These mood disruptions are a valuable environment for identifying behavioural effects and isolating sentiment-driven price dynamics, as they are plausibly exogenous to economic fundamentals (Shu, 2010; Kaustia and Rantapuska, 2016).

Empirical studies exploit a wide range of unconventional proxies. Prominent examples include weather conditions (Hirshleifer and Shumway, 2003; Goetzmann et al., 2015), sports outcomes (Edmans et al., 2007), aviation disasters and other tragic events (Kaplanski and Levy, 2010), terrorist attacks (Chen et al., 2020), and emotionally salient media events such as television finales, music consumption, or newspaper imagery (Lepori, 2015; Edmans et al., 2022; Obaid and Pukthuanthong, 2022). Despite their diverse nature, these studies consistently demonstrate that mood-related shocks have an impact on stock returns, volatility, and trading behaviour, providing compelling evidence that investor sentiment can influence asset prices through channels unrelated to fundamentals (Nofer and Hinz, 2015; Chang et al., 2008; Li and Ahn, 2024).

A key strength of unconventional sentiment measures lies in their identification strategy. This is because the underlying disruptions are largely orthogonal to macroeconomic fundamentals and firm-specific information (Yu et al., 2013). Consequently, it is challenging to reconcile observed market reactions with rational information-based explanations. Therefore, these metrics offer particularly compelling evidence for the existence of sentiment-driven pricing effects and serve as a complement to findings that are based on more traditional sentiment proxies.

Concurrently, unconventional sentiment metrics are subject to significant constraints. Typically, they are unrefined, episodic, and difficult to associate with specific firms or assets. Consequently, they are inadequately equipped to investigate high-frequency dynamics, firm-level sentiment formation, or cross-sectional return variation. Additionally, these metrics offer robust identification capabilities; however, they offer inadequate comprehension of the mechanisms by which sentiment is generated, disseminated, or reflected in market narratives.

In summary, unconventional sentiment measures are crucial for the establishment of the behavioural foundations of investor sentiment and the demonstration that financial markets can be influenced by non-fundamental factors. Nevertheless, their utility for applications that necessitate high-frequency sentiment measurement and firm-level alignment is limited by their indirect economic interpretation and limited granularity.

In this regard, they are most effectively perceived as supplementary evidence that bolsters the broader significance of sentiment in asset pricing, rather than as the primary instruments for empirical analysis.

2.1.5. Textual Sentiment

Textual sentiment measurements derive investor sentiment from language, written or spoken, rather than from actual market outcomes (Kearney and Liu, 2014). Textual approaches analyse the explicit expression of beliefs, evaluations, and affect embedded in narratives, in contrast to market-implied proxies, which indirectly recover sentiment from prices, trading activity, or corporate actions. Consequently, they offer a more direct perspective on investor psychology and belief formation (Puurtilinen, 2003; Kearney and Liu, 2014). Language not only communicates optimism or pessimism, but also framing, emphasis, and narrative structure, all of which influence expectations, perceived risks, and investment behaviour (Tversky and Kahneman, 1981; Rothman and Salovey, 1997; Slovic et al., 2002; Corvellec, 2011).

A significant advantage of textual sentiment is its high temporal resolution. Financial news articles, social media posts, and corporate communications are typically observed at daily or intraday frequencies, enabling empirical designs that examine short-horizon mechanisms such as immediate price pressure, liquidity effects, volatility responses, and attention spillovers (Barber and Odean, 2008). The rapid expansion of this literature reflects both the digitisation of financial communication and advances in NLP, which have made large-scale textual analysis computationally feasible within empirical finance.

The fundamental insight that underpins textual sentiment research is that language is not a neutral carrier of information. Investor beliefs and behaviours can be significantly influenced by variations in tone or narrative framing, even when factual content remains unaltered. This concept is formalized in early research that establishes a connection between media pessimism and market dynamics. Tetlock (2007) demonstrates that negative tone in a widely read Wall Street Journal column predicts short-term downward price pressure followed by reversals, a pattern consistent with

sentiment-driven trading rather than the efficient incorporation of fundamental information. Subsequent studies reinforce this interpretation by demonstrating that media tone predicts returns, volatility, and trading volume even after accounting for earnings announcements and macroeconomic news, and that sentiment effects are exacerbated during periods of increased uncertainty or binding limits to arbitrage (Tetlock et al., 2008; García, 2013). Collectively, these results establish textual sentiment as an economically significant channel by which beliefs influence asset prices.

A substantial body of research has been conducted on news media as a primary source of textual sentiment. News outlets are particularly influential in the formulation of market narratives because they typically combine new information with commentary and interpretation, which reaches a broad investor audience. Empirical studies investigate sentiment in newspapers (Hossain et al., 2021; Rahab et al., 2021; Shapiro et al., 2022; Chandra et al., 2025; Xi and Jiang, 2025), newswire services like Dow Jones and Reuters (Uhl, 2014; Allen et al., 2019; Madani, 2025), and broadcast media, including television news (Mazzotta, 2022; Möller and Reichmann, 2023). Furthermore, aggregation services or commercial sentiment indices, such as LexisNexis-based corpora (Okuneva et al., 2024) and the Thomson Reuters MarketPsych indicators (Loang, 2025), are employed by numerous studies to address scale and heterogeneity. Disentangling sentiment from information remains a persistent identification challenge in this literature, as negative tone may be indicative of adverse fundamentals, narrative framing, or transitory mood. Consequently, sentiment-driven effects are isolated in high-quality studies by utilizing reversal patterns, exploiting short-horizon timing, or employing meticulous controls for fundamental news (Yekrangi and Abdolvand, 2021).

A second major source of textual sentiment is social media and online investor forums, which capture sentiment voluntarily expressed by market participants. Early work by Antweiler and Frank (2004) analyses message-board postings on Yahoo! Finance and finds that online bullishness predicts trading volume and volatility, although return predictability is modest. Subsequent research has led to the development of more advanced textual pipelines and larger datasets, with a focus on platforms such as

Yahoo! Finance (Kucuktunc et al., 2012; Abayomi-Alli et al., 2022), Seeking Alpha (Chen et al., 2014), HotCopper (Hu and Tripathi, 2016), and StockTwits (Renault, 2017; Batra and Daudpota, 2018; Nousi & Tjortjis, 2021). One of the unique features of StockTwits is the user-provided Bullish and Bearish labels, which produce a large, continuously updated training set for supervised sentiment classification and alleviate the small-sample constraints that restricted early machine-learning applications (Renault, 2020).

In addition to finance-specific platforms, general social networks, particularly Twitter (now X), have been extensively utilized due to features such as hashtags and cashtags that facilitate firm and topic identification (Giachanou and Crestani, 2016; Albladi et al., 2025; Smith-Mutegi et al., 2025). Twitter sentiment metrics have been associated with volatility, short-horizon returns, and trading activity, although there are still significant concerns regarding demographic bias, strategic manipulation, and bots (Cresci et al., 2019). Recent research has expanded textual sentiment analysis to Reddit and Discord, particularly in the context of meme-stock episodes, where sentiment diffusion within tightly knit online communities can generate rapid and extreme price movements disconnected from fundamentals (Lee et al., 2025; Li and Li, 2026).

A third pillar of textual sentiment research investigates managerial communication and corporate disclosures. Corporate filings, earnings press releases, and earnings-call transcripts convey not only quantitative information but also managerial tone, emphasis, and forward-looking narrative (Ning et al., 2025; Kwanjai and Nettayanun, 2025). Feldman et al. (2008) demonstrate that the tone in the management discussion and analysis section of SEC filings contains incremental information, while Li (2010) demonstrates that forward-looking statements exhibit systematic linguistic structure with predictive content. Earnings calls have garnered significant attention due to their capacity to facilitate the communication of nuanced evaluations of performance and prospects by managers. Sentiment extracted from call transcripts has been shown to predict returns, volatility, and analyst behaviour in addition to earnings surprises (Price et al., 2012; Nagendra et al., 2024; Chin and Fan, 2023). Although these disclosures are disseminated at a lower frequency than news or social media, their perceived

credibility and informational authority render them particularly influential (Guest, 2021).

Closely related is a burgeoning macro-finance literature that investigates sentiment embedded in central bank and policy communications (Bennani, 2020; Celler, 2024; Czudaj and Nguyen, 2025). Central banks possess informational advantages regarding economic conditions and financial stability, and the tone of their communications can influence expectations independently of policy actions. The linguistic measures of optimism, risk, and uncertainty are predictive of asset prices, volatility, and broader financial-cycle dynamics, as evidenced by text-based analyses of Federal Open Market Committee statements, European Central Bank communications, and financial stability reports (Hubert and Fabien, 2017; Correa et al., 2021; Mullings, 2023).

An additional challenge in textual sentiment measurement concerns language diversity. Much of the early literature on financial text analysis was developed and validated primarily on English-language corpora, which limits direct comparability across linguistic settings. Applying textual sentiment models to other languages may introduce additional difficulties related to grammar, morphology, word segmentation, financial terminology, and the availability of high-quality annotated corpora. Although modern LLMs are increasingly described as multilingual, their performance may still vary across languages depending on training-data availability, linguistic complexity, and domain-specific coverage. This issue is particularly relevant for financial sentiment analysis because measurement error arising from language-specific model performance may affect both classification accuracy and subsequent economic inference.

To summarize, textual sentiment is a fundamental component of contemporary empirical finance. By analysing language, textual approaches enable researchers to investigate the ways in which investor behaviour and asset prices are influenced by narratives, tone, and emotion, particularly during periods of heightened uncertainty and short horizons. Simultaneously, the heterogeneity and richness of textual data present challenges in the areas of economic alignment, attribution, and measurement

accuracy, which are essential for the empirical application of textual sentiment in asset-pricing applications.

2.1.5.1. Textual Sentiment Methods

A central challenge in empirical sentiment analysis is the development of quantitative sentiment measures from unstructured text. Textual sentiment, in contrast to market-implied or survey-based sentiment, does not permit a natural or unique mapping from raw language to a scalar variable. Rather, sentiment must be inferred through methodological decisions that incorporate assumptions regarding semantics, context, and interpretation (Shi et al., 2019). Consequently, the literature has evolved along a variety of methodological paradigms, each reflecting trade-offs between measurement accuracy, scalability, domain adaptation, and interpretability. Figure 2.2 provides an overview of the evolution of textual sentiment analysis methods. Broadly, existing approaches can be classified into Lexicon-based methods, traditional supervised machine-learning classifiers, neural representation learning and deep learning models, discriminative transformer-based approaches, and, most recently, generative LLMs.

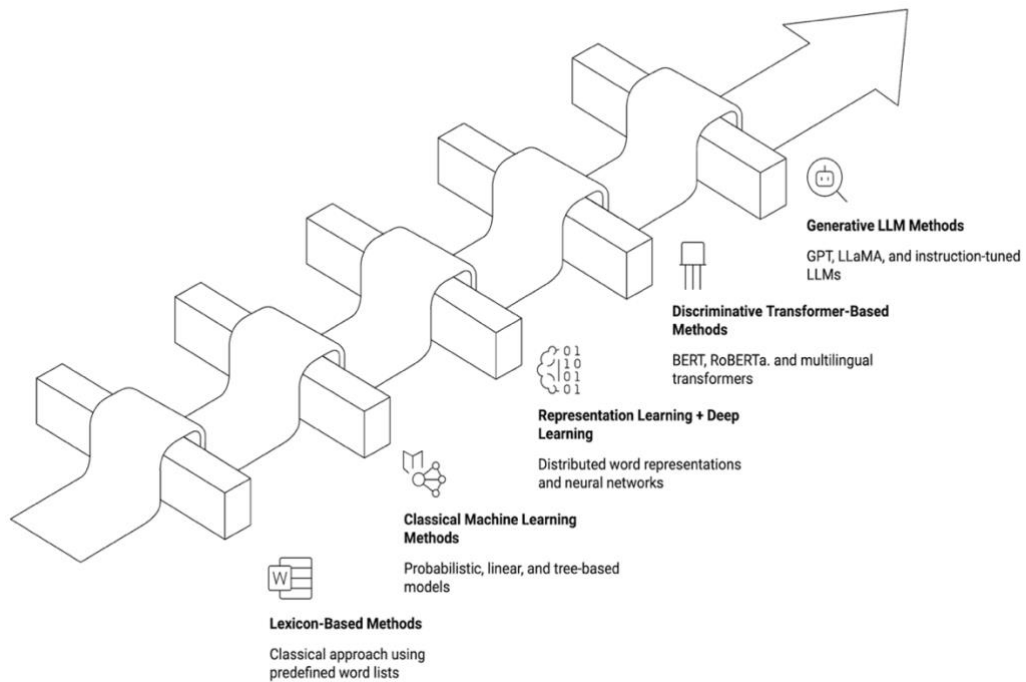


Figure 2.2: Evolution of textual sentiment analysis methods.

Early work in financial text analysis predominantly relies on lexicon-based (dictionary) methods. These methods quantify sentiment by counting the frequency of words allocated to predefined affective categories. Initially, studies often implement general-purpose psychological dictionaries, including the General Inquirer (Stone and Hunt, 1963) and DICTION (Hart, 2001), as well as subsequent rule-based tools such as VADER (Hutto and Gilbert, 2014). Lexicon-based methodologies are appealing due to their transparency, computational affordability, and simplicity of replication. They have been instrumental in influential early investigations of corporate disclosures and media tone (Tetlock, 2007; Tetlock et al., 2008; Kothari et al., 2009).

However, a well-documented limitation of general-purpose lexicons is their lack of domain specificity. Many words that are negatively valenced in everyday language, such as liability, tax, or capital, are economically neutral or even positively informative in financial texts. This insight motivates the development of finance-specific dictionaries, most notably the Loughran-McDonald (2011) lexicon. The Loughran-McDonald dictionary illustrates that misclassification caused by generic lexicons can significantly distort empirical inference and introduce sentiment dimensions beyond polarity, such as litigiousness, uncertainty, and modal strength. Subsequent research refines dictionary construction using semi-automated methods, including statistical text models that incorporate multinomial inverse regression (Taddy, 2013; Garcia et al., 2023) and seed-word expansion based on semantic similarity (Hu and Liu, 2004). Despite these improvements, lexicon-based methods are intrinsically restricted: they are unable to capture negation or context-dependent meaning, disregard semantic interaction, and abstract from word order, which restricts their efficacy in intricate financial narratives (Khoo and Johnkhan, 2018).

To address these shortcomings, an expanding body of literature has implemented supervised machine-learning classifiers that directly infer sentiment from labelled data. Early applications in finance rely on probabilistic models, such as Naïve Bayes, which are computationally efficient and relatively resilient to noise in textual features (Antweiler and Frank, 2004; Li, 2010; Sprenger et al., 2014). Support vector machines, maximum entropy models, and tree-based methods, such as random forests, are also frequently employed (Renault, 2017, 2020; Al-Nasser and Ali, 2018; Huang et al.,

2022). When high-quality labels are available, these models typically outperform dictionary-based approaches by utilizing nonlinear decision boundaries and richer feature representations.

Nevertheless, conventional supervised classifiers are significantly reliant on manual feature engineering, with bag-of-words or n-gram representations being the most common. They are also limited in their ability to model long-range dependencies or contextual meaning (Ahmad et al., 2017). Empirical comparisons in financial applications yield no universally dominant approach. Al-Nasseri and Ali (2018) demonstrate that tree-based methods exhibit superior performance in specific settings, while Renault (2020) demonstrates that comparatively simple classifiers can perform comparably to more complex alternatives when labels are scarce or noisy. These findings emphasize that the efficacy of conventional machine-learning methodologies is exceedingly susceptible to text length, domain characteristics, and labelling strategies employed.

The limitations of feature-engineered representations motivate the adoption of neural representation learning and deep learning models, which learn distributed semantic embeddings directly from data. Early developments in word embeddings, such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014), show that semantic relationships can be encoded in dense vector spaces, resulting in representations that are significantly more detailed than sparse bag-of-words features. Building on these embeddings, deep neural architectures, such as convolutional neural networks (CNNs) (Kim, 2014) and recurrent neural networks (RNNs) (Medsker and Jain, 2001), particularly long short-term memory (LSTM) models (Hochreiter & Schmidhuber, 1997), enable modelling of word order, compositional semantics, and sequential dependencies. In financial text analysis, CNN and LSTM-based models are shown to outperform traditional classifiers by capturing contextual dynamics in news articles, earnings calls, and social media discussions (Chen et al., 2014; Rehman et al., 2019; Dang et al., 2020; Li et al., 2020). Unlike earlier machine-learning approaches, these models learn feature representations and classification boundaries in an end-to-end framework, thereby reducing the need for manual feature construction and enhancing robustness to linguistic variation. However, pre-transformer deep learning

architectures continue to encounter difficulties in modelling very long documents and global context, and their performance often relies on substantial labelled data, paving the way for attention-based transformer architectures (Islam et al., 2024).

A major methodological shift occurs with the introduction of discriminative transformer-based models, which employ self-attention mechanisms to acquire contextual representations of language. BERT (Devlin et al., 2019) represents a significant milestone in the field of classification by facilitating the implementation of deep contextual embeddings, which significantly enhance downstream tasks. In financial applications, however, general-purpose transformers frequently underperform because of the prevalence of numerical and forward-thinking language, domain-specific syntax, and specialized vocabulary. This motivates the development of domain-adapted models, most notably FinBERT (Araci, 2019; Yang et al., 2020), which is pre-trained on large financial corpora and fine-tuned on labelled financial sentiment datasets. A burgeoning body of evidence indicates that FinBERT and related models consistently outperform lexicon-based methods and traditional classifiers in a diverse array of financial sentiment tasks (Huang et al., 2023; Wu and Gu, 2023; Rizinski et al., 2024).

Subsequent research refines transformer-based approaches along two dimensions. First, computationally efficient variants, such as DistilFinRoBERTa, trade model size for inference speed, thereby facilitating large-scale or near-real-time applications (Atak, 2023; Muhammad and Rospoche, 2025). Secondly, more expressive architectures, such as DeBERTa-v3, exhibit superior performance in target- and aspect-based sentiment analysis, which is especially critical for financial texts that reference multiple entities with heterogeneous sentiment (Jayakody et al., 2024). Although these models surpass previous methods in terms of classification accuracy and robustness, they continue to be computationally intensive and depend on labelled training data, which restricts their scalability across domains and tasks (Huang et al., 2022; Kirtac and Germano, 2024).

Generative LLMs are the most recent and swiftly expanding strand of the literature. In contrast to discriminative classifiers, which are trained to explicitly map text to

sentiment labels, LLMs model language through autoregressive or encoder-decoder architectures and can perform sentiment inference in zero-shot or few-shot settings. This capability significantly diminishes dependence on task-specific labelled datasets and facilitates adaptable deployment across domains, text lengths, and sentiment definitions (Krugmann and Hartmann, 2024; Biswas et al., 2024). Empirical evidence indicates that LLMs, ranging from general-purpose models such as ChatGPT, Gemini, and LLaMA variants to domain-specific models such as BloombergGPT, often match or exceed the performance of fine-tuned discriminative transformers in financial sentiment tasks (Leippold, 2024; Wu and Gu, 2023; Chumakov et al., 2023).

Importantly, the relative advantage of LLMs is most apparent in environments that necessitate document-level comprehension, context-sensitive interpretation, and entity-specific sentiment attribution, such as financial news articles and narrative disclosures that contain multiple firms or topics (Mun and Kim, 2025; Muhammad et al., 2025). These characteristics position LLMs as a qualitatively different measurement technology for extracting sentiment from intricate financial text. Nevertheless, their significance in the field of financial economics is contingent upon the economic significance of the sentiment signals they generate and their alignment with the underlying asset, rather than solely on classification accuracy.

To summarize, textual sentiment methods have progressed from simple dictionary-based counts to highly expressive neural language models. Although methodological advancements have significantly enhanced the capacity to extract sentiment from financial text, they have also brought up new concerns regarding the validity of measurements, attribution, and economic interpretation. These considerations motivate closer attention to the level at which sentiment is attributed within text, which is the focus of the following section.

2.1.5.2. Textual Sentiment Granularity

Sentiment granularity, which refers to the level at which sentiment is extracted and attributed within a text, is a critical but frequently overlooked aspect of textual sentiment measurement. As illustrated in Figure 2.3, sentiment analysis in NLP is

commonly performed at four hierarchical levels: document, sentence, target, and aspect. Each level represents a distinct trade-off between informational precision and scalability. The selection of granularity has first-order implications for empirical inference in finance, as it directly influences the alignment between textual signals and the pertinent economic object, as well as the accuracy of attribution and the degree of noise in firm-level signal construction (Kearney and Liu, 2014; Loughran and McDonald, 2016).

Early studies in finance predominantly rely on document-level sentiment, in which a single sentiment score is assigned to an entire text, such as a news article, corporate filing, or earnings call transcript (Tetlock, 2007; Tetlock et al., 2008; Loughran and McDonald, 2011; García, 2013). This method is computationally efficient and has facilitated large-scale empirical analysis, particularly in dictionary-based frameworks and related count-based measures. Therefore, document-level sentiment has been extensively employed to investigate market-wide sentiment dynamics, disclosure sentiment, and media tone (Kearney and Liu, 2014; Behdenna et al., 2018; Mao et al., 2022). Nevertheless, document-level measures implicitly presume a uniform tone and disregard the potential for a single text to convey heterogeneous, or even opposing, sentiments toward various entities (Žitnik et al., 2022).

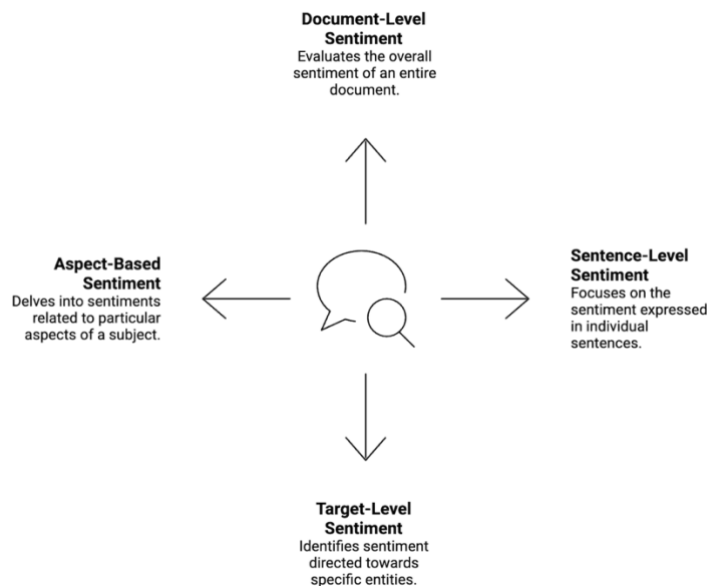


Figure 2.3: Textual sentiment granularity.

Sentence-level sentiment offers a partial refinement by estimating sentiment at the sentence level and aggregating it across multiple sentences. This method is particularly well-suited for the analysis of specific sections of corporate disclosures or news articles, as it reduces the noise introduced by boilerplate language and enables researchers to concentrate on sentiment-bearing statements (Feldman et al., 2008; Li, 2010; Tang et al., 2015). In earnings communication and managerial disclosures, sentence- or segment-level tone measures have been shown to be informative (Price et al., 2012). However, sentence-level sentiment is typically entity-agnostic, which means that it does not address the fundamental attribution issue that occurs when multiple firms, industries, or macroeconomic variables are discussed within the same text (Žitnik et al., 2022; Muhammad et al., 2025).

Financial environments are particularly susceptible to this attribution issue. Financial texts, including news articles, analyst reports, and earnings call transcripts, frequently reference multiple firms and economic actors, frequently expressing distinct or opposing evaluations within a single document. Aggregating sentiment across entities can therefore lead to coarser attribution, potential dilution of firm-specific signals, and reduced sensitivity in detecting firm-level return associations (Lee, 2012; Chakraborty et al., 2022; Pan et al., 2025). From an asset-pricing perspective, this difference in granularity may affect the economic coherence of the empirical specification, since asset prices are inherently firm-specific.

Target-level sentiment analysis directly addresses this limitation by explicitly attributing sentiment to specific entities (targets) mentioned in the text, such as individual firms, industries, or financial instruments (Hyun et al., 2019; Lu et al., 2022; Žitnik et al., 2022). Constructively, target-level sentiment preserves entity-specific evaluative content and enables the disentanglement of heterogeneous or opposing sentiment signals within the same document. This granularity is crucial for event studies, cross-sectional analyses, and short-horizon return predictability, as it enables a significantly more precise correspondence between textual information and asset-level outcomes.

Recent research in computational linguistics and finance has shown that methods that are capable of extracting return-relevant, entity-specific textual signals can outperform coarse, aggregate sentiment measures in both prediction and inference (Ke et al., 2019; Žitnik et al., 2022). Nevertheless, the implementation of target-level sentiment analysis in the financial sector has been historically hindered by methodological obstacles, such as the identification of named entities, the disambiguation of entities, and the establishment of a dependable connection between textual mentions and financial identifiers (Žitnik et al., 2022). These challenges have been significantly diminished as a result of advancements in NLP. Transformer-based language models facilitate the robust contextual representation of financial text, while domain-specific pretraining enhances performance in environments that are numerically rich and information-dense. The scalable extraction of sentiment in intricate financial narratives is facilitated by general-purpose encoders like BERT (Devlin et al., 2019) and finance-adapted variants like FinBERT (Araci, 2019; Yang et al., 2020).

Rigorous target-level modelling is further facilitated by the increasing availability of entity-level supervision. Entity-span annotations with sentiment labels for financial news are provided by datasets such as FinEntity, which also facilitate the systematic benchmarking of both pretrained transformer models and LLM-based baselines for entity-level sentiment classification (Tang et al., 2023). In addition to these advancements, recent research has assessed prompt-based and generative LLM approaches for target-level financial sentiment tasks, emphasizing both performance improvements and methodological sensitivities in LLM-driven pipelines (Muhammad and Rospocher, 2025; Muhammad et al., 2025). Collectively, these developments render target-level sentiment analysis practically feasible at scale in environments that were previously inaccessible.

Aspect-level sentiment analysis, the most comprehensive approach, further decomposes sentiment toward specific attributes of an entity, such as profitability, growth prospects, or risk exposure (Schouten and Frasincar, 2015). Aspect-based sentiment analysis (ABSA) is a well-established field in NLP, supported by shared-task benchmarks and mature methodological taxonomies (Pontiki et al., 2016; Zhang et al., 2022; Hua et al., 2024). Although aspect-level methods have the potential to

provide a more comprehensive economic interpretation, their use in finance is restricted by the necessity of domain-specific aspect definitions, extensive annotation, and the meticulous validation of economic meaning (Huang et al., 2022).

In summary, the financial literature's reliance on document- and sentence-level sentiment measures reflects historical computational and data constraints, rather than conceptual optimality (Kearney and Liu, 2014; Loughran and McDonald, 2016). As financial texts increasingly discuss multiple interconnected entities, low-granularity sentiment representations may be less closely aligned with firm-level asset-pricing applications. In this regard, sentiment granularity is not merely a technical detail, but a substantive modelling decision that must be consistent with the research question.

In line with this view, the present thesis adopts a target-level sentiment framework to capture the entity-specific informational content of financial texts. By aligning sentiment measurement with the firm-level economic object of interest, this approach enhances conceptual precision and enables sharper empirical inference regarding short-horizon return dynamics and economic relevance.

2.2. Investor Sentiment and Asset Pricing

Traditional asset-pricing theory, as formalised by the Efficient Market Hypothesis (EMH), posits that asset prices entirely and instantaneously reflect all publicly available information (Fama, 1970). Under this paradigm, price fluctuations arise from rational revisions in expectations about future cash flows or discount rates, and any predictable patterns in excess returns should be eliminated through arbitrage (Shleifer and Vishny, 1997). In this regard, variables such as investor sentiment should not generate systematic, risk-adjusted abnormal returns; at most, they may be associated with contemporaneous price movements reflecting rational information processing (Dash, 2016).

A substantial body of theoretical and empirical research challenges this perspective by relaxing the assumptions of fully rational investors and frictionless arbitrage. Early evidence on excess volatility (Shiller, 1981) and delayed market reactions motivates

behavioural asset-pricing models, in which sentiment-driven trading and heterogeneous beliefs play a central role. In particular, noise-trader models illustrate that mispricing can be induced by correlated shifts in investor sentiment when arbitrage is risky or constrained, even in the presence of rational, informed traders (De Long et al., 1990). Shleifer and Vishny (1997) further underscore that arbitrageurs may be unable to fully rectify sentiment-induced price distortions due to limitations to arbitrage, which include capital constraints, horizon risk, and agency frictions, resulting in the persistence of deviations from fundamental values.

Empirical evidence broadly supports the predictions of these models. Baker and Wurgler (2006) demonstrate that investor sentiment predicts the cross-section of stock returns, with the effects being concentrated among equities that are difficult to value or costly to arbitrage, such as small, young, volatile, or distressed firms. At the aggregate level, Brown and Cliff (2005) demonstrate that sentiment is significantly correlated with contemporaneous returns and volatility and possesses short-horizon forecasting power. However, it exhibits limited explanatory power for long-run fundamentals. Subsequent studies refine and extend these findings, illustrating that sentiment-based predictability is consistently robust across alternative sentiment constructions and is particularly evident when arbitrage limits are severe (Stambaugh et al., 2012; Huang et al., 2015).

A complementary strand of the literature links sentiment to information flows, media tone, and investor attention. Tetlock (2007) offers groundbreaking evidence that a pessimistic media tone predicts short-term downward price pressure followed by reversals. This pattern is consistent with sentiment-driven trading rather than efficient information incorporation. Da et al. (2011) demonstrate that spikes in investor attention, as proxied by internet search activity, predict short-run price pressure and subsequent reversals, while Antweiler and Frank (2004) demonstrate that sentiment expressed in online message boards is associated with contemporaneous returns, trading volume, and volatility. Together, these results indicate that sentiment and attention function as transient demand shocks that affect short-horizon return dynamics before being rectified through arbitrage and learning.

More recent research highlights the state-dependent nature of sentiment effects. García (2013) demonstrates that sentiment-related return predictability is more robust during economic downturns, when arbitrage capacity is diminished, and uncertainty is high. Ling et al. (2014) and Huang et al. (2015) offer additional evidence that sentiment-induced mispricing is exacerbated during periods of financial stress, low liquidity, or heightened funding constraints. This is consistent with behavioural models in which the price impact of belief-driven trading is amplified by limits to arbitrage. The increasing significance of sentiment and attention in the formation of short-horizon price dynamics in contemporary markets is further emphasized by evidence from modern, retail-dominated trading environments.

A related methodological issue concerns the potential bidirectional relationship between sentiment and market dynamics. Although sentiment is often modelled as a driver of asset prices, returns, and volatility, market movements may also shape investor beliefs, expectations, and expressed sentiment. Large price changes, earnings surprises, or volatility shocks can influence the tone of news coverage and investor commentary, generating feedback mechanisms between sentiment and prices. This simultaneity makes it difficult to distinguish predictive effects from contemporaneous reactions and complicates causal interpretation. The direction and strength of the sentiment-market relationship may also vary across horizons, with sentiment potentially reflecting market movements at very short horizons while contributing to gradual information diffusion over subsequent days.

A central challenge in interpreting sentiment-based return predictability is distinguishing true mispricing from compensation for systematic risk (Sharpe, 1964). Raw return predictability does not necessarily indicate abnormal performance, as sentiment-driven strategies may mechanically rely on known risk factors, such as tilting portfolios toward small, distressed, or high-growth firms. This is the reason why the empirical literature assesses the economic significance of sentiment by employing asset-pricing factor models and interprets performance through the lens of risk-adjusted abnormal returns (alpha). Early studies employ the Capital Asset Pricing Model (CAPM) (Sharpe, 1964; Lintner, 1965; Mossin, 1966), while most contemporary work employs multifactor benchmarks, particularly the Fama-French

three-factor and five-factor models, which account for systematic exposures related to market risk, size, value, profitability, and investment (Fama and French, 1993, 2015).

Within this framework, sentiment is deemed economically significant only if it generates statistically and economically significant alpha after accounting for established risk factors. Stambaugh et al. (2012) underscore that sentiment effects persist as abnormal returns when arbitrage is constrained, while Baker and Wurgler (2006) interpret sentiment-induced return predictability as evidence of mispricing precisely because it survives adjustment for conventional risk exposures. In contrast, predictability that is eliminated during factor adjustment is more consistent with risk premia or characteristic-based exposures than with sentiment-driven mispricing. As a result, risk-adjusted alpha has become the standard metric for economic validation in the sentiment literature.

Taken together, the literature demonstrates that investor sentiment can significantly impact asset prices, particularly over short horizons and in environments characterized by heightened uncertainty, limited arbitrage, and heterogeneous beliefs. Simultaneously, the interpretation and magnitude of sentiment effects are contingent upon the methods used to assess performance and sentiment. While aggregate sentiment measures have proven informative for studying broad market dynamics, asset-pricing applications necessitate sentiment indices that are aligned with the pertinent firm-level economic object and evaluated using rigorous risk-adjusted benchmarks. This insight motivates the focus of the present thesis on firm-specific narrative sentiment and its economic significance for short-horizon return dynamics, evaluated through factor-adjusted abnormal performance.

The literature reviewed in this chapter establishes that investor sentiment is an economically significant force in financial markets, particularly over short horizons and in settings characterized by limited arbitrage, heightened uncertainty, and heterogeneous beliefs. A variety of sentiment measures, such as market-implied, internet query-based, survey-based, textual, and unconventional mood-based proxies, have been devised to capture various aspects of investor psychology. Advancements in NLP have further broadened the scope of textual sentiment analysis, facilitating the

high-frequency examination of narratives embedded in news, social media, and corporate communication.

Despite this progress, our comprehension of the formation, transmission, and pricing of sentiment at the firm-level is still limited by a number of constraints. Many sentiment measures that are frequently employed are either highly aggregated or indirectly inferred from market outcomes, which restricts their applicability for asset-level analysis. While textual sentiment provides a more direct and contemporaneous perspective on belief formation, a significant portion of the finance literature continues to rely on sentiment representations that are agnostic or document-level, even when analysing firm-specific returns. In such settings, the economic correspondence between the sentiment measure and the firm whose return is being analysed may be less precise, potentially reducing the granularity of firm-specific informational signals and limiting empirical sensitivity.

Recent advances in machine learning and LLMs have the potential to mitigate some of these limitations by facilitating target-level sentiment extraction in intricate, multi-entity texts. Nevertheless, most current research assesses these models solely in terms of classification accuracy or by directly integrating them into trading or predictive frameworks. Consequently, the function of sentiment measurement quality, as opposed to model complexity or forecasting power, remains insufficiently disentangled in the literature. In particular, there is a scarcity of systematic evidence regarding the extent to which enhancements in sentiment measurement at the firm level result in economically significant improvements in risk-adjusted performance or return predictability.

Additionally, there is a lack of understanding regarding the extent to which the informational content of sentiment varies among textual sources. Although news headlines provide concise and timely signals, full-length news articles contain more detailed narratives, contextualization, and framing that may affect investor beliefs over longer short-horizon windows. The degree to which these sources offer sentiment signals that are either complementary or distinct for return dynamics remains an open empirical question.

Taken together, these considerations point to four interrelated research gaps. First, high-quality, reproducible target-level sentiment datasets for financial news remain scarce, limiting reliable evaluation of sentiment extraction methods. Second, the relative efficacy of lexicon-based approaches, discriminative transformer-based models, and generative LLMs for target-based financial sentiment analysis has not been systematically benchmarked in finance. Third, the differential informational value of sentiment extracted from headlines versus full-length articles is not well understood, particularly at short horizons. Fourth, numerous studies prioritise predictive accuracy or in-sample fit while neglecting rigorous economic validation, which raises the question of whether enhanced sentiment measurement generates statistically robust and economically significant risk-adjusted returns.

These gaps motivate the empirical analyses conducted in Chapters III-V. Chapter III addresses the measurement challenge by constructing an expert-annotated dataset of financial news headlines labelled at the firm level and benchmarking sentiment extraction methods in a target-based context. Chapter IV extends this analysis to full-length financial news articles, evaluating the performance of LLMs in more complex narrative environments. Chapter V evaluates the economic relevance of validated firm-level sentiment measures by integrating them into realistic short-horizon forecasting and trading frameworks and evaluating their capacity to generate statistically robust risk-adjusted alpha. In doing so, the thesis explicitly connects advances in sentiment measurement to economic outcomes, thereby addressing the central question posed in Chapter I.

CHAPTER III

ON ASSESSING THE PERFORMANCE OF LLMS FOR TARGET-LEVEL SENTIMENT ANALYSIS IN FINANCIAL NEWS HEADLINES

3.1. Introduction

The financial sector, known for its dynamic nature, is greatly impacted by the continuous flow of news and information that changes market dynamics and investor views (Adhikari et al., 2023). Every day, a large number of articles and microblogs containing detailed financial information about traded corporations are distributed online. This generates a substantial quantity of disorganized written information, posing a significant difficulty for companies, investors, and analysts involved in data analysis. The conventional approaches of manual data analysis are time-consuming and increasingly unfeasible due to the vast amount of data (Agarwal, 2020).

Automated sentiment analysis, a popular NLP task, is a viable solution that enables the automatic identification of emotional tones in texts, thereby offering valuable insights into investor behaviour and market trends (Pang and Lee, 2008; Deng et al., 2024). The decision-making processes of investors and market stakeholders are significantly impacted by the sentiments conveyed through financial news (Mishev et al., 2020). For example, positive news regarding a company's performance may increase investor confidence and increase stock prices, whereas negative news may incite dread and trigger selloffs.

Sentiment analysis in this field utilizes several strategies, such as lexicon-based approaches, conventional machine learning techniques, advanced deep learning models, and transformer-based architectures (Simmering and Huoviala, 2023). Generative LLMs like ChatGPT and Gemini have the potential to greatly impact sentiment analysis, as they can effectively work in situations where there is little or no training data available without requiring substantial manual annotation effort for

supervised training (Brauwers and Frasincar, 2022). Rigorously evaluating their performance in sentiment analysis for specific domains, such as the financial sector, is essential to assess their suitability for integration into domain-specific tools, including financial reporting software.

Sentiment analysis can be performed at four different levels of detail: document, sentence, target, and aspect level. The document level evaluates the overall sentiment of the entire text, while the sentence level analyses sentiment in individual sentences. The target level assesses sentiment toward specific entities or categories, referred to as targets within the text (Shirsat et al., 2017; Phan et al., 2023; Atandoh et al., 2023), and the aspect level focuses on sentiment toward different features or attributes of an entity. Prior research on financial sentiment analysis has predominantly focused on the document and sentence levels (Lutz et al., 2018; Husejinović and Mašetić, 2023; Du et al., 2023). Nevertheless, target-based analysis is essential for accurately interpreting and responding to the complex sentiments of financial news, as these documents frequently contain multiple sentiments associated with distinct entities (Deng et al., 2024). For example, a headline such as “Netflix shares dip despite Apple’s surge in streaming subscribers” illustrates divergent sentiments associated with specific companies, Netflix and Apple, within a single sentence, emphasizing the necessity of nuanced, target-based sentiment analysis.

Despite the financial sector’s significant relevance in target-level sentiment analysis, this domain is primarily underexplored, with limited exceptions such as implementations in economic and political news (Žitnik et al., 2022) and several commercial products (Ho et al., 2019). Additionally, the considerable interest in the integration of LLMs in various disciplines has prompted extensive investigation into their potential applications in a wide range of domains (Zhang et al., 2023; Minaee et al., 2024). However, their deployment within the stock market, particularly in investigating their effectiveness in conducting target-level financial sentiment analysis in zero-shot learning scenarios, has not yet been investigated. Furthermore, there is a significant lack of comparative performance analyses between traditional models and LLMs for financial sentiment analysis tasks.

The objective of this chapter is to fill these gaps by assessing the capability of LLMs to analyse sentiment at the target level in the financial field, with a specific focus on news related to the stock market. We believe that such an assessment is both timely and relevant, with the potential to influence the development of professional financial reporting software. We compare the effectiveness of generative LLMs, such as ChatGPT and Gemini, to traditional models, such as VADER, as well as to fine-tuned discriminative transformer-based models, including DistilFinRoBERTa and DeBERTa-v3-base-absa-v1.1, to identify and categorize sentiment at the target level within financial news headlines. More specifically, we address the following research questions:

RQ1: How effectively do zero-shot LLMs perform in target-level financial sentiment analysis, and how do they compare to other state-of-the-art sentiment analysis techniques?

RQ2: How do the performances of different versions of ChatGPT (specifically, ChatGPT 3.5, ChatGPT 4, and ChatGPT 4o) compare to those of Gemini models (including Gemini 1 Pro, Gemini 1.5 Flash, and Gemini 1.5 Pro) in target-level sentiment analysis tasks under zero-shot scenarios?

The work detailed in this chapter makes the following novel key contributions compared to existing studies:

(1) We present and publicly release a novel dataset that is specifically manually annotated for target-level sentiment analysis. This dataset consists of 1476 headlines from the financial sector. Studying target-level sentiment in news headlines is crucial because headlines shape immediate perceptions, influence public opinion, and dominate social media sharing, offering a concise and impactful representation of the news that often carries more bias and emotional weight than the full text. This resource is a critical instrument for the development and evaluation of sentiment analysis models that differentiate and evaluate sentiments directed at specific entities within the financial domain, as it uniquely addresses the nuanced requirements of target-level

analysis in financial texts. To the best of our knowledge, no other dataset with the given characteristics is currently available.

(2) Utilizing the contributed dataset, we present a thorough comparative analysis of LLMs compared to conventional sentiment analysis methods, emphasizing their strengths and weaknesses in the context of target-level financial sentiment analysis. This study is one of the first to systematically assess the efficacy of generative LLMs, such as different versions of Gemini and ChatGPT, in conducting target-level financial sentiment analysis within a zero-shot learning approach. Our work thus contributes to the development of NLP tools in the financial sector, enabling stakeholders to make more informed decisions by utilizing target-level sentiment analysis.

This chapter is based on research published in the journal *Algorithms*⁴, co-authored with Prof. Marco Rospocher.

The chapter is organized as follows: Section 3.2 reviews related literature. Section 3.3 delineates the empirical methodology and the dataset creation and annotation process. The findings and discussion are reported in Section 3.4. Finally, Section 3.5 concludes the chapter and offers possible future research directions in the field of financial sentiment analysis.

3.2. Literature Review

3.2.1. Financial Sentiment Analysis

Financial sentiment analysis is a distinct area of study in data analytics that specifically deals with the interpretation of emotions, opinions, and attitudes conveyed in financial texts (Yekranghi and Nikolov, 2023). These encompass many types of information, such as news stories, social media posts, analyst reports, and earnings call transcripts. The main objective is to assess the sentiment toward financial markets, specific

⁴ <https://doi.org/10.3390/a18010046>

companies, or broader economic conditions, as these factors significantly impact investment choices and market dynamics (Usmani and Shamsi, 2023).

The theory that public sentiment affects financial markets was initially introduced by Keynes (1937), who attributed market behaviours to “animal spirits”—alternating waves of optimism and pessimism. This concept has been substantiated by a multitude of studies that have shown substantial correlations between investor sentiment and market fluctuations in asset prices and trading volumes (Tetlock, 2007; Baker and Wurgler, 2007).

Tetlock (2007) made significant contributions to the field of sentiment analysis for financial texts utilizing lexicon-based techniques, including the Bag of Words (BoW) model, to identify underlying sentiments in financial news. This method entails creating word lists with each word labelled as either having a positive or negative sentiment. (Loughran and McDonald, 2011) created a financial emotion dictionary specifically designed for evaluating 10-K reports, emphasizing the need to have lexicons that are tailored to specific sectors. Although lexicon-based analysis is widely used, it has been criticized for its simplistic approach and lack of consideration for word context (Sohangir et al., 2018; Bonta et al., 2019; Taj et al., 2019; Shang et al., 2023).

To overcome these constraints, researchers have transitioned to employing machine learning methodologies. Machine learning frameworks, which are trained on labelled financial texts, offer more accurate sentiment assessments by adapting to word context, unlike static lexicon-based algorithms (Schumaker and Chen, 2009). Substantial advancements have been achieved in the field, such as the introduction of the Financial PhraseBank dataset and the utilization of machine learning for sentiment detection (Malo et al., 2014). Additionally, Dridi et al. (2019) suggested a competitive approach that relies on semantic characteristics.

The field has undergone further development following the integration of deep learning methods into NLP, resulting in considerable improvements in sentiment analysis capabilities. Transformer-based models such as BERT, presented by Devlin

et al. (2018), have shown outstanding performance in sentiment classification tasks. This is attributed to their capacity to understand context, capture long-term relationships, and decrease data complexity (Refaeli and Hajek, 2021). Furthermore, the implementation of FinBERT, which is a specialized version of BERT (Araci, 2019) for the financial domain, has enhanced the reliability and accuracy of financial sentiment analysis (Farimani et al., 2022; Leippold, 2023).

The latest advancements in LLMs have significantly improved their effectiveness for NLP. New standards have been established by innovative techniques such as instruction tuning (Minaee et al., 2024) and the retrieval-augmented framework for strengthening sentiment analysis (Du et al., 2024a). The models ChatGPT, Gemini, and LLaMAs are presently under evaluation to determine their ability to forecast market trends based on news headlines (Chen et al., 2022; Lopez-Lira and Tang, 2023). This suggests a positive outlook for LLMs in the field of financial analytics.

3.2.2. Evolution of Target and Aspect-Based Sentiment Analysis

The primary focus of conventional sentiment analysis was the assessment of the general sentiment at the sentence or document level. Nevertheless, this method was insufficient in capturing subtle opinions regarding particular elements or entities. To overcome this constraint, researchers have developed target-based and aspect-based sentiment analysis techniques. These approaches seek to offer a more detailed comprehension of sentiment specifically connected to particular aspects or entities (Žitnik et al., 2022). Target-level sentiment analysis (TLSA) is a method used to determine the general sentiment linked to a specific entity, which is known as the target. Conversely, aspect-based sentiment analysis (ABSA) concentrates on several facets of an item, such as its quality, robustness, or pricing (Yi et al., 2003).

The techniques used in target-level and aspect-based sentiment analysis vary from conventional machine learning methods to advanced deep learning and transformer-based models, which have greatly improved accuracy and contextual comprehension. Noble (2006), Webb et al. (2010), and Varghese and Jayasree (2013) employed conventional methods that involved the use of algorithms such as Naive Bayes,

Logistic Regression, and Support Vector Machines. These algorithms were used for extracting features and predicting sentiment in relation to specified entities or aspects.

The efficacy of TLSA and ABSA has been further enhanced by the emergence of deep learning. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which include Long Short-Term Memory (LSTM) networks and Gated Recurrent Units, have been instrumental in capturing intricate patterns and contextual information. The use of CNNs and LSTMs enhanced by attention mechanisms to concentrate more accurately on relevant text segments when assessing sentiment for specific aspects is highlighted in studies by Wang et al. (2021) and Sadr et al. (2021).

Transformer-based models, specifically BERT, have significantly transformed the fields of TLSA and ABSA by offering an unparalleled level of comprehension in terms of context and semantics (Rietzler et al., 2019; Karimi et al., 2021). These models achieve superior sentiment assessment performance by utilizing extensive pre-training on large datasets, which is followed by task-specific fine-tuning.

Although BERT-based models are effective, they encounter problems such as the need for pre-training that is appropriate to a particular domain and the management of ambiguous contexts (Silva and Marcacini, 2021). To address these constraints, large language models (LLMs), including LLaMa, ChatGPT, and Gemini, have been implemented for TLSA and ABSA tasks (Brauwers and Frasincar, 2022). Chumakov et al. (2023) conducted experiments with the SemEval-2016 for the Restaurant and Laptop domains to investigate GPT's capabilities for the Aspect Sentiment Triplet Extraction (ASTE) subtask. In a similar manner, Magdaleno et al. (2023) employed GPT to forecast sentiment scores for different elements derived from bakery reviews. Nevertheless, there is a scarcity of thorough comparison assessments on different LLMs in specific fields like finance.

Building on this literature review, it is clear that preliminary investigations into the application of LLMs for ABSA and TLSA demonstrate potential; however, the specificity and complexity of domains such as finance necessitate a more comprehensive examination. Our research represents a significant effort to assess the

efficacy of LLMs for TLFA in the financial sector. By conducting a comparative analysis of different LLMs, specifically ChatGPT versions 3.5, 4, and 4o, as well as the Gemini series, such as Gemini 1 Pro, Gemini 1.5 Flash, and Gemini 1.5 Pro, in conjunction with traditional methods like VADER, DistilFinRoBERTa, and DeBERTa-v3-base-ABSA-v1.1, our study assesses their capabilities in the complex context of financial sentiment analysis.

3.3. Methodology

This section outlines the methodology used to evaluate LLMs like ChatGPT and Gemini, as well as traditional financial sentiment analysis methods such as VADER, DistilFinRoBERTa, and DeBERTa-v3-base-ABSA-V1.1, in the context of target-level financial sentiment analysis (TLFSA). We elaborate on our methods for collecting data and the use of well-established methodologies for analysing financial sentiment. Afterward, we fine-tune the DeBERTa-v3-base-ABSA-V1.1 model using our curated dataset. Subsequently, we formulate a precise prompt tailored for the ChatGPT and Gemini models to assess the effectiveness of these LLMs in TLFSA tasks. In the end, a thorough examination is carried out to compare traditional approaches with LLMs in the task of TLFSA.

3.3.1. Dataset Creation and Annotation

To conduct a comprehensive examination in this research, news headlines pertaining to well-known stock companies, namely Amazon, Netflix, Nvidia, and Alphabet, were collected. These headlines were sourced from Bloomberg’s financial news service, Bloomberg Terminal. Bloomberg is a well-acknowledged platform that is extensively utilized for consolidating financial news data from many sources (Chen, 2021; Fedyk and Hodson, 2023; Fedyk, 2024). The dataset encompasses the time period from 1 September 2023 to 31 January 2024.

The dataset consists of 1476 distinct news headlines, which are each linked to a particular stock company (referred to as the “target”), a timestamp, and a URL. Keywords relevant to the targeted companies, such as “target-company” and “target-

company in headlines”, were used during the data-collecting phase from the Bloomberg Terminal. For example, when specifically analysing Amazon.com, Inc. as the subject company, keywords such as ‘Amazon.com Inc.’ and ‘Amazon in headlines’ were employed to aid in extracting pertinent articles. As a result, this method produced a collection of news articles that were pertinent to the pre-identified target companies. These articles had headlines that mentioned these companies. Subsequently, the gathered articles underwent processing to extract crucial information such as headlines, timestamps, and URLs, which were later recorded in a database, as seen in Figure 3.1.

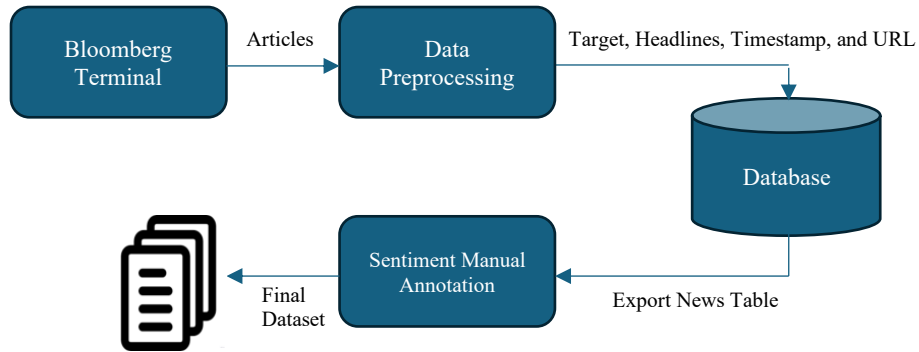


Figure 3.1: Dataset creation workflow.

To determine the dataset's dependability and confirm the experimental evaluation, each headline is manually annotated to evaluate the sentiment toward the considered target company. This technique involves examining the sentiment expressed in the headlines in relation to the set targets and categorizing these sentiments as positive, negative, or neutral. The annotations are performed by three annotators, experts in finance and economics, all having a high level of English proficiency (CEFR level C1).

The annotators are given comprehensive instructions that outline the tasks required for target-level annotation. Here are some examples that demonstrate the annotation guidelines given to the annotators:

- Assess the sentiment of each headline exclusively with respect to the target rather than considering the overall sentiment of the entire headline. Annotators should assume the viewpoint of an investor and evaluate the probable influence of the headline on investment decisions.
- Assign a sentiment score to a headline in relation to the target, using a scale: (0) for a neutral sentiment, (-1) for a negative sentiment, and (+1) for a positive sentiment.
- Formulate annotations based solely on the sentiment directly conveyed in the headline rather than relying on external information.
- Assign a neutral sentiment score of 0 to headlines that contain muddled sentiments, uncertainty, or ambiguity.
- Assign a neutral sentiment when there is no evident positive or negative sentiment regarding the target.

The manual annotation technique for news headlines involves an initial phase when the three annotators individually examine the same subset of 150 headlines. To quantify the level of agreement among the three coders and ensure the consistency of the manual labelling process, we employ Krippendorff's Alpha (Krippendorff, 2018) as a measure of inter-coder agreement. The calculated Krippendorff's Alpha score is 0.8188, which indicates a satisfactory level of agreement among the coders, thus confirming the reliability of the contributed annotations.

After ensuring the satisfactory level of intercoder agreement, the remaining headlines, which amount to 1326, are divided equally among the three raters. Each rater is assigned the task of personally annotating 442 headlines. Figure 3.2 reports the distribution of sentiment labels assigned by each annotator across the randomly distributed headline subsets. Because the remaining 1326 headlines were assigned randomly among annotators, the resulting label frequencies are broadly similar across annotators, suggesting the absence of substantial annotator-specific labelling bias.

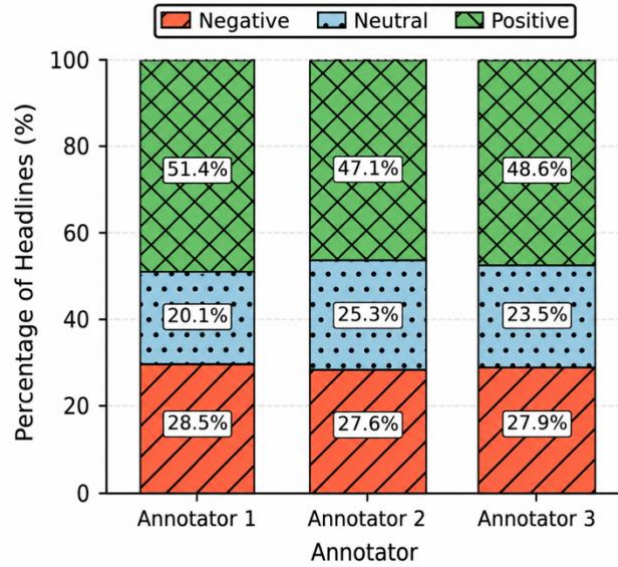


Figure 3.2: Distribution of sentiment labels assigned by annotators.

Table 3.1 presents examples of annotated headlines. Table 3.2 displays quantitative measures pertaining to our dataset, including the overall count of articles, as well as the standard deviation and average of daily articles, categorized by each target. Moreover, Figure 3.3 shows the distribution of sentiment specific to each target. Overall, the dataset is slightly unbalanced with roughly 49% of positive headlines, 28% of negative headlines, and 23% of neutral headlines.

Table 3.1: Examples of annotated headlines.

Target	Headline	Label
Amazon	Pioneer Disciplined Growth Buys More Nike Class B, Cuts Amazon	-1
Amazon	Amazon to Invest Over \$440m to Boost Delivery Drivers' Pay	1
Netflix	Netflix, Tarantino Win \$20 Million California Film Tax Credits	1
Netflix	Netflix Shares Dip Despite Apple's Surge in Streaming Subscribers	-1
Netflix	Netflix CFO Speaks at Bank of America Conference	0
Nvidia	Nvidia to Partner with India's Tata, Reliance for AI Development	1
Nvidia	Neuberger Berman Guardian Buys More Walmart, Cuts Nvidia	-1
Alphabet	Alphabet Hits 52-Week High at \$139.17	1

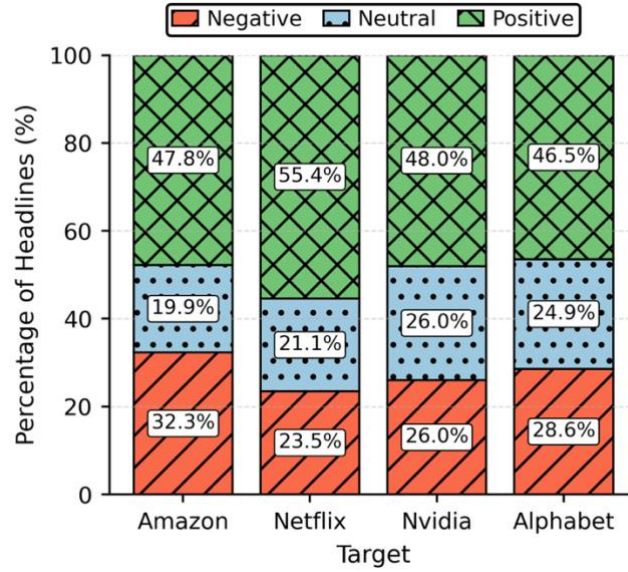


Figure 3.3: Distribution of sentiment in news headlines across targets.

Table 3.2: Dataset statistics.

Target	Articles	Daily articles
Amazon	573	5.07 (4.41)
Netflix	298	3.77 (5.14)
Nvidia	392	3.70 (4.28)
Alphabet	213	3.04 (3.95)
Total	1476	15.58 (17.78)

In order to enhance the research community and guarantee the reproducibility of our procedures, we publicly release our curated dataset⁵. This dataset, consisting of news headlines related to prominent stock companies and their relevant sentiments, is a substantial resource and is anticipated to be highly beneficial for researchers and practitioners who specialize in utilizing machine learning methods for target-level financial sentiment analysis. This work is expected to foster additional study in this field, thereby enhancing our collective comprehension of the complex connections between market behaviour, AI models, and financial news sentiment.

⁵ Due to copyright restrictions, this dataset only includes URLs of headlines; the actual text of the headlines cannot be released. However, the URLs can be used by users with access to a Bloomberg terminal to recreate the dataset. https://github.com/iftikharm895/Target-Level_Financial_Sentiment_Analysis

3.3.2. Target-Level Sentiment Classification Using Traditional Methods

This chapter utilizes three conventional models for financial sentiment analysis: VADER, DistilFinRoBERTa, and DeBERTa-v3-base-absa-v1.1. The VADER⁶ tool, developed by Hutto and Gilbert (2014), has been widely used by researchers in the field of financial sentiment analysis. It is a rule-based tool for performing document/sentence sentiment analysis. While specifically attuned to social media, VADER also performs well in other domains. Some notable studies that have utilized VADER in the financial domain include Sohangir et al. (2018), Agarwal (2020), Nemes and Kiss (2021), Padmanayana and Bhavya (2021), and Cristescu et al. (2022).

DistilFinRoBERTa⁷ is a lightweight variant of FinBERT, which is a model that was specifically pre-trained for understanding financial texts. This entails utilizing a financial collection of texts and refining the model for categorizing financial sentiment, incorporating resources like the in-house documents and annotations from John- Snow LAB and the Financial PhraseBank by Malo et al. (2014). DistilFinRoBERTa has been implemented in the financial sector for document/sentence sentiment analysis by scholars such as Rizinski et al. (2023) and Atak (2023).

DeBERTa-v3-base-ABSA-v1.1⁸ is a model specifically developed for ABSA. It uses the FAST-LCF-BERT model, which includes Microsoft/DeBERTa-v3-base and is sourced from PyABSA (Yang and Li, 2021; Yang et al., 2023). This model is trained using a large dataset of over 30000 ABSA samples. It is then fine-tuned with an additional 180000 examples from various ABSA datasets⁹, including SemEval-2014 Task 4 (Laptop14 and Restaurant14) (Kirange et al., 2014), MAMS (Jiang et al., 2019), SemEval-2016 Task 5 (Restaurant-16) (Pontiki et al., 2016), and other datasets specific to product categories such as T-shirts (Mukherjee et al., 2021; Rajapaksha and Ranathunga, 2022) and televisions (Cooray et al., 2021). This model has been utilized by various researchers for ABSA tasks, such as Boitel et al. (2023), Mahendru and

⁶ <https://github.com/cjhutto/vaderSentiment>

⁷ <https://huggingface.co/mrm8488/distilroberta-finetuned-financial-news-sentiment-analysis>

⁸ <https://huggingface.co/yangheng/deberta-v3-base-absa-v1.1>

⁹ <https://github.com/yangheng95/ABSADatasets>

Pandit (2024), and Mughal et al. (2024). We employ DeBERTav3- base-ABSA-v1.1 in two configurations: the available pre-trained model, and a fine-tuned version developed through 5-fold cross-validation¹⁰. For fine tuning, the model was trained on each fold for 10 epochs using default parameters. The entire 5-fold cross-validation process was completed in 8 min on a standard NVIDIA RTX 4090 GPU.

We remark that VADER and DistilFinRoBERTa provide sentence/document level sentiment analysis, while DeBERTa-v3-base-ABSA-v1.1 is developed to perform target/aspect-based sentiment analysis.

3.3.3. Target-Level Sentiment Classification with ChatGPT and Gemini

The recent rise of LLMs has been highly notable, showcasing their exceptional capacity to execute a wide range of text-oriented tasks (Yang et al., 2024). Presently, OpenAI’s ChatGPT¹¹ and Google’s Gemini¹² are two highly esteemed and widely used generative LLMs. The current study utilized multiple versions of the ChatGPT and Gemini models, such as ChatGPT-3.5, ChatGPT-4, ChatGPT-4o, Gemini 1 Pro, Gemini 1.5 Flash, and Gemini 1.5 Pro, to perform TLFSA. No fine-tuning of the models was conducted, as the objective of this study was to evaluate the zero-shot performance of these advanced generative AI tools on the given task. Interactions with ChatGPT-3.5, ChatGPT-4, and ChatGPT-4o were carried out using the ChatGPT user interface (UI) in its default configuration, accessible to all users, with a preset temperature of 0.7. Additionally, access to Gemini models (Gemini 1 Pro, Gemini 1.5 Flash, and Gemini 1.5 Pro) was obtained via the Gemini API. The recommended temperature settings provided by the API were followed: 0.9 for Gemini 1 Pro and 1.0 for both Gemini 1.5 Flash and Gemini 1.5 Pro.

¹⁰ 5-fold cross-validation tests the model on the entire dataset by ensuring every data point is used for both training and testing, but never at the same time. This helps avoid overfitting and provides a more reliable measure of model performance.

¹¹ <https://chatgpt.com/> (Versions available as of 30 June 2024).

¹² <https://gemini.google.com/app> (Versions available as of 30 June 2024).

To evaluate the performance of several ChatGPT and Gemini models in TLFSa, we utilized a zero-shot prompting method¹³ (Liu et al., 2023; Wei et al., 2023); that is, the model is instructed with an indication of how to perform a given task without providing any gold annotated example (i.e., a headline and the corresponding manual annotation in our context). More in detail, the following prompt was crafted:

Evaluate the sentiment conveyed by the headline with respect to the {target} from an investment perspective. Employ a three-tier scale for this assessment: allocate a score of (-1) for a negative sentiment, (0) for a neutral sentiment, and (+1) for a positive sentiment. Assign a neutral score of 0 if the headline is vague or does not clearly express positive or negative implications about the {target}.

By utilizing zero-shot prompting, we exploit the pre-existing training and comprehension of language context and patterns in ChatGPT and Gemini to produce desirable responses without the need for additional training or task-specific fine tuning. This method enables a straightforward evaluation of the intrinsic capacities of these models in sentiment analysis without any alterations, showcasing their potential for effortless integration and utilization in different contexts.

The performance of all the above models and methods was evaluated using a variety of well-established metrics for sentiment/classification tasks (Chicco and Jurman, 2020), such as precision, recall, F1-measure, and accuracy. Specifically, we report the aggregated performance across the three classes (Positive, Neutral, Negative) using both macro-averaging, where metrics are calculated for each class separately and averaged equally across all classes, and weighted averaging, where metrics are weighted by the number of samples in each class. These metrics provide a comprehensive evaluation of each model’s effectiveness in detecting target-level sentiments, even in the presence of a slightly imbalanced class distribution, as seen in our dataset (Chicco and Jurman, 2020; Rospocher and Eksir, 2023).

¹³ We also conducted preliminary experiments using a few-shot prompting strategy by incorporating a small number of gold-annotated headlines as in-context examples. These experiments did not yield substantial performance improvements over the zero-shot setting. A more comprehensive evaluation of few-shot learning strategies is provided in Chapter 4, where the analysis is extended to full-length news articles.

3.4. Results and Discussion

Table 3.3 reports the performance of each model on the novel dataset introduced in this study.

Table 3.3: Performance outcomes of target-level sentiment classification across models.

Model	Accuracy	Macro precision	Macro recall	Macro F1	Weighted precision	Weighted recall	Weighted F1
VADER	0.49	0.51	0.47	0.48	0.53	0.49	0.50
DistilFinRoBERTa	0.51	0.62	0.54	0.50	0.68	0.51	0.53
DeBERTa-v3	0.52	0.67	0.58	0.54	0.75	0.52	0.53
DeBERTa-v3 (ft)	0.87	0.85	0.84	0.84	0.87	0.87	0.87
ChatGPT-3.5	0.75	0.77	0.75	0.74	0.82	0.75	0.77
ChatGPT-4	0.86	0.84	0.83	0.83	0.86	0.86	0.86
ChatGPT-4o	0.84	0.82	0.83	0.82	0.86	0.84	0.84
Gemini 1 Pro	0.78	0.75	0.73	0.74	0.77	0.78	0.77
Gemini 1.5 Pro	0.83	0.82	0.83	0.81	0.86	0.83	0.84
Gemini 1.5 Flash	0.83	0.80	0.80	0.80	0.83	0.83	0.83

Notes: ft denotes the fine-tuned version of the model. Best scores are bold.

VADER exhibits the lowest performance across all criteria, highlighting its constraints in target-level sentiment analysis. VADER demonstrates a macro precision of 0.51, a macro F1-score of 0.48, a macro recall of 0.47, and an overall accuracy of 0.49. The weighted metrics provide more evidence of its insufficiency with a weighted F1-score of 0.50, a weighted precision of 0.53, and a weighted recall of 0.49. VADER, a lexicon-based technique originally designed for general sentiment analysis, particularly in social media contexts, faces difficulties in handling the subtle language and specialist terminology commonly found in financial literature. The system’s static and rule-based nature hinders its ability to accurately understand the changing linguistic nuances commonly seen in financial discussions. This limitation restricts its usefulness in dynamic and contextually sensitive situations.

Pre-trained models such as DistilFinRoBERTa and DeBERTa-v3-base-ABSA-v1.1 exhibit some improvement compared to VADER. DistilFinRoBERTa demonstrates a marginal improvement in accuracy, reaching a value of 0.51. This improvement is

accompanied by enhanced macro measures, including a recall of 0.54, a precision of 0.62, and an F1-score of 0.50. DeBERTa-v3-base-ABSA-v1.1 demonstrates improved performance, with an accuracy of 0.52, macro recall of 0.58, macro precision of 0.67, and macro F1-score of 0.54. These models perform better than VADER, which is likely because of their sophisticated structures and specialized training. DistilFinRoBERTa benefits from effective pre-training on various datasets in the financial domain, while DeBERTa-v3-base-ABSA-v1.1 uses a unique disentangled attention mechanism that is crucial for understanding target-level sentiments. Moreover, we recall that DeBERTa-v3-base-ABSA-v1.1 is specifically fine-tuned for target/aspect-level sentiment analysis, while VADER and DistilFinRoBERTa are meant for document/sentiment-level sentiment analysis.

The fine-tuned DeBERTa-v3-base-ABSA-v1.1 model stands out by achieving the highest recorded accuracy of 0.87 in this investigation. This model underwent thorough training and fine-tuning using our financial dataset, employing a 5-fold cross-validation approach. The model attains a macro recall of 0.84, macro precision of 0.85, and macro F1-score of 0.84. The weighted metrics also demonstrate its exceptional performance. The results underscore the significant advantages of refining this model on domain-specific data.

Among the LLMs, ChatGPT-4 achieves an outstanding performance, ranking second only to the highly optimized DeBERTa-v3-base-ABSA-v1.1. It achieves a macro average F1-score of 0.83 and an accuracy of 0.86. The performance of Gemini models, specifically Gemini 1.5 Pro and Gemini 1.5 Flash, is noteworthy, demonstrating significant improvements compared to previous versions such as Gemini 1 Pro.

To summarize, conventional models for document/sentence-level sentiment analysis, like VADER and DistilFinRoBERTa, proved inadequate for TLFSA, even if the latter is trained on financial text. This highlights the limitations of traditional sentiment analysis tools in handling the nuanced and context-specific requirements of target-level sentiment analysis, especially in specialized domains like finance. The pre-trained DeBERTa-v3-base-ABSA-v1.1, despite being built for the considered task, exhibits limitations when applied to the financial domain, which is likely due to the

different domain of its training data. This finding underscores the importance of domain adaptation in achieving optimal performance.

Fine-tuning DeBERTa-v3-base-ABSA-v1.1 on financial data significantly enhances its efficacy, as evidenced by the high performance achieved through 5-fold cross-validation. The success of this fine-tuning process illustrates the critical role of tailored training for domain-specific applications. This result aligns with prior research suggesting that pretrained models require domain-specific adaptation to handle specialized vocabulary, semantics, and contextual relationships effectively.

LLMs such as ChatGPT-4, ChatGPT-4o, and Gemini 1.5 Pro demonstrate substantial potential for the task. Despite not being explicitly fine-tuned on financial sentiment data, these models demonstrate performance levels comparable to those of fine-tuned DeBERTa-v3-base-ABSA-v1.1. This seems to suggest that LLMs have an inherent ability to generalize across tasks and domains, which is likely due to their extensive pre-training on diverse and large-scale datasets. Their capacity to perform well in TLFS without additional training makes them highly versatile and efficient alternatives for direct application in such contexts, reducing the need for resource-intensive manual annotation and fine-tuning processes. These outcomes are in line with the findings observed in different contexts and experiments, such as Brauwers and Frasincar (2022), Chumakov et al. (2023), and Magdaleno et al. (2023).

The evaluation framework adopted in this chapter primarily relies on standard multi-class classification metrics, including accuracy, precision, recall, and F1-score. While these measures provide a useful benchmark for comparing model performance, they treat sentiment categories as nominal rather than ordinal classes. Consequently, misclassifications between adjacent categories (e.g., negative versus neutral) receive the same penalty as misclassifications between opposite categories (e.g., negative versus positive). Future research could therefore complement the present evaluation with ordinal-sensitive measures, such as weighted Cohen's Kappa or confusion-matrix analysis, to provide additional insight into the structure and severity of classification errors. In addition, ensemble approaches combining the outputs of different sentiment models through majority-voting or weighted aggregation schemes may represent a

promising direction for improving robustness and classification performance in target-level financial sentiment analysis.

The current study on TLFSA reveals notable progress in sentiment analysis in the financial sector, specifically with the fine-tuned DeBERTa-v3-base-ABSA-v1.1 and LLMs like ChatGPT-4, ChatGPT-4o, and Gemini 1.5 Pro. These models exhibit very high effectiveness in capturing nuanced emotions, making them valuable instruments for businesses dependent on sentiment analysis for strategic decision making. These models can be incorporated into financial reporting and predictive systems through user-friendly APIs, allowing seamless integration with existing financial software. For financial technology firms, the integration of these models could result in more advanced financial reporting tools that combine sentiment data elicited from news headlines with other economic indicators, such as stock prices, trading volumes, interest rates, and macroeconomic trends, offering more comprehensive market predictions and supporting informed decision making. The versatility of LLMs like ChatGPT-4, which do not require extensive fine-tuning for financial data, reduces implementation time and costs, making them an appealing choice for both large institutions and smaller financial firms.

We recognize that integrating these models into real-world applications poses several challenges. A key issue is the interpretability of model outputs, as LLMs function as black boxes, generating outputs without transparent reasoning processes. This is particularly critical in financial contexts, where professionals require clear and justifiable explanations for sentiment predictions. Notably, ongoing research in Explainable AI (Zhao et al., 2024) aims to address this challenge. Additionally, the LLM models used in our work are maintained and controlled by external entities (e.g., OpenAI, Google), which frequently update them to enhance their capabilities. This dynamic necessitates regular performance evaluations to ensure their efficacy remains consistent. For example, periodic benchmarking on datasets such as the one contributed by this work can help monitor and validate their reliability for financial applications.

3.5. Conclusion

This study has performed a thorough assessment of target-level financial sentiment analysis (TLFSA), specifically examining the efficacy of conventional sentiment analysis techniques, such as VADER, DistilFinRoBERTa, and DeBERTa-v3-base-ABSA-v1.1, in comparison to advanced generative LLMs like ChatGPT and Gemini, within the domain of the stock market.

The study utilized a zero-shot prompting strategy to investigate the ability of generative LLMs to identify target-specific sentiments. This was accomplished by utilizing a novel, rigorously constructed dataset of annotated stock-related news headlines. The dataset is made available to the public to promote further research and innovation.

VADER and DistilFinRoBERTa exhibit reduced effectiveness for TLFSA, lacking the specialized capabilities to accurately detect and evaluate the sentiment toward specific targets within the text. Hence, their performance in sentiment analysis at the target level is lower than models specifically developed for this task, such as DeBERTa-v3-base-ABSA-v1.1. Indeed, the fine-tuned version of the DeBERTa-v3-base-ABSA-v1.1 model exhibits exceptional performance, attaining the top scores in all evaluation metrics. LLMs also achieve very competitive performance, with ChatGPT-4, ChatGPT-4.0, and Gemini 1.5 Pro demonstrating impressive abilities. These models obtained almost ideal outcomes without requiring any training at all, emphasizing their potential for direct practical use in TLFSA.

The findings obtained from this study are relevant for investors, financial analysts, and fintech companies, illustrating how both conventional approaches and advanced LLMs can improve decision-making processes. The existing performance leaderboard offers beneficial information for choosing models depending on individual performance requirements and the availability of resources. This study not only expands our understanding of advanced sentiment analysis tools but also lays the foundation for future advancements that could improve the accuracy of financial sentiment analysis, facilitating more strategic and informed decision making in financial markets.

Although the study has shown encouraging results, it also acknowledges limitations, specifically the limited range of data used, which could impact the applicability of our findings to a broader context. Future studies should (i) expand the dataset to include a broader range of financial documents and extend the data collection timeframe to better capture the diversity of market dynamics and (ii) incorporate additional existing datasets to evaluate the generalizability of the findings. Furthermore, it will be essential to evaluate the potential of other LLMs for financial sentiment analysis as generative AI technologies continue to evolve, including models such as BLOOM, LLaMA, and competitive small-scale models (e.g., ORCA, OLMO). A thorough assessment of their few-shot performance is also recommended to further corroborate our preliminary observations on this aspect.

CHAPTER IV

BENCHMARKING LARGE LANGUAGE MODELS FOR TARGET-BASED FINANCIAL SENTIMENT AND STOCK RETURN

4.1. Introduction

The financial sector, a pivotal pillar of the global economy, is increasingly influenced by vast amounts of unstructured textual data, including news articles, earnings call transcripts, regulatory filings, and analyst reports (Wu et al., 2024). These textual sources significantly impact investor decisions, market volatility, and strategic financial activities (Todd et al., 2024). The inadequacy of traditional manual methods for processing such extensive data has led to the adoption of automated procedures using NLP techniques (Cambria et al., 2014). Sentiment analysis, a crucial NLP tool, evaluates the emotional tone of the text, providing valuable predictive insights on investor sentiment and market movements (Nguyen et al., 2025).

Financial Sentiment Analysis (FSA), a specific subtask of NLP, identifies subjective tones in financial texts, offering insights for market forecasting, risk management, and the development of trading strategies (Du et al., 2024b). Methods for FSA range from conventional lexicon-based techniques and machine learning algorithms to advanced deep learning models, particularly transformer architectures (Rizinski et al., 2024). Recently, generative LLMs such as Llama, Gemma, ChatGPT, and DeepSeek have exhibited considerable promise in NLP tasks, especially in zero-shot and few-shot learning contexts, owing to their ability to reduce reliance on extensive manual annotations (Sivarajkumar et al., 2024; Lee et al., 2025). However, the efficacy of these models in specialized fields, such as finance, is still inadequately examined, underscoring the necessity for thorough assessment before their incorporation into practical applications like financial reporting software and market analysis systems.

A notably complex facet of sentiment analysis in financial texts is the recurrent presence of conflicting sentiments towards multiple entities within a single narrative (Wadawadagi et al., 2025). For example, the statement “Nvidia’s AI-driven growth overshadows Netflix’s subscriber stagnation” concurrently expresses positive and negative sentiments regarding two distinct entities. Conventional sentiment analysis methods at the sentence or document level frequently conflate these subtle perspectives, obscuring critical insights necessary for precise decision-making. To overcome this constraint, Target-Based Financial Sentiment Analysis (TBFSa) disaggregates sentiment at the entity level, facilitating a more detailed examination of specific financial instruments, business entities, or market segments (Deng et al., 2024; Zhang et al., 2025). Nonetheless, the capacity of LLMs to execute zero-shot and few-shot TBFSa tasks in financial markets remains insufficiently investigated. Furthermore, rigorous comparison analyses of lexicon-based tools, discriminative transformer-based approaches, and generative LLMs in this particular setting remain scarce.

This chapter aims to fill these significant gaps by evaluating the potential of LLMs to conduct target-specific sentiment analysis in financial news articles, with implications for financial text analysis and market monitoring. Specifically, we seek to answer the following research questions:

RQ1: How do zero-shot and few-shot generative LLMs perform in TBFSa compared to lexicon-based and discriminative transformer-based models?

RQ2: Does few-shot learning substantially improve the performance of LLMs compared to zero-shot methods in TBFSa?

RQ3: To what extent do sentiment scores derived from LLMs correlate with actual market returns, and how effectively do they align with observed market directionality?

Our contributions differ from previous studies in numerous significant aspects:

1. We develop and publicly release a novel, manually annotated TBFSa dataset comprising 1162 financial news articles categorized by target-specific sentiments. In contrast to current financial datasets (e.g., FiQA-2018¹⁴, Financial PhraseBank (Male et al., 2024)), our dataset distinctly encapsulates sophisticated target-level opinions within intricate financial narratives that exhibit conflicting sentiments.
2. Utilizing this dataset, we systematically evaluate generative LLMs (ChatGPT, Llama, Gemma, DeepSeek), conventional lexicon-based instruments (VADER, TextBlob), and discriminative transformer-based models (Finbert, DistilFinRoBERTa, Finbert-Tone, DeBERTa-v3-base-absa-v1.1), emphasizing the strengths and limitations of each approach specifically in the context of TBFSa. This extensive comparison investigation is among the first to critically evaluate advanced LLMs' performance in zero-shot and few-shot frameworks for target-level financial sentiment analysis.
3. Beyond traditional NLP evaluation measures, we conduct Pearson correlation analysis and directional accuracy tests to assess the alignment of sentiment scores from LLMs with stock market returns. This dual-metric methodology improves the practical relevance of our results, providing additional evaluation criteria for financial sentiment analysis and market monitoring applications.

This chapter presents the extended version of our work initially published in the *proceedings of the Eleventh Italian Conference on Computational Linguistics (CLiC-it 2025)*¹⁵, conducted in collaboration with Prof. Marco Rospocher, Prof. Dr. Slavko Žitnik, and Dr. Timotej Knez. The conference paper constitutes a preliminary version of this work, while the full version presented in this chapter is currently under review at the *International Journal of Data Science and Analytics*.

¹⁴ <https://sites.google.com/view/fiqa/home>

¹⁵ <https://aclanthology.org/2025.elicit-1.74/>

The subsequent sections of this chapter are structured as follows: Section 4.2 presents relevant literature on financial sentiment analysis. Section 4.3 delineates the establishment of our dataset, annotation processes, and methodological techniques. Section 4.4 delineates empirical findings and discussion, while Section 4.5 concludes the chapter and provides key implications and avenues for future research.

4.2. Related Work

The evolution of financial sentiment research over the past decade has been considerably influenced by developments in NLP and the growing accessibility of financial textual data. Previous research predominantly aligns with three main paradigms: lexicon-based approaches, discriminative transformer-based models, and generative LLMs.

4.2.1. Lexicon-Based Methods

Lexicon-based approaches, which form the foundation of financial sentiment analysis, initially drew from general-purpose instruments such as LIWC and SentiWordNet. However, these tools lacked domain-specific accuracy and contextual nuance (Ribeiro et al., 2016). Frameworks like VADER and TextBlob were then developed to incorporate contextual scoring and automatic lexicon enhancement (Hutto and Gilbert, 2014; Aljedaani et al., 2022). Numerous scholars have utilized VADER in the financial domain (Siek and Chandra, 2024; Saleem et al., 2024; Nagendra et al., 2024). Siek and Chandra (2024) confirmed the accuracy of VADER in sentiment classification for financial headlines, whereas Saleem et al. (2024) established a correlation between VADER-derived Reddit sentiment and Bitcoin price variations. Nagendra et al. (2024) similarly identified robust connections between Twitter sentiment in earnings announcements and subsequent equity fluctuations. However, it struggles to handle sector-specific terminology (Khandelwal et al., 2024), highlighting the need for more effective solutions. Similarly, TextBlob, which integrates predefined lexicons with a classifier trained on film reviews, allows for swift implementation in initial analyses. However, it falls short in complex financial scenarios due to its inadequate domain adaptation (Khandelwal et al., 2024).

While lexicon-based methods have been practical, they face significant challenges in deciphering complex linguistic patterns, domain-specific vocabulary, and contextual nuances (Zhao et al., 2024; Muhammad and Rospoche, 2025). These limitations have led to transformer-based models leveraging deep learning to capture semantic and contextual subtleties more effectively in financial texts.

4.2.2. Discriminative Transformer-Based Models

Transformer-based architectures, particularly BERT (Devlin et al., 2019), transformed NLP by employing self-attention techniques that effectively capture contextual relationships. Although general transformers excel at conventional NLP tasks, their effectiveness declines in financial contexts due to specialized lexicons and nuanced tone differences. As a result, domain-specific models fine-tuned on financial data have developed an increased sensitivity to the subtleties of financial language and numerical settings (Liu et al., 2021).

FinBERT, trained initially on financial documents like SEC filings and subsequently fine-tuned with the FiQA dataset, represented a notable progression in financial sentiment analysis (Shen et al., 2024). Studies conducted by Liu et al. (2021) and Mahendran et al. (2025) confirmed FinBERT's superiority compared to general-purpose models. Expanding on this, FinBERT-Tone implemented tonal analysis to discern subtle sentiment indications essential for market forecasting (Huang et al., 2023). Initiatives to improve efficiency, shown by DistilFinRoBERTa, tailored for real-time applications, have also garnered attention (Atak, 2023). Furthermore, sophisticated models like DeBERTa-v3-base-absa-v1.1 exhibited accuracy in aspect- and target-oriented sentiment analysis, adeptly interpreting intricate narratives in financial documents (Mughal et al., 2024; Muhammad and Rospoche, 2025).

Comparative assessments consistently demonstrate that optimized transformer-based models exceed traditional lexicon-based and machine-learning methodologies (Liu et al., 2021; Kim et al., 2023; Shen et al., 2024). Nevertheless, their demand for processing resources and extensive labelled datasets has initiated the exploration of

generative LLMs as viable alternatives that scale more effectively with fewer task-specific labels.

4.2.3. Generative Large Language Models

Recent developments in LLMs have shown exceptional proficiency in financial sentiment analysis, surpassing conventional lexicon-based and discriminative transformer-based methodologies (Kim et al., 2023; Mahendran et al., 2025). The intricate linguistic characteristics of financial texts have prompted the creation of specialized LLMs, such as BloombergGPT and FinVis-GPT, specifically tailored for the financial sector (Lee et al., 2024). Models such as InvestLM, especially fine-tuned for investing environments, have demonstrated effectiveness equivalent to commercial advice systems (Ramya Devi and Sasidharan, 2024).

Furthermore, recent research highlights the efficacy of smaller, computationally efficient models, attaining performance akin to larger LLMs via focused fine-tuning (Ma et al., 2024; Alizadeh et al., 2025). Methods like parameter-efficient tuning (e.g., LoRA) have enhanced their utilization in practical financial scenarios (Agarwal and Gupta, 2024). Notably, even general-purpose models such as ChatGPT have exhibited remarkable proficiency in financial sentiment analysis without the necessity for domain-specific fine-tuning (Fatouros et al., 2023; Mathebula et al., 2024; Kang et al., 2024).

Despite the significant progress in sentiment analysis, the focus has been mainly on general sentiment analysis tasks, leaving a gap in target-based sentiment analysis within financial contexts. The current chapter, aiming to fill this gap, conducts a comprehensive benchmarking of advanced generative LLMs, such as Llama 3-8B, Gemma 2 (9B and 27B), DeepSeek-R1, and various ChatGPT models (4, 4o, o1). We compare these models with conventional lexicon-based and discriminative transformer-based models for target-oriented financial sentiment analysis. Although the current study focuses on TBFS, prior research indicates that LLMs are applicable to a wider range of sentiment and financial text tasks, such as document-level sentiment analysis and mixed-domain corpora. Therefore, the insights gained from

TBFSA benchmarking may be applicable to tasks beyond the specific task under investigation.

4.3. Methodology

This section delineates the methodological framework utilized to assess the performance of generative LLMs, specifically Gemma, Llama, ChatGPT, and DeepSeek, in executing TBFSA. To effectively benchmark these LLMs, we utilized various lexicon-based sentiment analysis tools, specifically VADER and TextBlob, in conjunction with discriminative transformer-based models, including FinBERT, DistilFinRoBERTa, FinBERT-Tone, and DeBERTa-v3-base-absa-v1.1. We began by outlining our methodology for dataset collecting and annotation, a meticulous process that ensured high reliability and validity criteria. Subsequently, we fine-tuned the benchmark discriminative transformer-based models utilizing this dataset to achieve optimal alignment with the specific requirements of financial sentiment analysis. To thoroughly assess the generative LLMs, we designed precise, task-oriented prompts appropriate for TBFSA. Thereafter, we conducted a comprehensive comparative study to evaluate the efficacy and robustness of LLMs compared to the benchmark models for TBFSA. Finally, to evaluate the financial significance of the sentiment outputs, we assessed LLMs employing Pearson correlation and Directional Accuracy.

4.3.1. Dataset Construction and Annotation

To establish a thorough evaluation framework, we obtained historical daily closing stock prices from Yahoo Finance and news articles from the Bloomberg Terminal regarding four prominent stock companies (Alphabet, Amazon, Netflix, and Nvidia), covering the period from September 4, 2023, to January 30, 2024. The assembled dataset comprises 1,170 articles. Each article was systematically analysed to extract critical information, including the timestamps, news text (excluding headlines), and URLs. Non-text elements (e.g., embedded charts, disclaimers, and advertisements) were removed, and duplicate articles were eliminated to mitigate bias. The refined dataset was subsequently stored in a structured database (as depicted in Figure 4.1).

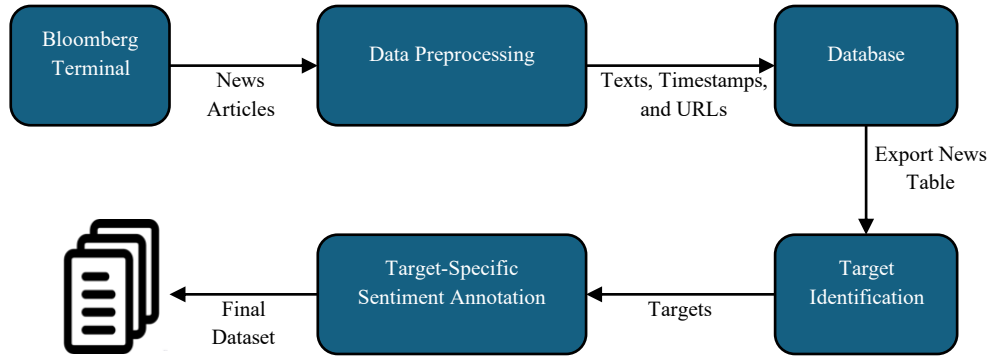


Figure 4.1: Dataset construction process.

In contrast to the headline-based dataset introduced in Chapter III, the full-length articles examined in this chapter contain substantially richer and more complex narrative structures. Many articles reference multiple firms simultaneously, include contextual qualifications and evolving discussions, and may express heterogeneous or conflicting evaluations across different sections of the text. Consequently, the dataset provides a more demanding environment for target-level sentiment analysis, requiring models to attribute sentiment accurately to specific firms within longer and contextually nuanced financial narratives.

Each article was meticulously annotated for sentiment concerning the target companies to ensure data quality and confirm the experimental evaluation. The annotation was carried out by three annotators with extensive expertise in finance and economics, all possessing advanced English competence (CEFR level C1). Their annotations were guided by comprehensive guidelines aimed at standardizing target identification and sentiment assessment. A concise summary of these guidelines entails:

- A thorough examination of each article to identify direct references to the target entities: Alphabet, Amazon, Netflix, and Nvidia.
- Identification of multiple target entities within a single article, where applicable.
- Labelling articles devoid of explicit target references as “no target.”

- Evaluation of sentiment from an investor's viewpoint, relying exclusively on the textual content.
- Sentiment classification as positive (1), negative (-1), or neutral (0).
- Identification of prevailing sentiment in instances of mixed expressions.
- Neutral labelling for vague, ambiguous, or passing references.

The annotating procedure was organized into two separate phases. The annotators initially conducted target identification individually across all 1170 articles. Eight articles were excluded as having "no target" by consensus. Inter-annotator reliability for target identification yielded a Krippendorff's alpha (Krippendorff, 2018) of 0.96 and a percentage agreement (Artstein and Poesio, 2008) of 98.95% for the remaining 1162 articles, signifying consistent annotations. Texts with majority-agreed targets were forwarded for sentiment annotation, yielding 1334 unique annotation cases due to multiple target references within specific articles.

In the second phase, sentiment annotation was performed for all identified target entities. Annotators used a defined scale to assign sentiments: '1' for positive, '-1' for negative, and '0' for neutral sentiment. To ensure consistency, annotators collaboratively annotated a shared subset of 150 texts, resulting in satisfactory inter-annotator reliability (Krippendorff's Alpha of 0.8142; percentage agreement of 83%). The sentiment labels for the 150 texts were established by majority consensus, and the remaining 1184 texts were allocated evenly among annotators for individual sentiment labelling. Figure 4.2 reports the distribution of sentiment labels assigned by each annotator across the randomly distributed text items following the initial overlapping annotation phase. Because the remaining 1184 text items were assigned randomly among annotators, the resulting label frequencies are broadly similar across annotators, suggesting the absence of substantial annotator-specific labelling bias.

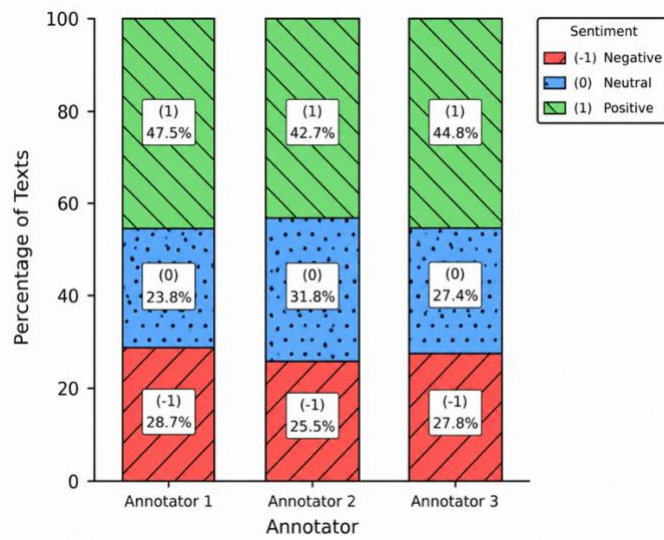


Figure 4.2: Distribution of sentiment labels assigned by annotators.

The final annotated dataset consists of 1334 texts; each explicitly associated with a target entity and an annotated sentiment label. The dataset demonstrates a moderate class imbalance, with positive sentiments accounting for 45%, negative sentiments for 27%, and neutral sentiments for 28%. Table 4.1 presents annotated instances, whereas Figure 4.3 visually represents the sentiment distribution. Additional quantitative parameters, including the total number of news texts, average daily texts, average text length (measured in tokens), and average target mentions, are outlined in Table 4.2.

Table 4.1: Instances of annotated texts.

Target	Text	Label
Amazon	Amazon's CEO Worldwide Amazon Stores Douglas J Herrington reported an insider stock transaction to the US Securities and Exchange Commission. Filing included 10(b)5-1 transactions. Amazon shares fell 3% in the past month.	-1
Amazon	Amazon Japan says it will build its first "sort center" in Japan in Shinagawa, Tokyo, located ~3.5km from Haneda International Airport. Expects to create ~1000 new jobs. Will handle as many as 750000 items/day.	1
Netflix	Netflix co-CEO Ted Sarandos says talks with striking actors broke down after the union asked for a "levy" on streaming customers. Sarandos speaks at the first-ever Bloomberg Screentime conference in Los Angeles.	-1
Netflix	Jamie Lumley, Senior Analyst at Third Bridge, discusses the latest earnings from Netflix and outlook for the streaming services. He speaks with Annabelle Droulers and Paul Allen on ""Daybreak Australia".	0
Nvidia	The projected ex-date for Nvidia's dividend moved to Dec. 6 from Nov. 30, according to an updated Bloomberg Dividend Forecast. The new ex-date falls after the Dec. 1 option expiry.	0
Nvidia	Nvidia Corp. shares are on pace for their longest streak of advances since a record-setting dash in December 2016, as the company announced updates to its artificial intelligence processors. Simone Foxman reports on Bloomberg Television.	1
Alphabet	Alphabet Inc. shares tumbled the most in a year on Wednesday after the Google parent reported a smaller than expected profit in cloud computing, raising concerns about its position in a market critical to its future. Ed Ludlow reports.	-1

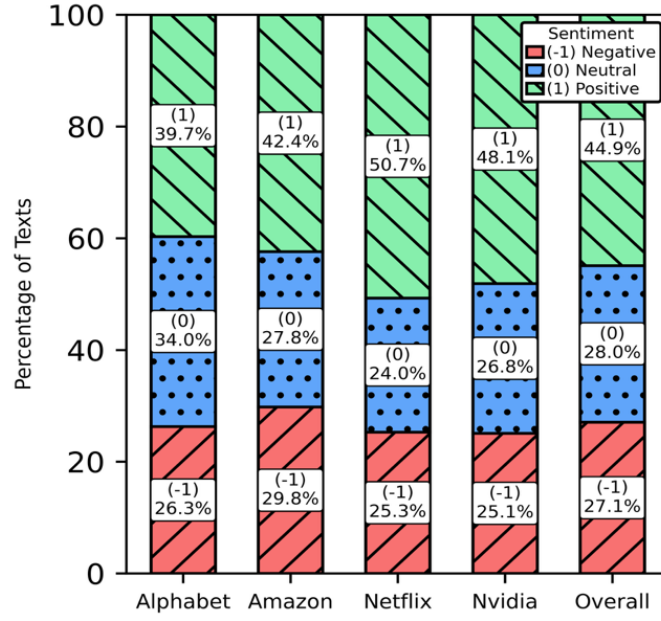


Figure 4.3: Sentiment distribution.

Table 4.2: Dataset statistics.

Target	No of texts	Daily texts	Text tokens	Target mentions
Alphabet	247	2.84 (2.59)	456.85 (574.30)	3.29 (4.43)
Amazon	496	4.82 (3.13)	538.31 (590.15)	5.19 (6.48)
Netflix	233	3.11 (3.31)	245.92 (259.06)	3.18 (3.13)
Nvidia	358	3.81 (3.47)	381.97 (427.39)	3.32 (3.47)
Total	1334	14.58 (12.50)	430.20 (511.70)	3.99 (5.00)

Notes: The values in parentheses denote standard deviations.

We publicly release our curated dataset¹⁶ to assist the academic community and guarantee methodological transparency and reproducibility. The dataset is limited to metadata due to copyright restrictions, encompassing publication dates, timestamps, specified target entities, and associated source URLs, while omitting the complete content of the news articles. Researchers with Bloomberg Terminal access can retrieve the full article content using the provided URLs, subject to licensing constraints.

¹⁶ https://github.com/iftikharm895/Target-Based_Sentiment_Analysis_in_Financial_News

4.3.2. Baseline Models

To meticulously assess generative LLMs in TBFSa, we have conducted a comprehensive comparison of their efficacy with established benchmarks: lexicon-based instruments (TextBlob, VADER) and discriminative transformer architectures (FinBERT, FinBERT-Tone, DistilFinRoBERTa, DeBERTa-v3-absa-v1.1).

TextBlob¹⁷, an open-source Python library developed on the Natural Language Toolkit (NLTK) and Pattern libraries, assigns sentiment polarity scores ranging from -1 to $+1$ and has been widely utilized for financial texts, including stock news and market reports (Sohangir et al., 2018; Nemes and Kiss, 2021; Khandelwal et al., 2024). VADER¹⁸, developed by Hutto & Gilbert (2014), a rule-based framework, incorporates lexical, grammatical, and syntactic heuristics, validated against LIWC and ANEW, and has also been extensively employed in financial contexts (Patil et al., 2024; Hernández-Pérez et al., 2024; Muhammad and Rospoche, 2025; El Idrissi et al., 2025).

Discriminative transformer-based baselines comprise: (1) DistilFinRoBERTa¹⁹, a distilled variant of RoBERTa fine-tuned on financial datasets (e.g., Financial PhraseBank and John Snow Labs) for three-class sentiment analysis (Atak, 2023); (2) FinBERT²⁰, a BERT adaptation pre-trained on earnings calls news articles and regulatory filings and fine-tuned on Financial PhraseBank (Malo et al., 2014); (3) FinBERT-Tone²¹ (Huang et al., 2023), which enhances FinBERT to identify tonal nuances (e.g., optimism, uncertainty, and legal implications), fine-tuned on SEC filings, earning reports, and financial news; and (4) DeBERTa-v3-absa-v1.1²², builds upon the DeBERTa-v3 architecture, has been fine-tuned for Aspect-Based Sentiment Analysis (ABSA) through the FAST-LCF-BERT framework (Yang et al., 2023). It is trained on an extensive dataset, comprising 30000 ABSA-specific samples and further

¹⁷ <https://textblob.readthedocs.io/en/dev/>

¹⁸ <https://github.com/cjhutto/vaderSentiment>

¹⁹ <https://huggingface.co/mrm8488/distilroberta-finetuned-financial-news-sentiment-analysis>

²⁰ <https://huggingface.co/ProsusAI/finbert>

²¹ <https://huggingface.co/yiyanghkust/finbert-tone>

²² <https://huggingface.co/yangheng/deberta-v3-base-absa-v1.1>

fine-tuned on an additional 180000 annotated examples from a variety of datasets²³, including SemEval (Pontiki et al., 2016), MAMS (Jiang et al., 2019), and multiple domain-specific corpora (Cooray et al., 2021; Mukherjee et al., 2021; Rajapaksha and Ranathunga, 2022). These discriminative transformer-based models have been extensively employed in financial sentiment research (Atak, 2023; Voigt et al., 2024; Dmonte et al., 2024; Nach, 2025).

The current study involved fine-tuning DistilFinRoBERTa, FinBERT, and FinBERT-Tone using a learning rate of 3×10^{-5} , 10 training epochs, and a batch size of 32. For DeBERTa-v3-absa-v1.1, we utilized a 5-fold cross-validation approach, with each fold trained for 10 epochs utilizing default hyperparameters on an NVIDIA RTX 4090 GPU.

4.3.3. Evaluated Generative LLMs

Recent improvements in LLMs have garnered significant academic interest owing to their proven effectiveness in several text-based tasks (Yang et al., 2024). Notable and widely utilized models include OpenAI's ChatGPT²⁴, Gemma²⁵—a series of open models based on Google's Gemini architecture, Meta's LLaMA²⁶, and DeepSeek²⁷.

The current study assessed the efficacy of various advanced generative LLMs within the framework of TBFSA. The evaluated models include ChatGPT-4, ChatGPT-4o, ChatGPT-o1, LLaMA 3 8B, Gemma 2 9B, Gemma 2 27B, and DeepSeek-R1. All models were assessed in their default configurations, without any additional fine-tuning, to evaluate their zero-shot and few-shot capabilities in executing the specified task. Interactions with ChatGPT variations were executed via OpenAI's standard web interface, utilizing a temperature setting of 0.7. The Gemma models were accessed via the Gemini API, which suggests a temperature setting of 1.0 for both the 9B and 27B variants. DeepSeek-R1 was accessed via its public chat interface, employing its

²³ <https://github.com/yangheng95/ABSADatasets>

²⁴ <https://chatgpt.com/>. The ChatGPT variants analyzed in this chapter are limited to those available during the research period.

²⁵ <https://gemini.google.com/app>

²⁶ <https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

²⁷ <https://www.deepseek.com/>

standard temperature setting. To interact with the LLaMA model, we utilized a local instance of the Meta-Llama-3-8B-Instruct model, running under the Ollama²⁸ application. For testing purposes, we used default hyperparameters and advanced optimization techniques, including 4-bit quantization (Huang et al., 2024), to efficiently execute this model on consumer-grade GPU systems.

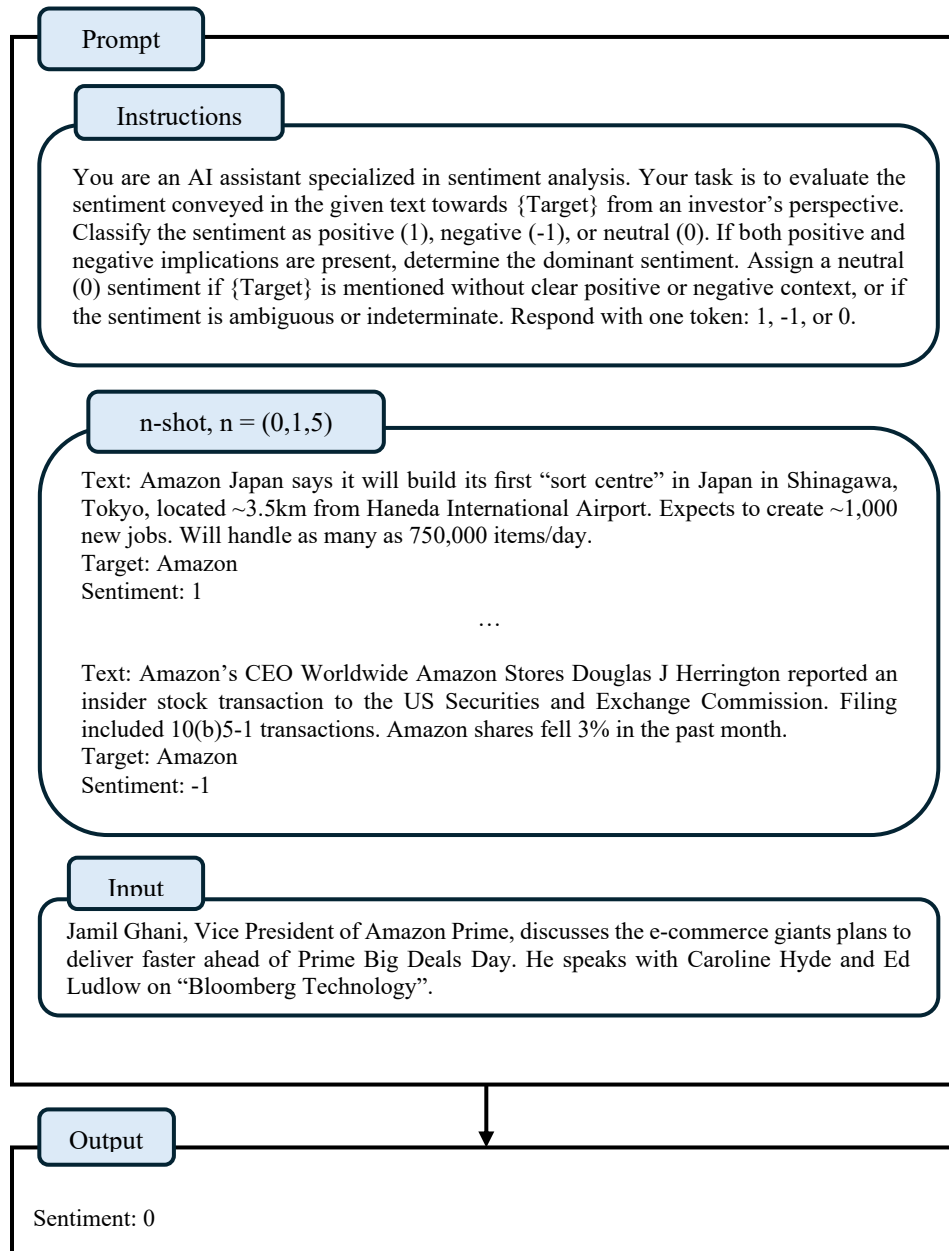


Figure 4.4: Prompt used in zero/few-shot learning approach for LLMs.

²⁸ <https://ollama.com/library/llama3>

We utilized zero-shot and few-shot prompting strategies to assess generative LLMs' performance in TBFSA. In the zero-shot context, models were given task instructions without illustrative examples. In few-shot contexts, prompts were augmented by either one (1-shot) or five (5-shot) annotated examples for contextual grounding. Figure 4.4 depicts the prompts employed in both zero-shot and few-shot contexts. This approach utilizes LLMs' inherent language and contextual reasoning abilities, enabling performance evaluation without requiring task-specific training or model adaptation. The method offers a clear assessment of model generality and adaptability, enhancing their suitability for effortless implementation in diverse practical applications.

The evaluation of model performance utilized recognized criteria for sentiment categorization, including precision, accuracy, recall, and F1-score (Chicco and Jurman, 2020). The metrics were calculated across three sentiment categories (negative, neutral, and positive), utilizing both macro-averaging (equal weight across classes) and weighted averaging (weighted by class sample size) to ensure robustness amid moderately imbalanced class distributions (Rospocher and Eksir, 2023).

To evaluate the financial significance of the sentiment outputs, we further assessed selected LLMs (ChatGPT-4, ChatGPT-4o, ChatGPT-o1, and DeepSeek-R1) employing Pearson correlation and Directional Accuracy (DA). For each target, sentiment scores were aggregated on a daily basis by averaging the scores of all $N_{d,T}$ texts associated with that target within a given day to obtain the overall sentiment score, $S_{d,T}$, for target T on day d :

$$S_{d,T} = \frac{\sum_{i=1}^{N_{d,T}} s_{i,T}}{N_{d,T}} \quad (4.1)$$

Where:

$S_{d,T}$ denotes the overall sentiment score for target T on day d .

$N_{d,T}$ represents the number of texts associated with target T on day d .

$s_{i,T}$ refers to the sentiment score assigned to the i th text associated with target T .

To evaluate possible alignment with market performance, the resulting daily sentiment scores, $S_{d,T}$, were correlated with the actual daily returns, $R_{d,T}$, of the corresponding targets using Pearson's correlation coefficient:

$$R_{d,T} = \left(\frac{P_{d,T}}{P_{d-1,T}} - 1 \right) \times 100 \quad (4.2)$$

Where:

$R_{d,T}$ denotes the percentage return for target T on day d .

$P_{d,T}$ represents the closing price of target T on day d .

$P_{d-1,T}$ represents the closing price of target T on day $d-1$.

Daily sentiment scores were computed by aggregating all news items associated with a given target that were published on calendar day d according to Bloomberg timestamps and averaging their sentiment scores as shown in Eq. (4.1). Daily stock returns were computed on a close-to-close basis for the same trading day using Eq. (4.2). We acknowledge that alternative alignment schemes, such as aggregating news between the previous trading day's market close (16:00) and the current day's close, may affect correlation and directional accuracy estimates. However, the objective of this investigation is to conduct a comparative evaluation under a consistent alignment protocol, rather than to optimize temporal alignment for return predictability.

Directional Accuracy (DA) measures the percentage of days where the forecasted sentiment direction aligns with the actual market return direction (Blaskowitz and Herwartz, 2011). If the model accurately forecasts market direction on 70 out of 100 days, it attains a directional accuracy of 70%. To contextualize these findings, we established a naive baseline that consistently forecasts each target's most frequent sentiment direction and the DA that was attained with ground-truth sentiment annotations. The former establishes a fundamental standard, especially pertinent due to class imbalances in real-world data, whilst the latter measures the intrinsic correlation between sentiment and market fluctuations within the dataset.

This dual evaluation framework, consisting of linguistic performance metrics and financial alignment indicators, offers a comprehensive assessment of the models' utility for financial sentiment analysis and market monitoring tasks.

4.4. Results and Discussion

The efficacy of TBFSA is assessed through various modelling techniques, demonstrating significant discrepancies influenced mainly by differences in model architecture and overall capability. Table 4.3. presents the outcomes for all models evaluated on the novel dataset introduced in this research.

Table 4.3: Performance outcomes of target-based sentiment classification across models.

Model	Shot	Accuracy	Macro precision	Macro recall	Macro F1	Weighted precision	Weighted recall	Weighted F1-score
TextBlob	-	0.46	0.40	0.39	0.35	0.41	0.46	0.39
VADER	-	0.50	0.48	0.41	0.37	0.48	0.50	0.41
FinBERT	-	0.56	0.53	0.54	0.54	0.58	0.56	0.57
DistilFinRoBERTa	-	0.61	0.56	0.57	0.57	0.61	0.61	0.61
FinBERT-Tone	-	0.63	0.61	0.62	0.62	0.66	0.63	0.63
Deberta-v3-absa-v1.1	-	0.68	0.67	0.67	0.66	0.68	0.68	0.68
Llama 3 8B	0	0.68	0.75	0.62	0.63	0.72	0.68	0.66
Gemma 2 9B	0	0.66	0.69	0.65	0.66	0.71	0.66	0.67
Gemma 2 27B	0	0.69	0.70	0.68	0.69	0.71	0.69	0.70
ChatGPT-4	0	0.79	0.78	0.77	0.77	0.79	0.79	0.79
ChatGPT-4o	0	0.78	0.78	0.77	0.77	0.79	0.78	0.78
ChatGPT-o1	0	0.81	0.81	0.80	0.80	0.82	0.81	0.81
DeepSeek-R1	0	0.82	0.84	0.81	0.81	0.83	0.83	0.82
Llama 3 8B	1	0.52	0.60	0.44	0.44	0.54	0.40	0.43
Gemma 2 9B	1	0.66	0.70	0.66	0.66	0.72	0.66	0.67
Gemma 2 27B	1	0.71	0.72	0.71	0.71	0.73	0.71	0.72
ChatGPT-4	1	0.80	0.79	0.79	0.79	0.81	0.80	0.80
ChatGPT-4o	1	0.81	0.81	0.80	0.80	0.82	0.81	0.81
ChatGPT-o1	1	0.85	0.85	0.84	0.84	0.85	0.85	0.85
DeepSeek-R1	1	0.83	0.84	0.82	0.83	0.83	0.83	0.83
Llama 3 8B	5	0.64	0.69	0.61	0.63	0.64	0.64	0.63
Gemma 2 9B	5	0.65	0.68	0.65	0.65	0.71	0.65	0.66
Gemma 2 27B	5	0.72	0.71	0.71	0.71	0.73	0.72	0.72
ChatGPT-4	5	0.82	0.81	0.80	0.80	0.82	0.82	0.82
ChatGPT-4o	5	0.82	0.82	0.81	0.82	0.83	0.82	0.83
ChatGPT-o1	5	0.86	0.86	0.85	0.86	0.86	0.86	0.86
DeepSeek-R1	5	0.87	0.88	0.86	0.87	0.87	0.87	0.87

Lexicon-based approaches, such as VADER and TextBlob, exhibit consistently subpar performance across all evaluation metrics. For example, VADER achieves a macro-precision of 0.48, a macro-recall of 0.41, a macro F1-score of 0.37, and an overall accuracy of 0.50. The weighted metrics (F1: 0.41; precision: 0.48; recall: 0.50) underscore its constrained effectiveness. TextBlob attains a macro-precision of 0.40, macro-recall of 0.39, macro F1-score of 0.35, and accuracy of 0.46, with weighted metrics (F1: 0.39; precision: 0.41; recall: 0.46) indicating similar deficiencies. These models are limited by their dependence on static, general-purpose sentiment lexicons that do not incorporate domain-specific financial language, in addition to their document-level emphasis and rigid rule-based architecture. As a result, they fail to capture the contextual intricacies and entity-specific sentiment differentiations necessary for effective TBFSA.

Conversely, discriminative transformer-based models optimized for financial sentiment analysis tasks substantially exceed the performance of lexicon-based models. FinBERT attains an accuracy of 0.56, accompanied by macro recall, precision, and F1-scores of 0.54, 0.53, and 0.54, respectively. DistilFinRoBERTa enhances these outcomes, achieving an accuracy of 0.61 and macro metrics (recall: 0.57; precision: 0.56; F1-score: 0.57). FinBERT-Tone demonstrates progressive improvements, achieving an accuracy of 0.63 and macro scores (recall: 0.62; precision: 0.61; F1-score: 0.62). DeBERTa-v3-base-ABSA-v1.1 has superior performance among fine-tuned transformers, attaining a macro-F1 score of 0.66 via five-fold cross-validation on the financial dataset. The model's exceptional performance is due to its advanced architecture. It incorporates disentangled attention mechanisms tailored for target/aspect-level sentiment tasks, facilitating precise token-level sentiment attribution while reducing the overfitting risks prevalent in general-purpose models.

A significant distinction arises in the design objectives of the assessed baseline models: DeBERTa-v3-base-ABSA-v1.1 is developed explicitly for target/aspect-based sentiment analysis, while lexicon-based methods (VADER, TextBlob), and other discriminative transformer-based models, such as FinBERT, FinBERT-Tone, and DistilRoBERTa, predominantly focus on sentence- or document-level sentiment classification.

Among the generative LLMs assessed under zero-shot settings, DeepSeek-R1 demonstrates the highest performance, with an accuracy of 0.82 and a macro-F1 score of 0.81. Subsequently, ChatGPT-o1 exhibits an accuracy of 0.81 and a macro-F1 score of 0.80, followed by ChatGPT-4 with an accuracy of 0.79 and a macro-F1 score of 0.77, and ChatGPT-4o with an accuracy of 0.78 and a macro-F1 score of 0.77. Performance is generally enhanced with few-shot prompting. In the 1-shot situation, ChatGPT-o1 attains the highest accuracy (0.85) and macro-F1 score (0.84), closely followed by DeepSeek-R1 (accuracy and macro-F1: 0.83). Performance is optimal in the 5-shot configuration, with DeepSeek-R1 achieving an accuracy and a macro-F1 score of 0.87, surpassing ChatGPT-o1 (accuracy and macro-F1 score: 0.86). These findings underscore the importance of few-shot learning, especially in improving contextual adaptation capacities. However, smaller models such as Llama 3 8B demonstrate significant instability in few-shot scenarios. The zero-shot accuracy is 0.68 (macro-F1: 0.63), but performance deteriorates with 1-shot prompting (accuracy: 0.52; macro-F1: 0.44), showing only a modest recovery in the 5-shot scenario (accuracy: 0.64; macro-F1: 0.63). This deterioration underscores the difficulties smaller models face in contextual adaptability and example-based learning. Their restricted parameter count limits their ability to clarify ambiguities, prioritize task-specific signals, or disentangle indirect sentiment cues associated with targets. Larger models such as ChatGPT and DeepSeek, on the other hand, utilize scale, advanced attention mechanisms, and RLHF-aligned training to enhance prediction accuracy, highlighting the inherent trade-offs between computational efficiency and nuanced reasoning.

In summary, lexicon-based sentiment analysis methods like VADER and TextBlob are insufficient for TBFSA because they fail to capture contextual financial semantics. Discriminative transformer-based models such as DistilFinRoBERTa, FinBERT, and FinBERT-Tone provide quantifiable enhancements but remain inadequate regarding precision and entity-level interpretability. Domain-adapted models like DeBERTa-v3-base-ABSA-v1.1, although tailored for target/aspect-based tasks, are surpassed by generative LLMs such as ChatGPT variants and DeepSeek-R1. The consistent success of ChatGPT-4, ChatGPT-4o, ChatGPT-o1, and DeepSeek-R1 on the TBFSA task demonstrates the efficacy of comprehensive pre-training, which equips these LLMs to

perform exceptionally in zero/few-shot scenarios and generalize across several domains without requiring task-specific fine-tuning. Their consistent superiority over conventional lexicon-based systems and discriminative transformer-based models underscores a significant transition towards generative LLMs that integrate high adaptability with robust domain-agnostic generalization, thus providing an efficient substitute for resource-intensive supervised methods in specialized tasks such as TBFSa. These findings are consistent with previous studies in related fields (Chumakov et al., 2023; Magdaleno et al., 2024; Krugmann and Hartmann, 2024).

Moreover, we analyse the correlation between sentiment outputs and real market performance to evaluate the practical utility of generative LLMs in financial settings. We specifically compare the sentiment scores generated by ChatGPT-4, ChatGPT-4o, ChatGPT-o1, and DeepSeek-R1 with the daily stock returns of the corresponding target companies. Section 4.3 delineates the methodology for calculating aggregate sentiment scores. Pearson correlation coefficients and directional accuracy (DA), as detailed in Section 4.3, are utilized to assess the congruence between sentiment predictions and actual financial outcomes.

The correlation between market returns and sentiment scores is vital for assessing model efficacy. This linkage is measured using Pearson correlation coefficients spanning from -1 to $+1$. Accordingly, higher positive correlation indicates stronger sample-specific alignment between aggregated sentiment signals and contemporaneous stock returns; however, such correlation should not be interpreted as a universal measure of market reaction, causality, or tradable predictability. Rather, correlation values offer complementary evidence regarding the degree to which sentiment measures are in close alignment with observed market outcomes within the constraints of the studied dataset.

Figure 4.5 displays a correlation matrix from pairwise correlations obtained from daily aggregated datasets, as outlined in Section 4.3. This matrix depicts the linkages between different generative LLMs, annotated sentiment data, and daily market returns. The strongest significant correlation (0.87) is between true sentiment and DeepSeek-R1, indicating that DeepSeek-R1 most accurately corresponds with human-

annotated sentiment labels. ChatGPT-o1 exhibits a robust correlation with true sentiment (0.80), signifying a reasonable degree of reliability. ChatGPT-4 and ChatGPT-4o demonstrate a significant reciprocal correlation of 0.90, indicating internal consistency. Nonetheless, the correlations between ChatGPT models and DeepSeek-R1 are relatively weaker (e.g., ChatGPT-4 at 0.67), indicating discrepancies in sentiment interpretation among the models. Moreover, the correlation between true sentiment and daily market returns is approximately 0.32, indicating a modest linear relationship. This suggests that, although sentiment signals may partially explain variations in short-horizon stock returns, a substantial portion of return dynamics is likely influenced by additional market, firm-specific, and macroeconomic factors. While the reported correlations provide useful evidence regarding sentiment-market alignment, future research could further strengthen the analysis through formal significance testing and multiple-testing adjustments across targets and model specifications.

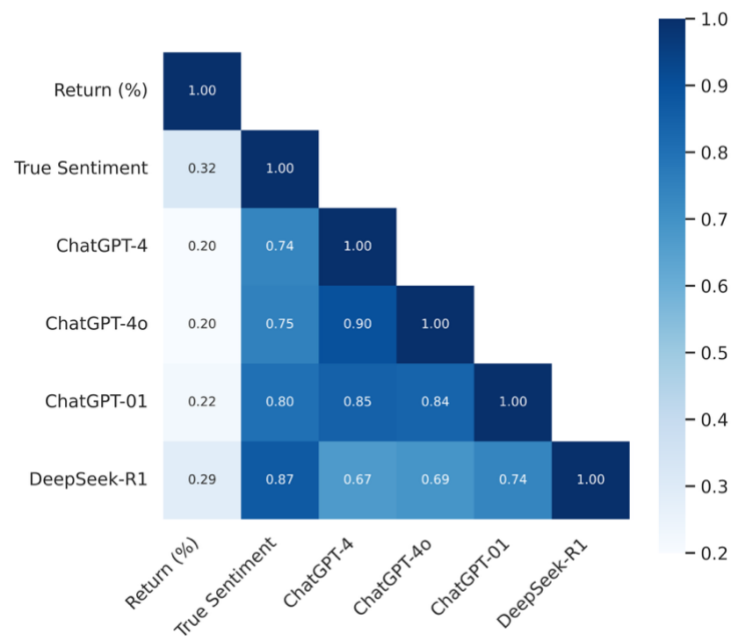


Figure 4.5: Correlation matrix of predicted sentiments and stock returns.

Table 4.4 presents a comprehensive correlation analysis between predicted sentiment scores and stock returns across the best-performing generative LLMs, elucidating their efficacy across various targets. The findings demonstrate that DeepSeek-R1

consistently surpasses all ChatGPT variants across various shot configurations, attaining its peak correlation of 0.29 in the five-shot setting. The most significant overall correlation is noted for true sentiment at 0.32, highlighting the disparity between model-derived sentiment predictions and actual market behaviour. ChatGPT-o1 demonstrates notable performance among the ChatGPT variants, especially in the five-shot context, attaining an overall correlation of 0.22. DeepSeek-R1 (five-shot) shows the strongest correlation for individual stocks, including Amazon (0.13), Nvidia (0.39), and Netflix (0.33). However, correlation strength varies significantly across firms. Netflix consistently demonstrates the strongest alignment between sentiment scores and market returns, whereas Amazon demonstrates the weakest correlation across models and for the true sentiment signal. This implies that Amazon’s short-horizon daily returns during the sample period are less aligned with news-based sentiment, which may be attributed to changes in investor attention, broader market-wide influences, or firm-specific information environments. This highlights firm-level heterogeneity and reinforces the need for firm-specific interpretation of correlation results.

Table 4.4: Correlation between predicted sentiments and stock returns for best-performing generative LLMs.

Model	Shot	Amazon	Netflix	Nvidia	Alphabet	Overall
ChatGPT-4	0	0.05	0.30	0.26	0.22	0.21
ChatGPT-4o	0	0.03	0.31	0.18	0.18	0.18
ChatGPT-o1	0	0.03	0.32	0.24	0.18	0.21
DeepSeek-R1	1	0.11	0.29	0.35	0.21	0.25
ChatGPT-4	1	0.03	0.28	0.23	0.21	0.19
ChatGPT-4o	1	-0.01	0.33	0.18	0.19	0.19
ChatGPT-o1	1	-0.04	0.31	0.27	0.32	0.21
DeepSeek-R1	1	0.11	0.31	0.36	0.26	0.27
ChatGPT-4	5	0.03	0.29	0.24	0.20	0.20
ChatGPT-4o	5	0.05	0.32	0.21	0.19	0.20
ChatGPT-o1	5	0.05	0.32	0.24	0.20	0.22
DeepSeek-R1	5	0.13	0.33	0.39	0.26	0.29
True Sentiment	-	0.09	0.35	0.46	0.36	0.32

The findings emphasize the capacity of generative LLMs to deliver insightful indicators for financial market analysis while also revealing diversity in performance among models and targeted companies. DeepSeek-R1 emerges as the most promising model for estimating true sentiment trends.

Table 4.5 presents the top-performing generative LLMs’ directional accuracy (DA), along with the naïve baseline and the DA based on true sentiment labels. The naïve method, which forecasts the predominant sentiment direction for each target, attains a baseline DA of 54.04%. Conversely, the DA utilizing true sentiment attains 69.08%.

Table 4.5: Directional accuracy of best-performing generative LLMs in predicting stock return movements.

Model	Shot	Amazon	Netflix	Nvidia	Alphabet	Overall
ChatGPT-4	0	42.72	42.67	58.51	50.57	48.75
ChatGPT-4o	0	45.63	44.00	55.32	47.13	48.19
ChatGPT-o1	0	49.51	52.00	55.32	42.53	49.86
Deepseek-R1	0	56.31	56.00	64.89	47.13	56.27
ChatGPT-4	1	42.72	46.67	56.38	48.28	48.47
ChatGPT-4o	1	47.57	48.00	50.00	45.98	47.91
ChatGPT-o1	1	44.66	50.67	59.57	47.13	50.42
Deepseek-R1	1	56.31	56.00	69.15	51.72	58.50
ChatGPT-4	5	43.69	46.67	58.51	49.43	49.58
ChatGPT-4o	5	46.60	49.33	57.45	43.68	49.30
ChatGPT-o1	5	51.46	54.67	58.51	43.68	52.09
Deepseek-R1	5	60.19	58.67	71.28	52.87	61.00
True Sentiment	-	66.02	66.67	76.60	66.67	69.08
Naïve Sentiment	-	52.43	48.00	60.64	54.02	54.04

Among the assessed models, DeepSeek-R1 consistently attains the highest DA across all shot configurations (56.27% in zero-shot, 58.50% in one-shot, and 61.00% in five-shot). The ChatGPT models exhibit increased variability, with ChatGPT-o1 surpassing ChatGPT-4 and ChatGPT-4o in most settings. Specifically, ChatGPT-o1 achieves a 52.09% DA in the five-shot arrangement. Although increasing the number of shots leads to performance enhancements, DeepSeek-R1 exhibits the most pronounced gains, indicating superior few-shot learning capabilities. Conversely, ChatGPT-4 and ChatGPT-4o demonstrate diminished improvements with additional shots.

Notwithstanding these enhancements, all LLMs exhibit subpar performance compared to the true sentiment benchmark (69.08%). Furthermore, the naïve baseline outperforms specific ChatGPT models in particular cases, such as predictions for Nvidia and Alphabet, underscoring the difficulties in accurately capturing intricate financial sentiment.

An investigation at the target level indicates that Nvidia sentiment is the most precisely predicted, with DeepSeek-R1 attaining a 71.28% DA in the five-shot configuration, approaching the benchmark of 76.60%. Although the DA for Amazon, Netflix, and Alphabet is somewhat steady, the highest precision attained by DeepSeek-R1 for these companies (60.19%, 58.67%, and 52.87%, respectively) falls short of the benchmark of 66%. The results indicate that while LLMs exhibit potential in financial sentiment analysis, they remain inadequate in capturing complex sentiment dynamics relative to human-labelled data.

Figure 4.6 presents a comparative examination of target-based sentiment prediction performance among the best-performing generative LLMs (ChatGPT-4, ChatGPT-4o, ChatGPT-o1, and DeepSeek-R1) across various shot learning configurations (0-shot, 1-shot, and 5-shot). The findings demonstrate that all assessed models exhibit advantages from few-shot learning, with performance improvements noted across all sentiment categories compared to the zero-shot condition. Significantly, ChatGPT-o1 demonstrates the highest gain in negative sentiment during the transition from 0-shot to 5-shot, whereas DeepSeek-R1 shows the most substantial improvement in neutral sentiment, especially in the 0-shot to 5-shot interval. Regarding positive sentiment, ChatGPT-4 demonstrates the most significant overall improvement, particularly during the transitions from 0-shot to 5-shot. These findings collectively highlight the efficacy of few-shot learning in improving sentiment prediction accuracy across various LLMs.

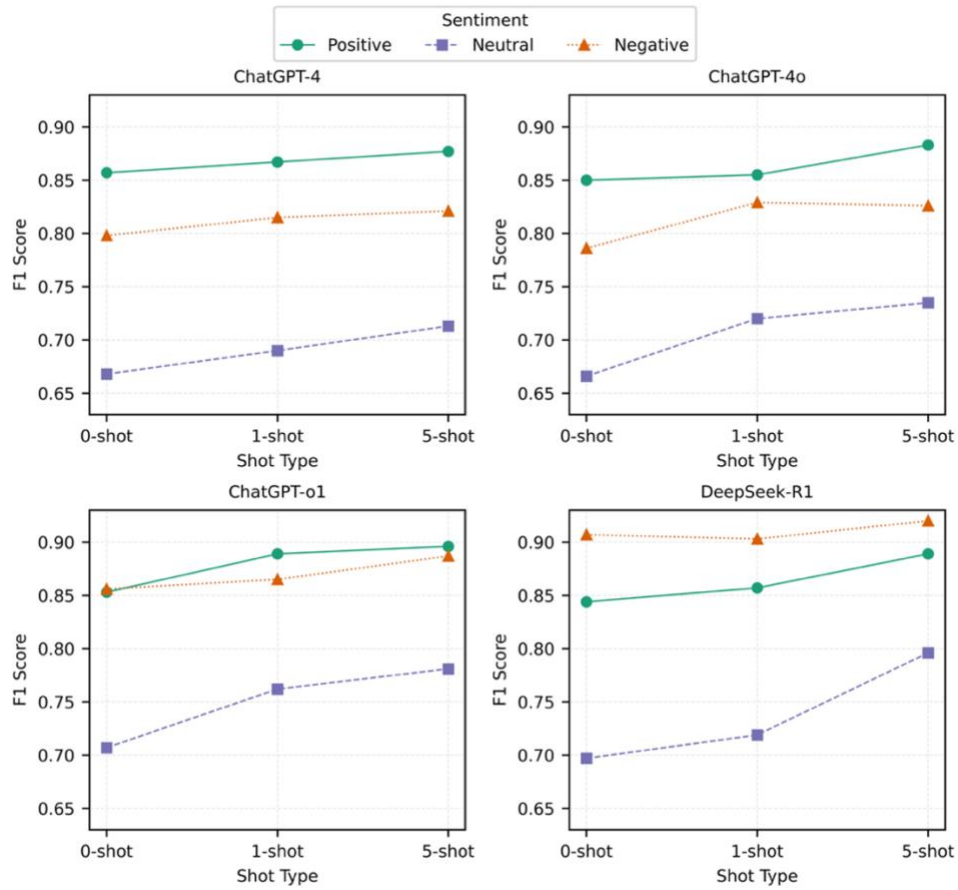


Figure 4.6: Comparison of 0-shot and few-shot performance of generative LLMs across sentiment classes.

As discussed in Chapter III, the sentiment categories considered in this study possess an ordinal structure, implying that misclassifications between adjacent classes are less severe than errors between opposite sentiment categories. Future research could therefore extend the present evaluation using ordinal-sensitive metrics, such as weighted Cohen’s Kappa, as well as ensemble prediction approaches that combine complementary strengths across different sentiment models.

To summarize, the study’s findings highlight the efficacy of advanced LLMs, namely ChatGPT-4, ChatGPT-4o, ChatGPT-o1, and DeepSeek-R1, in discerning nuanced financial sentiment from stock-related articles. ChatGPT-o1 and DeepSeek-R1 demonstrated superior performance in target-level sentiment analysis, reflecting their strong ability to identify contextual financial sentiment that closely correlates with stock market movements. Such granular, entity-specific sentiment interpretation holds

substantial implications for investors, financial analysts, and algorithmic trading systems. These advanced models allow stakeholders to participate in more informed decision-making, potentially improving portfolio management techniques and optimizing market timing decisions. Furthermore, the observed association between sentiment scores and stock returns indicates that sentiment signals may act as predictive market performance indicators. Moreover, the directional accuracy of sentiment-driven predictions corresponds with stock returns and accurately represents market trends. These findings confirm the efficacy of sentiment analysis as a significant enhancement to conventional financial analysis for market prediction and risk evaluation.

However, implementing LLMs in financial markets presents obstacles. Significant processing complexity and inference latency limit their applicability in ultra-high-frequency trading, where execution times are quantified in milliseconds. Moreover, regulatory issues arise from the intrinsic opacity of LLM decision-making, which contradicts compliance requirements such as MiFID II and SEC Rule 15c3-5 that necessitate model interpretability for audit and risk governance. These limitations underscore the need for transdisciplinary innovation. Strategies like model distillation for developing lightweight, domain-adapted LLMs and establishing explainability frameworks by regulatory norms are crucial for reconciling prediction accuracy with practical feasibility. The effective incorporation of LLMs into financial analytics will likely rely on hybrid architectures that combine language capabilities with conventional econometric models. These hybrid architectures hold the potential to revolutionize financial analytics, balancing traditional financial metrics' interpretability with AI's adaptive learning capabilities, and thereby mitigating the risks linked to opaque algorithmic decision-making. Furthermore, long-term research is crucial for assessing the durability of sentiment-driven alpha in diverse market conditions, ascertaining if its predictive efficacy is maintained or declines as market players adjust to LLM-driven signals. Resolving these complexities necessitates collaboration among AI researchers, economists, and regulatory authorities to ensure that innovations, such as federated learning for data privacy and synthetic financial text generation for enhanced training robustness, are implemented ethically and effectively.

4.5. Conclusion

The current chapter offers a comprehensive evaluation of target-based financial sentiment analysis (TBFSA), systematically comparing the effectiveness of cutting-edge generative LLMs, including ChatGPT, DeepSeek, LLaMA, and Gemma, with conventional lexicon-based methods (VADER, TextBlob) and discriminative transformer-based models (FinBERT, DistilFinRoBERTa, FinBERT-Tone, and DeBERTa-v3-base-ABSA-v1.1). The research employs a novel, meticulously manually annotated dataset of stock-related Bloomberg news articles, highlighting the transformative potential of LLMs in capturing nuanced financial sentiment via zero-shot and few-shot prompting frameworks.

Although lexicon-based techniques such as VADER and TextBlob provide computational efficiency, they demonstrate restricted effectiveness in target-specific sentiment tasks because of their inflexible, rule-based frameworks. Conversely, discriminative transformer-based models fine-tuned on financial corpora, specifically FinBERT, FinBERT-Tone, and DistilFinRoBERTa, exhibited significantly enhanced performance using domain-specific semantic comprehension. DeBERTa-v3-base-ABSA-v1.1 achieved superior results compared to both lexicon-based approaches and generative LLMs (e.g., LLaMA-3-8B and Gemma-2-9B) due to architectural enhancements such as a disentangled self-attention mechanism, which encodes each token with distinct content and relative-position vectors, facilitating precise modelling of target-opinion interactions and the subtle sentiment indicators associated with them.

The standout finding of this chapter is the exceptional efficacy of LLMs, especially ChatGPT variants (notably ChatGPT-o1) and DeepSeek-R1, surpassing all baseline models in target-level sentiment analysis. These models attained the highest directional accuracy and demonstrated statistically significant correlations with stock market returns. Their capacity to deduce implicit sentiment, adapt to financial terminology, and function efficiently without task-specific fine-tuning makes them scalable, ready-to-deploy solutions for practical applications like algorithmic trading and real-time risk assessment. These findings bear immediate implications for financial institutions, fintech developers, and analysts seeking to incorporate

sentiment-driven insights into investing and risk management processes. LLMs like DeepSeek-R1 and ChatGPT-o1 obviate the necessity for resource-demanding fine-tuning, thereby providing scalable and economical options appropriate for swift implementation under fluctuating market conditions.

Despite the promising findings, the current chapter acknowledges numerous limitations. The investigation is confined to news articles from four prominent technological firms (Alphabet, Amazon, Netflix, and Nvidia), potentially constraining the generalizability of the findings to other industries or smaller market-cap companies where sentiment-return dynamics may differ. Additionally, the focus on large-cap technology companies suggests a degree of sectoral concentration that may affect the sentiment characteristics observed and limit direct extrapolation to other economic sectors. The chapter encompasses a short time frame (September 4, 2023, to January 30, 2024), offering preliminary insights while potentially neglecting long-term patterns, seasonal fluctuations, and macroeconomic changes; an extended analysis period could yield more substantial conclusions. As this period largely coincides with a generally positive market environment, the reported sentiment-return relationships should be interpreted as conditional on the sampled market regime rather than as universally robust effects. Although zero- and few-shot prompting strategies were utilized for generative LLMs, the prompts were not entirely optimized, indicating that variations in prompt engineering could affect model outputs and necessitate further investigation. The sole dependence on news articles neglects other vital data sources, such as social media sentiment, earnings reports, and macroeconomic indicators, which could enhance the research. Moreover, while useful for assessing contemporaneous sentiment-return alignment, Pearson correlation and directional accuracy are descriptive measures that do not capture predictive performance, tradability, or potential causal mechanisms. In addition, they do not account for potential lagged market responses to news sentiment.

To address these constraints, future research could broaden the analysis to encompass various sectors and global markets, integrate additional data sources, and prolong the study over several years to assess LLM performance over various market regimes, including bull and bear market cycles. Additional robustness checks, including

alternative forecasting horizons, lagged return structures, alternative sentiment aggregation procedures, portfolio construction methods, and trading-based assessments incorporating transaction costs and execution frictions, also represent promising directions for future research. Enhancing prompt designs via automated techniques, investigating time-lagged sentiment effects, and improving the interpretability of LLM outputs signify promising avenues for attaining more robust, comprehensible, and sector-agnostic applications of LLM-driven financial sentiment research.

In conclusion, the current chapter's findings underscore the significance of LLMs in sentiment extraction and in improving alignment between sentiment signals and observed market movements within the studied sample, representing a notable advancement in financial NLP. As financial markets progressively incorporate AI-driven insights, further investigation into sectoral generalization, prompt optimization, and multi-source sentiment aggregation will enhance the robustness, interpretability, and broader applicability of LLM-based models in real-world financial decision-making.

CHAPTER V

FORECASTING RISK-ADJUSTED STOCK RETURNS WITH FIRM-SPECIFIC NEWS SENTIMENT: EVIDENCE FROM LARGE LANGUAGE MODELS

5.1. Introduction

Financial markets are intricate, information-driven systems in which asset prices continuously incorporate fundamentals, investor expectations, and evolving market sentiment (Chiu et al., 2018; Wang et al., 2023). Although traditional asset-pricing models focus on fundamental factors influencing returns, substantial evidence indicates that investor perceptions and emotions, driven by digital news, social media, and analyst commentary, are integral to price formation and short-term mispricing (Brown and Cliff, 2005; Baker and Wurgler, 2006; Sakariyahu et al., 2024; Sathya and Prabhavathi, 2024; Gorzon et al., 2024). Behavioural biases, including optimism, pessimism, and overconfidence, can lead to deviations of prices from their intrinsic values, highlighting the importance of understanding how sentiment propagates through markets for theories of price discovery, risk management, and the development of trading strategies.

The rapid proliferation of digital financial media has generated unprecedented volumes of textual data that increasingly influence investor perceptions and market responses (Benhamou et al., 2025). In this context, sentiment encompasses not only the positive or negative tone conveyed in text but also, more broadly, investor sentiment, specifically, the beliefs, expectations, and attitudes inferred from financial discourse that may affect trading behaviour and asset prices. Advances in NLP facilitate the extraction of such sentiment from news articles, earnings reports, and analyst commentary on a large scale, with progressively enhanced contextual understanding (Claus and Stella, 2022; Žitnik et al., 2022; Kim et al., 2024). However, conventional lexicon-based methods frequently fall short in capturing the semantic intricacies of financial language (Loughran and McDonald, 2011; Rizinski et al.,

2024), and topic models often neglect the deeper contextual relationships vital for conducting firm-level sentiment analysis (Bybee et al., 2020). Transformer-based architectures and financial-domain models such as FinBERT have enhanced contextual comprehension (Araci, 2019), and recent generative large language models (LLMs) such as ChatGPT and DeepSeek provide further gains through the generation of context-aware, entity-specific sentiment signals (Nakagawa et al., 2024; Fan et al., 2025; Muhammad et al., 2025). Nevertheless, despite these advancements, the predictive and economic significance of such fine-grained sentiment for short-term stock return forecasting remains inadequately examined.

Moreover, existing research often relies on document- or sentence-level sentiment analysis, which may obscure diverse signals directed at multiple firms within the same text (Žitnik et al., 2022; Muhammad et al., 2025). Furthermore, the informational content of textual sources, such as headings versus full-length articles, may variably affect return predictability, a distinction that has received limited systematic examination. Although machine learning models frequently surpass linear benchmarks in predicting asset returns (Gu et al., 2020; Basher and Sadorsky, 2022), limited understanding exists regarding whether these improvements correspond to economically meaningful, risk-adjusted performance measures such as statistically significant alpha.

To bridge these gaps, this chapter combines firm-specific sentiment derived from both domain-specific models (FinBERT) and generative LLMs (ChatGPT-o3, DeepSeek-R1) with lagged returns and technical indicators to predict one-, two-, and three-day-ahead returns for leading U.S. technology firms. Predictive models, including logistic regression, Random Forest and XGBoost, are assessed through both statistical performance indicators, including accuracy and F1-score, and economic metrics, such as Sharpe ratios, maximum drawdowns, win rates, and Fama-French five-factor model alpha.

In this chapter, we focus on three research questions:

RQ1: To what extent does firm-specific sentiment derived from headlines and full-length articles using domain-specific and generative LLMs provide incremental information for predicting short-horizon stock returns?

RQ2: How do different predictive modelling approaches, specifically linear classifiers versus nonlinear ensemble methods, influence the accuracy and robustness of sentiment-augmented return forecasts?

RQ3: Do sentiment-enhanced ensemble models generate statistically significant, risk-adjusted alpha, indicating meaningful economic value beyond improvements in predictive accuracy?

The current chapter offers several contributions. First, we construct fine-grained, firm-specific sentiment measures derived from both headlines and full articles, facilitating a systematic assessment of the impact of textual depth on return predictability. Second, we elucidate horizon-dependent patterns in the predictive efficacy of sentiment signals, offering new evidence on the diffusion of narrative information into prices over short intervals. Third, we analyse linear and nonlinear predictive frameworks that utilize LLM-derived sentiment, illustrating the enhanced efficacy of ensemble approaches in capturing nonlinear sentiment-return relationships. Finally, we evaluate the economic importance of predictive enhancements by explicitly analysing risk-adjusted returns and statistically significant alpha, thereby connecting predictive modelling with actionable trading outcomes. Collectively, these contributions enhance comprehension of sentiment-driven return dynamics and provide additional evidence regarding the economic benefits of incorporating advanced LLM-based sentiment with machine learning methodologies in short-horizon trading strategies.

This chapter is based on joint work with Prof. Marco Minozzo, Prof. Marco Rospocher, and Prof. Chris Biemann. The manuscript is currently under review at the *Journal of Forecasting*.

The subsequent sections of this chapter are structured as follows. Section 5.2 reviews the pertinent literature. Section 5.3 delineates the datasets, feature engineering, and methodology. Section 5.4 presents the empirical findings, encompassing predictive and economic assessments. Section 5.5 concludes with key insights and directions for future research.

5.2. Literature Review

Investor sentiment and behavioural biases have long been acknowledged as significant factors influencing asset price volatility, posing challenges to traditional models grounded in rational expectations and market efficiency (De Long et al., 1990; Shleifer and Vishny, 1997; Brown and Cliff, 2005; Baker and Wurgler, 2006; Ahmad et al., 2017). Sentiment-driven mispricing occurs when optimistic or pessimistic expectations temporarily cause prices to deviate from their fundamental values, subsequently reverting to the mean as markets incorporate new information (Bhatia and Bredin, 2016; Ding et al., 2019). These effects exhibit significant temporal variability: intraday and overnight returns react differently to sentiment owing to disparities in liquidity, investor composition, and trading frictions (Lou et al., 2019; Hendershott et al., 2020; Wang, 2023). Sentiment may serve as a momentum indicator over short timeframes but tends to reverse over extended periods (Brown and Cliff, 2004; Han and Li, 2017; Cheema et al., 2020), with these effects becoming more pronounced during recessions or times of increased uncertainty (García, 2013; Huang, 2019; Hanna et al., 2020).

Text-based sentiment analysis has become a fundamental aspect of empirical asset pricing. Early research indicates that the tone of news, corporate disclosures, and online commentary provides predictive information regarding returns (Tetlock, 2007; Heston and Sinha, 2017; Li et al., 2014; Sul et al., 2017). News obtained from conventional media outlets and social platforms predicts market trends, frequently with minimal reversals over short horizons (Gu and Kurov, 2020; Gan et al., 2020). Moreover, methodological advancements have progressively improved the extraction of information from financial texts. Conventional lexicon-based methods (Loughran and McDonald, 2011) face challenges in accurately capturing financial context,

resolving linguistic ambiguity, and addressing domain-specific nuances, thereby restricting their predictive effectiveness (Malo et al., 2014; Rizinski et al., 2024). Topic models such as latent dirichlet allocation (LDA) enhance scalability but are limited in their ability to capture long-range dependencies and complex semantic relationships (Bybee et al., 2020; Moreno-Pérez and Minozzo, 2024). Transformer-based architectures provide significant advancements. Financial-domain models such as FinBERT offer contextualized representations tailored to specialized terminology (Araci, 2019; Huang et al., 2023), whereas recent generative LLMs, including ChatGPT, Gemini, and DeepSeek, exhibit robust abilities in detecting subtle, entity-specific sentiment and reasoning patterns (Muhammad and Rospocher, 2025; Nakagawa et al., 2024; Mo et al., 2025). Empirical research indicates that sentiment derived from LLMs can surpass traditional sentiment indices in both predictive accuracy and economic performance metrics (Lopez-Lira and Tang, 2023; Ahmed et al., 2024).

A parallel body of literature investigates the incorporation of sentiment with market-related features such as returns, volatility, and technical indicators. Previous research integrating price and news data indicates that sentiment-enhanced models predict market movements more accurately than models relying solely on price data (Lavrenko et al., 2000; Akita et al., 2016). Recent multimodal learning methodologies that integrate news, social media indicators, and technical variables significantly enhance predictive accuracy and trading profitability by effectively capturing intricate cross-modal interactions (Avramelou et al., 2024; Farimani et al., 2024; Taylor and Ng, 2024). High-frequency analyses additionally demonstrate that swift changes in sentiment, particularly negative ones, induce immediate and frequently asymmetric price reactions, notably among firms experiencing significant arbitrage constraints (Broadstock and Zhang, 2019; Seok et al., 2019; Gao and Liu, 2020; Yoon and Takahashi, 2025).

Furthermore, machine learning models have become highly effective instruments for financial forecasting owing to their capacity to model nonlinear relationships and complex high-dimensional feature interactions. Tree-based ensemble models such as Random Forest and XGBoost often outperform linear benchmarks, especially when

incorporating textual or alternative data sources (Gu et al., 2020; Basher and Sadorsky, 2022; Bouteska et al., 2024). Nonetheless, only a limited number of studies evaluate whether enhancements in predictive accuracy correspond to economically significant, risk-adjusted returns. Gu et al. (2020) demonstrate that ensemble models and neural networks achieve significantly higher Sharpe ratios compared to linear models, while Daul et al. (2022) reveal that boosted trees effectively capture interaction effects that linear models consistently overlook, resulting in enhanced portfolio performance. These findings emphasize the importance of evaluating predictive models within rigorous asset-pricing frameworks that account for risk-adjusted alpha rather than prediction alone.

Despite these advancements, significant gaps remain. First, a significant portion of the current literature depends on sentiment analysis at the document or sentence level, thereby neglecting the diverse signals directed toward individual firms within multi-entity news narratives. Second, limited research investigates whether sentiment extracted from varying levels of textual detail, such as headlines compared to entire articles, provides differing predictive utility over short-term horizons. Third, the combination of detailed, firm-specific sentiment derived from advanced LLMs with nonlinear ensemble models has not been systematically assessed concerning both predictive accuracy and risk-adjusted economic benefits.

Motivated by these gaps, this chapter combines firm-specific sentiment derived from both discriminative and generative LLMs with market-based features, evaluates predictive performance over one-, two-, and three-day horizons, and examines whether these signals produce statistically significant risk-adjusted alpha. By integrating detailed textual sentiment analysis, ensemble-based forecasting, and rigorous asset-pricing assessment, this study enhances comprehension of sentiment-driven return behaviours and offers novel evidence regarding the economic significance of LLM-enhanced machine learning models in short-term trading.

5.3. Data and Methodology

5.3.1. Data

The empirical analysis examines four prominent large-cap U.S. technology firms: Amazon (AMZN), Netflix (NFLX), Nvidia (NVDA), and Alphabet (GOOGL). These firms are chosen due to their high liquidity, significant market capitalization, and widespread coverage in financial news media. Daily adjusted closing prices are sourced from Yahoo Finance²⁹. Moreover, this chapter utilizes two complementary textual datasets developed in the previous chapters: (i) a firm-specific headline dataset curated in Chapter 3, and (ii) a full-length financial news article dataset constructed in Chapter 4. Both datasets are sourced from the Bloomberg Terminal³⁰. All textual elements are subject to standardized preprocessing procedures to eliminate duplicate entries, advertisements, disclaimers, and other non-text elements. The final sample spans September 22, 2023, to January 29, 2024. Table 5.1 presents the distribution of textual items across the selected firms.

Table 5.1: Distribution of news items across the selected firms.

Firm	Headlines	Articles
Amazon	493	441
Netflix	257	202
Nvidia	337	312
Alphabet	189	215
Total	1276	1170

Textual data are mapped to firm-day observations based on publication date. All news items related to firm T and published on calendar day d are utilized to develop daily sentiment indicators for that firm on day d . When multiple headlines or articles are available for a given firm-day, sentiment signals are aggregated at the daily level before the model estimation. These firm-day predictors are then used to predict stock return directions over subsequent horizons, defined as close-to-close returns realized after day d .

²⁹ <https://finance.yahoo.com/>

³⁰ <https://www.bloomberg.com/>

To assess risk-adjusted performance, we augment the return and news databases with the daily-French five-factor series (Fama and French, 2015), including the market excess return (MKT), size (SMB), value (HML), profitability (RMW), and investment (CMA) factors, alongside the daily risk-free rate (RF). All factor data are sourced from the Kenneth R. French Data Library³¹ and integrated with strategy returns by calendar date for the alpha estimations detailed in Section 5.3.4.

5.3.2. Feature Construction

Predictive features incorporate short-term technical indicators, sentiment derived from news items, and firm-specific fixed effects. The empirical specification is estimated on pooled panel data, where each observation corresponds to firm T on day d . Firm heterogeneity is controlled for through firm fixed effects implemented via dummy variables.

Technical indicators encompass the 14-day Relative Strength Index (RSI), which measures short-term momentum and identifies overbought-oversold conditions, and the 10-day Simple Moving Average (SMA), which reflects recent price trends:

$$RSI_{d,T} = 100 - \frac{100}{1 + RS_{d,T}}, \quad (5.1)$$

where $RS_{d,T} = \frac{\text{Average Gain over 14 days}}{\text{Average Loss over 14 days}}$,

$$SMA_{d,T} = \frac{1}{10} \sum_{k=0}^9 P_{d-k,T}, \quad (5.2)$$

where $P_{d,T}$ denotes the adjusted closing price of firm T on day d . Short-term momentum is represented by one-day lagged returns:

$$R_{d,T} = \frac{P_{d,T} - P_{d-1,T}}{P_{d-1,T}} \times 100. \quad (5.3)$$

³¹https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html?utm_source=chatgpt.com

Firm-level investor sentiment features are derived from one discriminative model (FinBERT³²) and two generative LLMs (ChatGPT-o3³³ and DeepSeek-R1³⁴). FinBERT (Araci, 2019) is utilized without fine-tuning, whereas the generative models function in zero-shot mode (ChatGPT-o3 via the OpenAI API with a temperature set to 0.7; DeepSeek-R1 employing its default configurations). For each textual item, the generative models are queried using a standardized prompt, as shown in Figure 5.1.

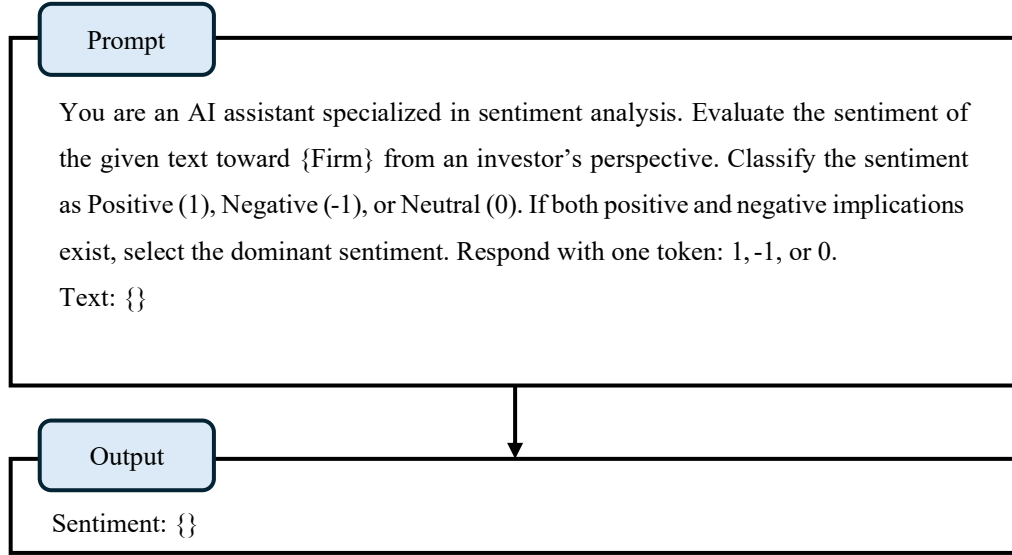


Figure 5.1: Prompt used for generative LLMs.

Each textual item is assigned a sentiment label $s_{j,T} \in [-1, 0, 1]$, representing negative, neutral, and positive sentiment, respectively. Daily average sentiment is computed as:

$$\bar{S}_{d,T} = \frac{1}{N_{d,T}} \sum_{j=1}^{N_{d,T}} s_{j,T}, \quad (5.4)$$

where $N_{d,T}$ represents the number of news items for firm T on day d . From these daily averages, three-day sentiment momentum and volatility are computed as:

$$M_{d,T} = \frac{1}{3} \sum_{k=0}^2 \bar{S}_{d-k,T}, \quad (5.5)$$

$$Vol_{d,T} = \sqrt{\frac{1}{3} \sum_{k=0}^2 (\bar{S}_{d-k,T} - M_{d,T})^2}. \quad (5.6)$$

³² <https://huggingface.co/ProsusAI/finbert>

³³ <https://chatgpt.com/>. ChatGPT variants used in this study correspond to those available during the research period.

³⁴ <https://www.deepseek.com/>

The daily average sentiment reflects the contemporaneous tone of investor-relevant information released for firm T on day d . Sentiment momentum encapsulates the short-run persistence of sentiment by smoothing daily sentiment over recent days, whereas sentiment volatility quantifies the short-term dispersion and variability in sentiment, indicating uncertainty or disagreement in news-based signals. Collectively, these measures encompass complementary aspects of investor sentiment dynamics that could affect short-horizon return behaviour. Moreover, Firm-specific heterogeneity is absorbed using one-hot-encoded ticker dummies (D_T), enabling pooled estimation while controlling for time-invariant structural differences across firms.

Table 5.2 provides pooled descriptive statistics for the variables used in the empirical analysis, including Fama-French risk factors, stock returns, technical indicators, and sentiment-based measures.

5.3.3. Model Specification

The study forecasts the direction of stock returns over horizons $h \in \{1, 2, 3\}$. The h -day-ahead return is defined as:

$$R_{d+h,T} = \frac{P_{d+h,T} - P_{d,T}}{P_{d,T}} \times 100. \quad (5.7)$$

and the binary prediction target is:

$$y_{d,h,T} = \begin{cases} 1, & \text{if } R_{d+h,T} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (5.8)$$

All predictors are developed using information available no later than the market close of day d . Technical indicators and lagged returns are derived from historical prices up to and including day d . Daily sentiment is computed by aggregating all firm-specific news items published on day d . Sentiment momentum and volatility are calculated from rolling windows of daily sentiment concluding on day d . Returns are then assessed over subsequent horizons, ensuring that no information dated after day d

influences the prediction phase and thereby preventing look-ahead bias. Figure 5.2 depicts the distribution of return directions across the prediction horizons.

Table 5.2: Pooled descriptive statistics across the firm-day panel.

Variable	Mean	Std Dev	Min	Max	N
MKT (%)	0.12	0.78	-1.57	2.18	302
SMB (%)	-0.03	0.90	-1.93	3.23	302
HML (%)	0.01	0.61	-1.27	2.03	302
RMW (%)	-0.04	0.54	-1.27	1.49	302
CMA (%)	-0.04	0.32	-0.97	0.77	302
RF (%)	0.02	0.00	0.02	0.02	302
1-Day-Ahead Return (%)	0.41	2.16	-9.51	16.05	302
2-Day-Ahead Return (%)	1.02	13.33	-78.76	136.77	300
3-Day-Ahead Return (%)	1.55	16.37	-78.80	138.83	299
1-Day Lagged Return (%)	0.00	0.02	-0.10	0.16	302
SMA	288.27	163.36	126.72	588.71	302
RSI	57.56	17.86	6.51	96.90	302
FinBERT Headline Sentiment	0.04	0.55	-1.00	1.00	302
FinBERT Headline Sentiment Momentum	0.04	0.32	-0.85	0.77	302
FinBERT Headline Sentiment Volatility	0.47	0.29	0.00	1.15	302
DeepSeek-R1 Headline Sentiment	0.21	0.65	-1.00	1.00	302
DeepSeek-R1 Headline Sentiment Momentum	0.21	0.40	-1.00	1.00	302
DeepSeek-R1 Headline Sentiment Volatility	0.54	0.30	0.00	1.15	302
ChatGPT-o3 Headline Sentiment	0.18	0.59	-1.00	1.00	302
ChatGPT-o3 Headline Sentiment Momentum	0.18	0.37	-0.79	1.00	302
ChatGPT-o3 Headline Sentiment Volatility	0.49	0.28	0.00	1.15	302
FinBERT Article Sentiment	-0.09	0.92	-7.00	5.00	302
FinBERT Article Sentiment Momentum	-0.10	0.52	-2.67	1.67	302
FinBERT Article Sentiment Volatility	0.66	0.64	0.00	6.00	302
DeepSeek-R1 Article Sentiment	0.22	0.65	-2.00	2.00	302
DeepSeek-R1 Article Sentiment Momentum	0.22	0.41	-0.90	1.33	302
DeepSeek-R1 Article Sentiment Volatility	0.50	0.35	0.00	2.08	302
ChatGPT-o3 Article Sentiment	0.25	0.72	-2.00	3.00	302
ChatGPT-o3 Article Sentiment Momentum	0.24	0.44	-1.04	1.67	302
ChatGPT-o3 Article Sentiment Volatility	0.56	0.40	0.00	2.08	302

Notes: MKT is the excess return on the market portfolio. SMB (Small Minus Big), HML (High Minus Low), RMW (Robust Minus Weak), and CMA (Conservative Minus Aggressive) are the size, value, profitability, and investment factors, respectively, from the Fama-French five-factor model. RF denotes the daily risk-free rate. RSI refers to the 14-day Relative Strength Index, and SMA to the 10-day Simple Moving Average.

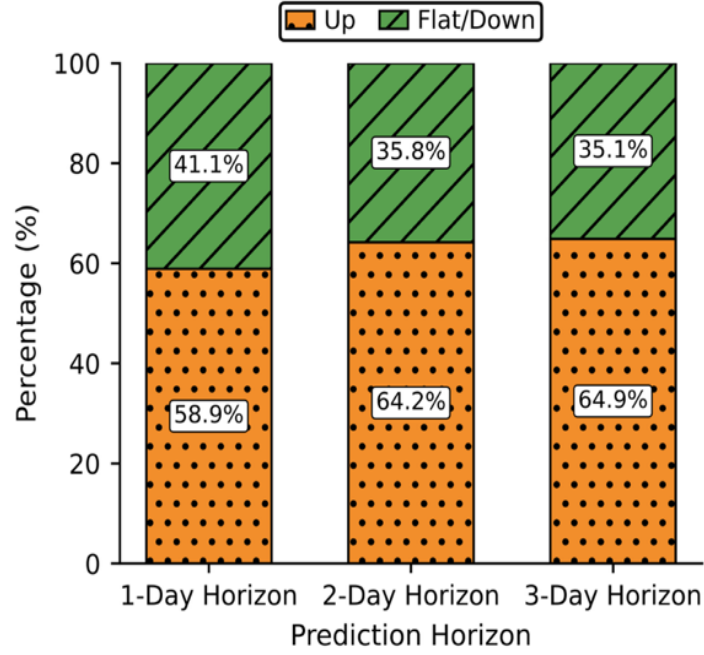


Figure 5.2: Distribution of return directions across forecast horizons.

Predictive models include Logistic Regression (LR), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). Using balanced class weights and up to 500 iterations, LR serves as the linear benchmark and estimates the probability of an upward movement as:

$$\Pr(y_{d,h,T} = 1 \mid X_{d,T}, D_T) = \Lambda(\beta_0 + X'_{d,T}\beta + D'_T\gamma), \quad (5.9)$$

where $X_{d,T}$ represents the vector of predictors, β is the vector of coefficients, D_T denotes firm fixed effects implemented via ticker indicators, and $\Lambda(\cdot)$ is the logistic cumulative distribution function. RF predictions are derived through majority voting across $B = 200$ bootstrap-sampled trees:

$$\hat{y}_{d,h,T} = \text{mode} \{f_1(X_{d,T}), f_2(X_{d,T}), \dots, f_B(X_{d,T})\}. \quad (5.10)$$

XGBoost builds an additive ensemble of trees through iterative gradient boosting:

$$F_m(X_{d,T}) = F_{m-1}(X_{d,T}) + \eta f_m(X_{d,T}), \quad (5.11)$$

with the predicted labels provided by:

$$\hat{y}_{d,h,T} = 1 \{ \sigma (F_M(X_{d,T})) > 0.5 \}, \quad (5.12)$$

where η is the learning rate, M is the number of boosting iterations, and $\sigma (\cdot)$ is the logistic function.

Logistic Regression was estimated with balanced class weights and a maximum of 500 iterations. Random Forest was implemented with 200 bootstrap-sampled trees, balanced class weights, and a fixed random seed of 42, while all remaining hyperparameters were retained at their default scikit-learn settings. XGBoost was implemented using the log-loss evaluation criterion and a fixed random seed of 42, with remaining hyperparameters retained at their default XGBoost configurations. No additional hyperparameter optimization or early stopping procedure was applied, as the objective of the study was comparative evaluation under a consistent modelling framework rather than model-specific hyperparameter maximization. All models were evaluated using a five-fold expanding-window cross-validation design that preserved temporal ordering, thereby mimicking real-time forecasting conditions. Predictors were standardized within each training fold for Logistic Regression and XGBoost to avoid information leakage between training and testing periods.

5.3.4. Evaluation Strategy

Model performance is evaluated through both statistical metrics and trading-related indicators. Classification accuracy and F1-score are calculated as:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}, F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (5.13)$$

$$\text{where Precision} = \frac{TP}{TP+FP} \text{ and Recall} = \frac{TP}{TP+FN}.$$

Because the distribution of return-direction classes exhibits moderate imbalance across forecasting horizons, as illustrated in Figure 5.2, the use of a fixed 0.5

probability threshold may affect the balance between sensitivity and specificity. While the present analysis focuses on Accuracy and F1-score for consistency with the implemented trading-rule framework, future research could extend the evaluation using threshold-independent measures such as ROC curves and the Area Under the Curve (AUC), as well as alternative threshold-selection procedures based on criteria such as the Youden index.

Economic value is assessed using a directional trading strategy. Each day d , the model forms a position for horizon h based on the predicted label $\hat{y}_{d,h,T}$. Strategy returns are:

$$r_{d,h,T}^{strat} = \begin{cases} R_{d+h,T}, & \text{if } \hat{y}_{d,h,T} = 1, \\ -R_{d+h,T}, & \text{if } \hat{y}_{d,h,T} = 0. \end{cases} \quad (5.14)$$

From these returns, the annualized Sharpe ratio, maximum drawdown (MDD), and win rate are computed as:

$$\text{Sharpe Ratio} = \frac{E[r_{d,h,T}^{strat}]}{\sigma(r_{d,h,T}^{strat})} \sqrt{252}, \quad (5.15)$$

$$\text{MDD (\%)} = \max_t \left(\frac{\text{Peak}_t - \text{Through}_t}{\text{Peak}_t} \right) \times 100, \quad (5.16)$$

$$\text{Win Rate (\%)} = \frac{\sum 1\{r_{d,h,T}^{strat} > 0\}}{N} \times 100. \quad (5.17)$$

Risk-adjusted performance is further evaluated through Fama-French alpha. Excess returns are defined as:

$$R_{d,h,T}^{\text{excess}} = R_{d,h,T}^{\text{strat}} - R_{f,d} \quad (5.18)$$

and the Fama-French five-factor model is estimated as:

$$R_{d,h,T}^{\text{excess}} = \alpha + \beta_{\text{MKT}}(R_{\text{Mkt},d} - R_{f,d}) + \beta_{\text{SMB}}\text{SMB}_d + \beta_{\text{HML}}\text{HML}_d + \beta_{\text{RMW}}\text{RMW}_d + \beta_{\text{CMA}}\text{CMA}_d + \varepsilon_{d,h,T} \quad (5.19)$$

where the intercept α measures the abnormal return attributable to a specific model-feature combination after accounting for standard risk exposures. We estimate α separately for each prediction horizon, model class, and feature set to quantify the incremental economic value of incorporating technical indicators and sentiment-based predictors. Statistical significance is evaluated employing robust standard errors. Moreover, out-of-sample performance is assessed through a five-fold expanding-window cross-validation approach that maintains the temporal sequence. With each iteration, the training window expands, and the following period is designated for evaluation, thereby simulating real-time forecasting scenarios. All metrics are combined across folds to derive reliable estimates of predictive and economic performance. Figure 5.3 illustrates the framework for LLM-derived firm-level sentiment and short-horizon risk-adjusted return prediction.

5.4. Results and Discussion

This section presents the empirical and economic implications of enhancing short-term return prediction models through the incorporation of firm-specific sentiment extracted from both headlines and comprehensive news articles. We evaluate whether sentiment enhances directional accuracy, whether nonlinear ensemble methods outperform linear benchmarks, and whether predictive improvements result in economically and risk-adjusted significant trading performance.

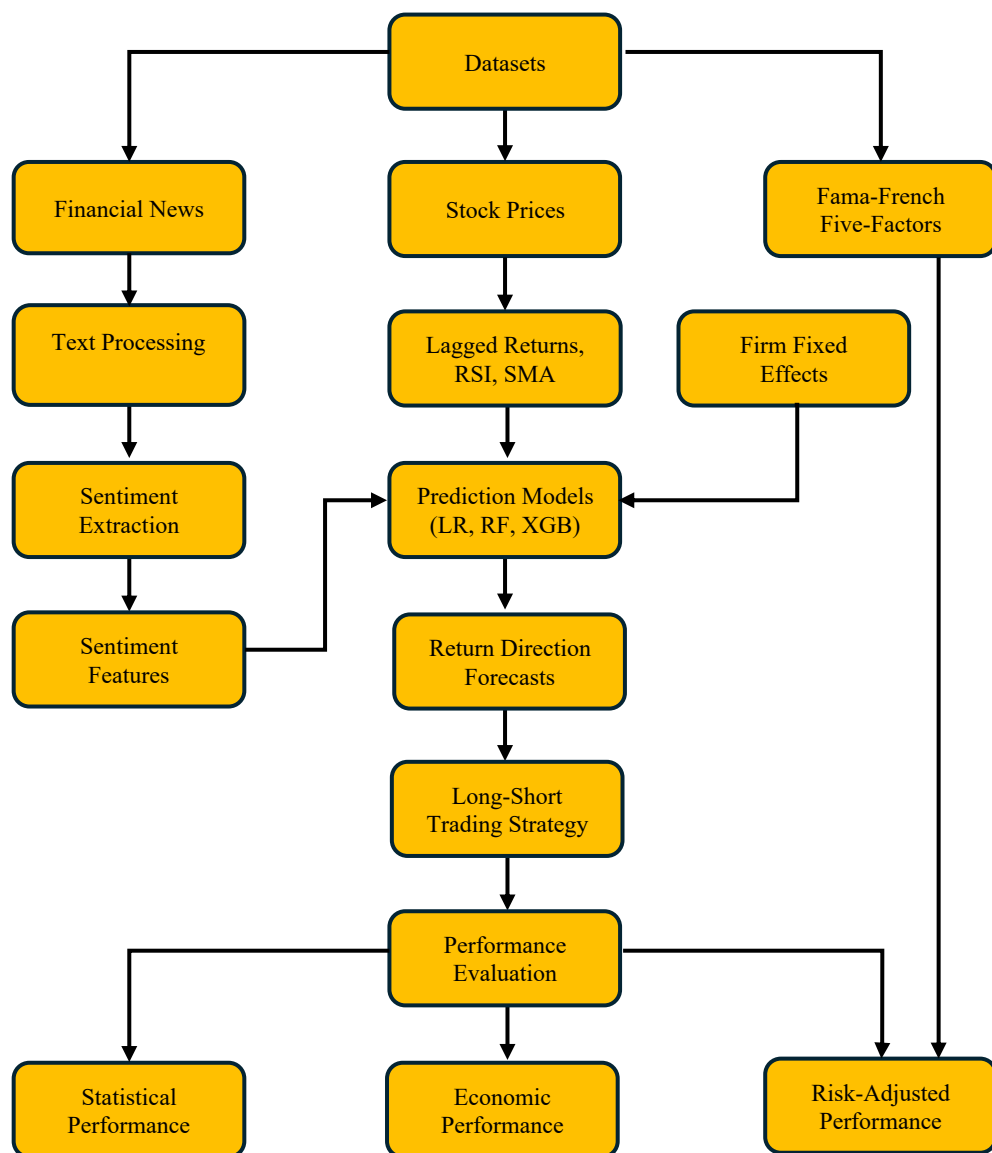


Figure 5.3: Framework for LLM-derived firm-level sentiment and short-horizon risk-adjusted return prediction.

5.4.1. Predictive Performance

Table 5.3 presents the pooled out-of-sample classification accuracy and F1-scores for all model-feature-horizon combinations. The reported metrics reflect pooled predictive performance across firms obtained from the pooled panel framework rather than firm-specific estimates. Firm-level heterogeneity is controlled for through ticker dummy variables included in all model specifications, allowing the models to exploit common predictive structures across firms while accounting for time-invariant cross-sectional differences. Several robust patterns emerge.

Table 5.3: Out-of-sample classification performance across models, feature sets, and forecast horizons.

Features	Model	1-day horizon		2-day horizon		3-day horizon	
		Acc.	F1	Acc.	F1	Acc.	F1
Returns + D_T (Baseline)	LR	0.46	0.39	0.49	0.50	0.46	0.41
Returns + D_T (Baseline)	RF	0.48	0.48	0.54	0.63	0.53	0.62
Returns + D_T (Baseline)	XGB	0.49	0.48	0.57	0.67	0.58	0.69
Returns + TIs + D_T	LR	0.47	0.39	0.53	0.53	0.42	0.34
Returns + TIs + D_T	RF	0.50	0.49	0.57	0.67	0.59	0.70
Returns + TIs + D_T	XGB	0.50	0.49	0.58	0.65	0.61	0.71
News headline sentiment							
Returns + TIs + D_T + FinBERT	LR	0.45	0.27	0.50	0.51	0.46	0.43
Returns + TIs + D_T + ChatGPT-o3	LR	0.47	0.29	0.52	0.52	0.47	0.46
Returns + TIs + D_T + DeepSeek-R1	LR	0.48	0.31	0.51	0.52	0.48	0.47
Returns + TIs + D_T + FinBERT	RF	0.47	0.45	0.57	0.68	0.58	0.70
Returns + TIs + D_T + ChatGPT-o3	RF	0.49	0.46	0.60	0.72	0.62	0.72
Returns + TIs + D_T + DeepSeek-R1	RF	0.48	0.50	0.58	0.69	0.60	0.71
Returns + TIs + D_T + FinBERT	XGB	0.47	0.49	0.57	0.64	0.55	0.65
Returns + TIs + D_T + ChatGPT-o3	XGB	0.50	0.50	0.57	0.66	0.56	0.66
Returns + TIs + α_F + DeepSeek-R1	XGB	0.51	0.53	0.58	0.67	0.60	0.68
News article sentiment							
Returns + TIs + D_T + FinBERT	LR	0.46	0.32	0.48	0.51	0.41	0.40
Returns + TIs + D_T + ChatGPT-o3	LR	0.47	0.34	0.50	0.50	0.42	0.41
Returns + TIs + D_T + DeepSeek-R1	LR	0.50	0.40	0.49	0.51	0.50	0.50
Returns + TIs + D_T + FinBERT	RF	0.50	0.52	0.62	0.73	0.60	0.72
Returns + TIs + D_T + ChatGPT-o3	RF	0.54	0.60	0.63	0.74	0.63	0.73
Returns + TIs + D_T + DeepSeek-R1	RF	0.52	0.53	0.62	0.73	0.60	0.73
Returns + TIs + D_T + FinBERT	XGB	0.52	0.53	0.58	0.66	0.56	0.65
Returns + TIs + D_T + ChatGPT-o3	XGB	0.54	0.57	0.58	0.69	0.62	0.73
Returns + TIs + D_T + DeepSeek-R1	XGB	0.53	0.54	0.65	0.74	0.59	0.70

Note: Acc. and F1 denote accuracy and F1-score, respectively.

First, models that depend solely on lagged returns demonstrate limited predictive power. Logistic regression (LR) performs near random performance, with accuracies ranging from 0.46 to 0.49 and consistently low F1-scores across all horizons. This aligns with the extensively documented challenge of predicting short-term return directions solely through linear models and historical pricing data (Lo & MacKinlay, 2011; Welch & Goyal, 2008). Ensemble learners, namely, Random Forests (RF) and XGBoost (XGB), demonstrate a modest improvement, attaining accuracies of approximately 0.57 to 0.58 over multi-day horizons. These enhancements strengthen

the evidence that nonlinear models are capable of leveraging complex interactions within financial data that linear models are unable to effectively capture (Gu et al., 2020; Rapach and Zhou, 2013; Healy et al., 2024).

Second, the inclusion of technical indicators (TIs) significantly enhances predictive accuracy exclusively for nonlinear models. RF and XGB demonstrate significant improvements. For example, RF's accuracy increases from 0.53 to 0.59 over the three-day horizon, whereas LR shows minimal or no enhancement and occasionally declines. This pattern indicates that technical indicators predominantly offer predictive value through nonlinear effects, aligning with recent research emphasizing nonlinear momentum and reversal structures (Li and Tam, 2018; Chen et al., 2024).

Third, sentiment derived from headlines offers incremental but moderate improvements in predictive accuracy. RF with ChatGPT-o3 headline sentiment achieves 0.60 accuracy ($F1 = 0.72$) at the two-day horizon, indicating a modest improvement over technical-only features; XGB exhibits similar patterns. However, in some configurations, the incremental enhancements are minimal. For instance, F1-scores rise from approximately 0.70 to 0.72 for RF, indicating that sentiment derived from headlines frequently provides limited additional information beyond what is already encompassed by price-based technical indicators. More broadly, although sentiment-based features reliably enhance predictive accuracy, the extent of these improvements differs across various model types and forecasting horizons. In contexts where nonlinear models and technical indicators already account for a significant portion of short-term price movements, the additional impact of headline-based sentiment tends to be limited.

Fourth, sentiment derived from comprehensive articles yields the most substantial and systematic enhancements. Article-level sentiment, especially when derived through generative LLMs, reflects narrative framing, causal reasoning, and contextual dependencies, all of which significantly improve return predictability. RF combined with ChatGPT-o3 article sentiment obtains an accuracy of 0.63 at both the two- and three-day horizons, with F1 scores of 0.73 and 0.74, respectively, while XGB paired with DeepSeek-R1 sentiment attains the highest overall accuracy of 0.65 ($F1 = 0.74$).

These findings reinforce and expand upon existing evidence that more comprehensive textual representations and supervised sentiment measures derived from news sources improve predictive accuracy compared to headline-only signals (Heston and Sinha, 2017; Ke et al., 2019).

Fifth, predictive accuracy demonstrates a modest improvement when extending the forecasting horizon from one to two days and subsequently stabilizes or exhibits a minor decline at the three-day horizon. This pattern aligns with the process of gradual information dissemination, whereby qualitative information influences returns with a short delay prior to being entirely reflected in prices (Tetlock, 2007; Da et al., 2015).

Finally, generative LLMs consistently outperform FinBERT in both headline and article tasks. Although FinBERT offers valuable sentiment indicators, generative models, benefiting from substantially larger parameter counts and broader pretraining, appear more adept at interpreting intricate financial terminology, resolving ambiguous sentiments, and inferring context-dependent meaning. Recent research similarly indicates that sentiment measures derived from generative LLMs outperform dictionary-based approaches and domain-specific transformer models in return prediction and trading applications (Lopez-Lira and Tang, 2023; Kirtac and Germano, 2024).

Figure 5.4 summarizes the incremental contribution of technical indicators and LLM-derived sentiment to predictive accuracy. While technical indicators provide modest improvements, especially for nonlinear models, headline sentiment yields additional gains, and article-level sentiment produces the largest and most consistent improvements across horizons. These results highlight the importance of both textual depth and model flexibility in short-horizon return prediction.

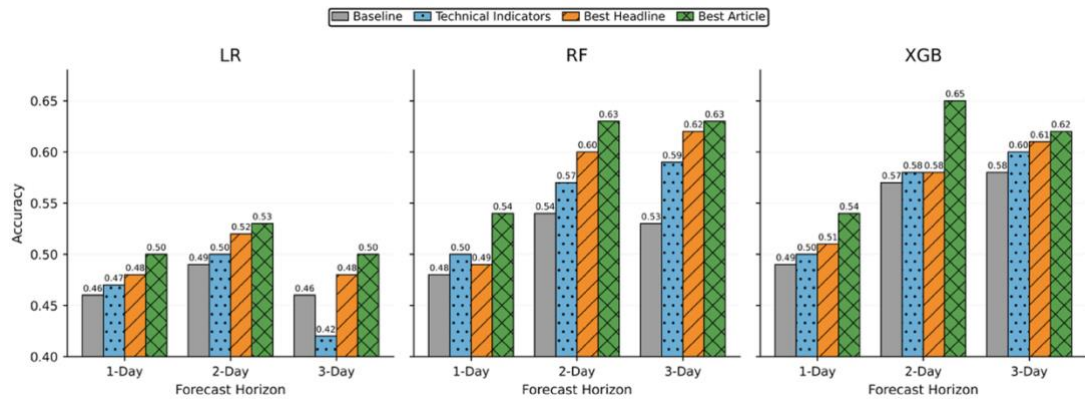


Figure 5.4: Incremental predictive performance across feature sets and forecast horizons.

In summary, the findings suggest that lagged returns alone have limited informational value; technical indicators are primarily significant in nonlinear settings. Sentiment improves predictability, with effect sizes influenced by textual richness. Furthermore, generative LLMs extract considerably more comprehensive information compared to FinBERT. These findings extend previous evidence on nonlinear return predictability (Gu et al., 2020; Chen et al., 2024) and the significance of advanced sentiment measures in forecasting returns (Heston and Sinha, 2017).

5.4.2. Economic Performance

Table 5.4 presents the pooled economic value of model-implied directional strategies, assessed through annualized Sharpe ratios, maximum drawdowns, and win rates.

First, baseline strategies that depend only on lagged returns generate minimal or negative Sharpe ratios and experience significant drawdowns. Even the highest-performing baseline ensemble model (XGB at the three-day horizon) attains a Sharpe ratio of merely 0.05. These findings reinforce the restricted tradability of short-term signals based solely on historical price data (Henkel et al., 2011; Farmer et al., 2023).

Second, incorporating technical indicators (TIs) into models substantially enhances the economic performance of nonlinear learners. RF and XGB demonstrate significant enhancements in Sharpe ratios and reductions in drawdowns. For instance, XGB

increases from 0.05 to 0.22 at the three-day horizon. Conversely, LR continues to be economically inefficient. These findings emphasize the significance of nonlinear architectures in harnessing technical signals for monetization and align with recent research indicating that machine learning can render technical analysis economically valuable under specific conditions (Li and Tam, 2018; Rigamonti, 2024).

Table 5.4: Economic performance of directional strategies across models, feature sets, and forecast horizons.

Features	Model	1-day horizon			2-day horizon			3-day horizon		
		SR	MDD	Win	SR	MDD	Win	SR	MDD	Win
Returns + D_T (Baseline)	LR	-0.08	-19.39	0.48	0.12	-37.35	0.56	-0.08	-52.30	0.46
Returns + D_T (Baseline)	RF	-0.01	-18.34	0.48	0.00	-58.34	0.54	0.02	-69.99	0.53
Returns + D_T (Baseline)	XGB	-0.08	-19.62	0.52	0.00	-63.02	0.57	0.05	-42.62	0.58
Returns + TIs + D_T	LR	-0.12	-19.55	0.45	0.06	-51.47	0.53	-0.07	-65.09	0.42
Returns + TIs + D_T	RF	-0.05	-17.32	0.51	0.16	-29.34	0.56	0.18	-38.07	0.59
Returns + TIs + D_T	XGB	-0.02	-18.17	0.48	0.10	-51.30	0.57	0.22	-35.30	0.61
News headline sentiment										
Returns + TIs + D_T + FinBERT	LR	-0.09	-19.27	0.46	0.11	-51.99	0.53	0.11	-59.68	0.50
Returns + TIs + D_T + ChatGPT-o3	LR	-0.12	-20.05	0.45	0.12	-56.60	0.51	0.04	-56.60	0.47
Returns + TIs + D_T + DeepSeek-R1	LR	-0.10	-18.88	0.46	0.11	-59.43	0.51	0.07	-88.70	0.48
Returns + TIs + D_T + FinBERT	RF	0.00	-18.18	0.50	0.13	-25.63	0.60	0.18	-41.17	0.58
Returns + TIs + D_T + ChatGPT-o3	RF	0.02	-14.44	0.53	0.18	-22.64	0.60	0.21	-31.29	0.61
Returns + TIs + D_T + DeepSeek-R1	RF	-0.07	-18.53	0.48	0.18	-24.62	0.57	0.19	-32.99	0.60
Returns + TIs + D_T + FinBERT	XGB	0.01	-14.16	0.51	0.10	-58.75	0.57	0.13	-31.53	0.61
Returns + TIs + D_T + ChatGPT-o3	XGB	-0.08	-19.90	0.46	0.14	-31.12	0.57	0.13	-36.15	0.56
Returns + TIs + D_T + DeepSeek-R1	XGB	-0.02	-14.70	0.50	0.18	-22.68	0.55	0.20	-37.70	0.60
News article sentiment										
Returns + TIs + D_T + FinBERT	LR	-0.09	-17.08	0.48	0.06	-62.10	0.48	-0.07	-61.20	0.46
Returns + TIs + D_T + ChatGPT-o3	LR	-0.09	-19.09	0.45	0.10	-46.10	0.50	-0.04	-56.83	0.42
Returns + TIs + D_T + DeepSeek-R1	LR	-0.06	-19.27	0.48	0.06	-47.50	0.49	0.06	-58.88	0.50
Returns + TIs + D_T + FinBERT	RF	0.03	-15.48	0.50	0.24	-24.88	0.62	0.24	-32.86	0.60
Returns + TIs + D_T + ChatGPT-o3	RF	0.09	-11.06	0.53	0.21	-23.71	0.63	0.25	-29.82	0.60
Returns + TIs + D_T + DeepSeek-R1	RF	0.09	-11.72	0.51	0.16	-22.89	0.62	0.18	-33.08	0.60
Returns + TIs + D_T + FinBERT	XGB	0.03	-17.08	0.51	0.22	-28.67	0.58	0.19	-36.44	0.56
Returns + TIs + D_T + ChatGPT-o3	XGB	0.02	-15.12	0.53	0.10	-47.57	0.58	0.24	-33.91	0.63
Returns + TIs + D_T + DeepSeek-R1	XGB	0.08	-11.18	0.51	0.24	-18.85	0.65	0.22	-30.48	0.59

Notes: SR, MDD, and W denote the Sharpe ratio, maximum drawdown (%), and win rate (%), respectively.

Third, sentiment derived from headlines yields modest yet consistent economic benefits. RF with ChatGPT-o3 headlines achieves Sharpe ratios ranging from 0.02 to 0.21 across various horizons, accompanied by diminished drawdowns compared to the technical-only benchmark; similarly, XGB with DeepSeek-R1 headlines demonstrates comparable enhancements. However, due to the limited contextual depth of headlines, these improvements remain modest, aligning with the perspective that brief, event-driven text encompasses only a portion of the information pertinent to pricing (Barber et al., 2022).

Fourth, article-level sentiment, especially when derived through generative LLMs, produces the most substantial economic enhancements. These strategies achieve higher Sharpe ratios, increased win rates, and significantly reduced drawdowns. XGB with DeepSeek-R1 article sentiment attains a Sharpe ratio of 0.24 and the lowest drawdown (-18.85%) at the two-day horizon, while RF with ChatGPT-o3 article sentiment obtains a Sharpe ratio of 0.25 at the three-day horizon. These findings affirm that a more comprehensive narrative context offers economically actionable insights and enhances recent evidence indicating that LLM-derived textual embeddings and sentiment metrics can be effectively utilized in trading strategies (Ke et al., 2019).

Finally, economic gains demonstrate a clear horizon dependence. One-day horizons produce Sharpe ratios approaching zero across all models, indicative of market microstructure noise and the challenges associated with exploiting qualitative information within a single day. Two- and three-day strategies, particularly those that incorporate technical indicators and sentiment at the article level, yield significantly higher Sharpe ratios and reduced drawdowns, aligning with the delayed incorporation of narrative information as documented in previous research (Tetlock, 2007; Da et al., 2015).

Figure 5.5 provides a consolidated visualization of these patterns, clearly illustrating the differences between headline- and article-based sentiment across models and horizons. While headline-based sentiment yields moderate improvements, article-level sentiment produces consistently higher Sharpe ratios, particularly for nonlinear models. The strongest performance is achieved by ensemble methods (RF and XGB)

combined with generative LLM sentiment, with gains most pronounced at the two- and three-day horizons.



Figure 5.5: Sharpe ratios across models and forecast horizons: headline- versus article-level sentiment.

Overall, the economic findings indicate that return-only strategies are not financially sustainable, nonlinear models are necessary to effectively leverage technical information, and article-level sentiment derived from generative LLMs produces the most significant and consistent enhancements in risk-adjusted performance. These patterns support recent findings that machine learning models produce economically significant trading signals when provided with comprehensive predictor sets (Gu et al., 2020) and that modern NLP models reveal sources of alpha inaccessible to conventional sentiment analysis methods (Lopez-Lira and Tang, 2023).

While the reported economic metrics demonstrate that sentiment-enhanced models improve predictive and trading performance, they do not fully identify the marginal contribution or directional effect of individual sentiment variables within the forecasting process. In particular, the present analysis does not directly distinguish whether negative sentiment contributes asymmetrically to subsequent negative returns or whether the economic effects vary systematically across firms and forecasting horizons. Future research could therefore complement the current framework with additional interpretability analyses, including coefficient-based interpretation for

linear models, feature-importance measures for ensemble learners, or SHAP-based approaches for nonlinear models, to provide deeper insight into the economic mechanisms underlying sentiment-driven return predictability.

Table 5.5 presents Fama-French alphas derived from the Fama-French five-factor model.

Table 5.5: Fama-French five-factor alphas of model-implied trading strategies across forecast horizons.

Features	Model	1-day horizon		2-day horizon		3-day horizon	
		Alpha	t-stat	Alpha	t-stat	Alpha	t-stat
Returns + D_T (Baseline)	LR	-0.002	-2.91	-0.010	0.26	0.006	0.91
Returns + D_T (Baseline)	RF	0.000	0.37	0.002	0.48	0.007	1.20
Returns + D_T (Baseline)	XGB	0.002	2.89	-0.004	-0.79	0.015	2.45
Returns + TIs + D_T	LR	-0.004	5.08	-0.005	-1.01	-0.018	-2.88
Returns + TIs + D_T	RF	-0.002	-2.78	0.017	3.56	0.029	4.74
Returns + TIs + D_T	XGB	-0.001	-1.53	0.007	1.47	0.032	5.14
News headline sentiment							
Returns + TIs + D_T + FinBERT	LR	-0.003	-4.54	0.007	1.36	-0.002	-0.35
Returns + TIs + D_T + ChatGPT-o3	LR	-0.003	-4.37	0.006	1.31	0.009	1.43
Returns + TIs + D_T + DeepSeek-R1	LR	-0.003	-4.54	0.007	1.36	-0.002	-0.35
Returns + TIs + D_T + FinBERT	RF	0.000	0.38	0.013	2.66	0.029	4.74
Returns + TIs + D_T + ChatGPT-o3	RF	-0.003	-3.56	0.019	3.87	0.029	4.76
Returns + TIs + D_T + DeepSeek-R1	RF	0.000	0.38	0.013	2.66	0.029	4.74
Returns + TIs + D_T + FinBERT	XGB	-0.003	-4.17	0.015	3.01	0.027	4.36
Returns + TIs + D_T + ChatGPT-o3	XGB	-0.001	-2.32	0.020	4.14	0.030	4.77
Returns + TIs + D_T + DeepSeek-R1	XGB	-0.003	-4.17	0.015	3.01	0.027	4.36
News article sentiment							
Returns + TIs + D_T + FinBERT	LR	-0.003	-4.30	0.004	0.83	-0.015	-2.40
Returns + TIs + D_T + ChatGPT-o3	LR	-0.003	-3.58	0.005	-1.07	0.002	0.33
Returns + TIs + D_T + DeepSeek-R1	LR	-0.003	-4.54	0.007	1.36	-0.002	-0.35
Returns + TIs + D_T + FinBERT	RF	0.001	-1.54	0.020	4.10	-0.033	5.27
Returns + TIs + D_T + ChatGPT-o3	RF	0.003	3.17	0.021	4.28	0.030	4.76
Returns + TIs + D_T + DeepSeek-R1	RF	0.000	0.38	0.013	2.66	0.029	4.74
Returns + TIs + D_T + FinBERT	XGB	0.000	-0.65	0.017	3.43	0.028	4.49
Returns + TIs + D_T + ChatGPT-o3	XGB	0.003	3.57	0.026	5.52	0.031	4.98
Returns + TIs + D_T + DeepSeek-R1	XGB	-0.003	-4.17	0.015	3.01	0.027	4.36

Note: Alpha values with $|t\text{-stat}| \geq 1.96$, indicating statistical significance at the 5% level, are shown in bold.

Several patterns emerge. First, baseline models offer essentially no indication of risk-adjusted profitability. LR produces negative or statistically insignificant alphas, whereas RF and XGB generate only slightly positive alphas with limited statistical significance. These findings reflect the broader conclusion that signals based solely on price offer minimal alpha once standard risk factors are accounted for (Welch and Goyal, 2008).

Second, technical indicators significantly enhance alpha for nonlinear models. RF and XGB generate statistically significant and economically substantial alphas at the two- and three-day horizons. For instance, RF achieves $\alpha = 0.017$ ($t = 3.56$) at the two-day horizon and $\alpha = 0.029$ ($t = 4.74$) at the three-day horizon, while XGB attains $\alpha = 0.032$ ($t = 5.14$) at the three-day horizon. These findings offer risk-adjusted evidence that nonlinear learners uncover structural information from technical variables that remains inaccessible to linear models.

Third, headline sentiment contributes to the enhancement of alphas in ensemble models, though to a limited extent. LR demonstrates minimal progress, whereas RF and XGB display statistically significant yet comparatively modest enhancements relative to article-level sentiment. For example, the RF model utilizing ChatGPT-o3 headline sentiment attains an alpha of 0.019 ($t = 3.87$) at the two-day horizon and 0.029 ($t = 4.76$) at the three-day horizon. Similarly, the XGBoost model with ChatGPT-o3 headline sentiment yields an alpha of 0.020 ($t = 4.14$) and 0.030 ($t = 4.77$) at the two- and three-day horizons, respectively. This pattern aligns with the perspective that headlines frequently lack comprehensive nuance or narrative context (Barber & Odean, 2008; Engelberg & Parsons, 2011).

Fourth, article-level sentiment produces the most significant alphas across all models. RF and XGB, when combined with ChatGPT-o3 or DeepSeek-R1 article sentiment, generate substantial and statistically significant alphas over multi-day horizons. RF with ChatGPT-o3 article sentiment yields an alpha of 0.021 ($t = 4.28$) at the two-day horizon and 0.030 ($t = 4.76$) at the three-day horizon, while XGB with ChatGPT-o3 article sentiment attains an alpha of 0.026 ($t = 5.52$) and 0.031 ($t = 4.98$) at the two- and three-day horizons, respectively. FinBERT-based article sentiment also produces

positive alphas; however, the magnitudes and t-statistics are consistently lower, thereby underscoring the enhanced capacity of generative models to accurately capture nuanced and predictive sentiment (Ke et al., 2019; Lopez-Lira and Tang, 2023).

Finally, alphas are generally minimal at the one-day horizon, reflecting both the predictive and economic findings. This indicates that sentiment information necessitates a minimum of one trading day to be partially reflected in pricing, aligning with gradual information assimilation and constrained investor attention (Tetlock, 2007; Da et al., 2015).

Figure 5.6 provides a consolidated visualization of these results, highlighting the cross-sectional and horizon-dependent variation in Fama-French Five-Factor alphas across models and sentiment specifications. The figure clearly shows that article-level sentiment yields consistently higher and more statistically significant alphas than headline-based sentiment, particularly for nonlinear models. The strongest performance is concentrated in RF and XGB models combined with generative LLM sentiment, with gains most pronounced at the two- and three-day horizons.



Figure 5.6: Fama-French alphas across models and forecast horizons: headline- versus article-level sentiment.

In summary, the alpha analysis indicates that lagged returns do not produce abnormal performance, technical indicators contribute positive alpha only when modelled

nonlinearly, and article-level LLM-derived sentiment generates economically meaningful and statistically significant alpha over multi-day horizons.

The findings of the chapter indicate that short-term return predictability is fundamentally contingent upon the quality of information and the adaptability of the predictive model. Generative LLMs serve as advanced measurement tools that transform intricate narrative content into detailed sentiment signals with genuine predictive and economic significance. The robust performance of article-based sentiment demonstrates that a more comprehensive textual context is vital for effectively capturing the informational nuance embedded in financial news. From an asset-pricing standpoint, the horizon dependence of the returns indicates progressive information assimilation and constrained investor focus, even within highly liquid technology stocks. For practitioners, the results signal that simple sentiment proxies are insufficient in modern markets, whereas integrating LLM-derived article sentiment into nonlinear forecasting models can materially enhance Sharpe ratios, reduce drawdowns, and improve trading robustness.

5.5. Conclusion

This chapter investigates whether firm-specific sentiment derived from financial news through both domain-specific and generative LLMs improves short-term stock return predictability and yields economically meaningful, risk-adjusted performance. Utilizing data from four leading U.S. technology firms, we incorporate lagged returns, technical indicators, and sentiment derived from headlines and full articles into both linear and nonlinear predictive models. Through a thorough assessment of predictive accuracy, trading performance, and Fama-French five-factor model alphas, several clear conclusions emerge.

First, the findings indicate that conventional price-based indicators and linear models offer limited utility in predicting short-term return directions. Although technical indicators provide modest enhancements, they offer economically meaningful insights primarily when integrated with nonlinear ensemble techniques, highlighting the crucial role of adaptable functional forms in capturing complex market behaviours.

Second, sentiment derived from financial news significantly improves predictive accuracy, with the depth and richness of the textual source serving as a crucial factor. Headline sentiment yields modest improvements; however, article-level sentiment, particularly when derived from advanced generative LLMs, delivers the most significant enhancements in both statistical and economic metrics. These findings suggest that modern LLMs function as advanced information-processing systems, capable of discerning subtle, forward-looking signals from narrative texts. Finally, the economic assessment demonstrates that the predictive gains derived from LLM-based article sentiment result in significant enhancements in Sharpe ratios, drawdowns, and win rates. Risk-adjusted performance analysis verifies the existence of statistically significant alpha at two- and three-day horizons, aligning with models of gradual information dissemination and constrained investor attention. Together, these findings underscore the financial significance of contextually rich textual data and position generative LLMs as important instruments for converting unstructured news into actionable quantitative indicators.

Several limitations present avenues for future research. The analysis focuses on a relatively short sample period and a limited selection of firms; expanding the framework to encompass a wider range of stocks, including less widely covered firms where textual data are sparse, alternative asset classes, and longer time horizons would enhance the generalizability of the findings. Some of the employed methods, including FinBERT and generative LLM-based approaches, involve inherent stochasticity, suggesting that repeated runs and the reporting of averaged performance measures could further improve the robustness of comparative assessments. Moreover, the models are evaluated on data from a fixed historical window, and their performance may deteriorate when applied to more recent data due to concept and language drift; systematically assessing such degradation and the relative robustness of various modelling approaches represents a promising avenue for future work. Finally, considering transaction costs and execution risks, together with a systematic comparison of alternative model architectures, fine-tuning strategies, prompt design choices, sentiment embeddings, and multimodal data integration, would further clarify real-world applicability and potentially enhance predictive and economic performance.

In summary, this chapter offers new evidence that modern LLMs significantly enhance the measurement of firm-level sentiment and that, when integrated into nonlinear machine learning models, these signals produce substantively meaningful improvements in short-term return forecasting. The findings highlight the increasing significance of AI-driven information processing within financial markets and point toward a future in which narrative-based models play a pivotal role in empirical asset pricing and quantitative investment strategies.

CHAPTER VI

GENERAL DISCUSSION AND CONCLUSION

This thesis addressed a central question in empirical finance: how investor sentiment, embedded in financial narratives, can be measured in a manner that is economically meaningful at the firm level, and whether improvements in such measurement alter conclusions about short-horizon return predictability and risk-adjusted performance. Across Chapters I to V, the dissertation developed a coherent research programme progressing from conceptual framing and literature synthesis to the construction and benchmarking of firm-specific sentiment measures, and finally to predictive and asset-pricing validation. Evaluated jointly, the findings support a consistent conclusion: investor sentiment embedded in financial news narratives contains economically relevant short-horizon information; however, the strength and visibility of that information depend critically on measurement alignment, modelling flexibility, and rigorous economic evaluation.

The conceptual foundation established in Chapter I framed sentiment not as a residual behavioural anomaly but as an evaluative representation of how investors interpret information under uncertainty. Asset prices reflect heterogeneous expectations about future firm performance; sentiment in this dissertation is therefore treated as a structured summary of narrative interpretation rather than as an irrational deviation from fundamentals. The empirical challenge was to determine whether firm-aligned narrative sentiment, particularly when extracted using modern language models, exhibits incremental predictive content and generates economically meaningful abnormal returns at short horizons.

A central insight emerging from Chapter II is that measurement alignment is not a technical detail but a first-order empirical concern. Much of the prior literature relies on aggregated or document-level proxies that assign a single polarity to multi-entity narratives, potentially obscuring heterogeneous firm-specific evaluations. Target-level extraction explicitly resolves this attribution structure by linking evaluative language to the relevant firm. While the thesis does not observe latent “true” sentiment directly,

subsequent empirical results demonstrate that alignment between textual measurement and the firm-level unit of analysis materially affects predictive and economic conclusions.

Chapters III and IV operationalized and benchmarked this measurement strategy. Firm-level sentiment was shown to be both conceptually coherent and empirically reproducible, supported by strong annotation reliability. Lexicon-based approaches and document-level transformer models struggled with contextual attribution and mixed-sentiment structures, whereas generative LLMs more effectively captured comparative framing, beneficiary-loser distinctions, and implied evaluative shifts. In full-length financial news articles, where narrative complexity, causal explanation, and forward-looking language are more pronounced, generative LLMs demonstrated clear advantages over lexicon-based and conventional transformer-based models. Few-shot prompting further enhanced attribution fidelity for larger models, indicating that both model scale and prompting design influence extraction quality.

Chapter V subjected these measurement advances to rigorous economic evaluation. Firm-specific sentiment derived from headlines and full articles was incorporated into both linear and nonlinear forecasting frameworks to predict one-, two-, and three-day-ahead returns. Linear models captured limited structure, whereas nonlinear ensemble methods revealed conditional interactions between sentiment, technical indicators, and prior returns. Article-level sentiment consistently outperformed headline-level sentiment, reinforcing the importance of contextual richness in expectation formation.

The horizon structure of results provides additional insight. Predictive gains were modest at the one-day horizon but strengthened at two and three days, consistent with the gradual incorporation of complex narrative information into prices. These findings do not imply persistent inefficiency; rather, they suggest that interpretive complexity may generate short-lived delays in information assimilation even in highly liquid markets.

Most importantly, nonlinear models incorporating article-level LLM-derived sentiment generated positive and statistically significant risk-adjusted alpha under the

Fama-French five-factor framework at multi-day horizons, whereas price-only benchmarks did not. This risk-adjusted evidence indicates that firm-specific narrative sentiment provides incremental predictive information beyond conventional factor exposures when evaluated under realistic out-of-sample conditions.

The thesis does not claim that narrative measurement fully captures underlying investor beliefs, nor that sentiment-based strategies yield persistent arbitrage opportunities. Instead, it demonstrates that the empirical strength of narrative sentiment in short-horizon asset pricing depends on careful alignment between textual measurement and the economic unit of analysis, appropriate modelling architecture, and disciplined economic validation.

6.1. Answers to the Research Questions

This section provides direct and integrated answers to the research questions introduced in Chapter I.

RQ1: To what extent can LLMs reliably extract firm-level (target-specific) sentiment from financial news, and how does their performance compare with lexicon-based and fine-tuned transformer-based approaches?

The evidence from Chapters III and IV demonstrates that contemporary LLMs reliably extract firm-level sentiment from financial news and, in most empirical configurations examined, outperform lexicon-based and conventional transformer-based approaches.

At the headline level, generative models such as ChatGPT-4 and Gemini 1.5 Pro achieve performance comparable to domain-adapted aspect-based architectures such as DeBERTa-v3-absa-v1.1, despite operating without task-specific fine-tuning. While specialized transformer models remain competitive when explicitly trained on financial data, LLMs achieve similar levels of performance in zero-shot settings. In contrast, lexicon-based tools such as VADER consistently underperform, reflecting their limited ability to resolve contextual ambiguity and entity-specific attribution in financial text.

When extended to full-length financial news articles, where narrative complexity and multi-entity attribution are more pronounced, larger generative models, particularly DeepSeek-R1 and ChatGPT-o1, consistently outperform lexicon-based methods and transformer models, including FinBERT, FinBERT-Tone, and DistilFinRoBERTa. Even aspect-adapted architectures such as DeBERTa-v3-absa-v1.1 are surpassed by these generative LLMs in out-of-the-box configurations.

Taken together, the findings support RQ1. State-of-the-art generative LLMs reliably perform firm-level sentiment extraction and generally exceed lexicon-based and conventional transformer approaches without requiring additional supervised training.

RQ2: Does few-shot prompting improve the accuracy and robustness of LLM-based target-level financial sentiment analysis relative to zero-shot approaches?

Chapter IV shows that structured few-shot prompting strengthens the performance of leading generative models in target-based financial sentiment analysis. For large instruction-tuned architectures such as ChatGPT-o1 and DeepSeek-R1, few-shot configurations enhance robustness and attribution consistency, particularly in multi-entity and mixed-sentiment narratives.

However, the benefits are not uniform across all models. Smaller architectures, such as Llama 3 8B, exhibit instability under few-shot conditions, indicating that model capacity and pretraining scale influence responsiveness to prompt-based adaptation.

These findings suggest that the prompting strategy is a substantive methodological choice rather than a superficial implementation detail. Accordingly, RQ2 is supported: few-shot prompting materially enhances the robustness of advanced generative LLMs in firm-level financial sentiment tasks.

RQ3: Does LLM-derived firm-specific sentiment exhibit a statistically significant association with short-horizon stock returns?

Chapter IV evaluates the financial relevance of LLM-derived sentiment by examining its association with short-horizon stock returns. Firm-level daily sentiment signals generated by ChatGPT-4, ChatGPT-4o, ChatGPT-o1, and DeepSeek-R1 display a positive and statistically meaningful alignment with contemporaneous return movements within the studied sample.

While the strength of this association varies across firms, reflecting heterogeneity in information environments and investor attention dynamics, LLM-derived signals consistently outperform naïve baselines in directional accuracy tests. In several cases, generative models approximate the alignment observed in manually annotated benchmark sentiment, although they do not fully replicate it.

Importantly, this evidence reflects contemporaneous alignment rather than causal identification or direct tradability. Nevertheless, the findings support RQ3: LLM-derived firm-specific sentiment captures economically relevant signals that are statistically associated with short-horizon stock returns.

RQ4: How do methodological choices, such as textual information depth (headlines versus full-length articles) and predictive modelling frameworks (linear versus nonlinear methods), affect the accuracy and robustness of sentiment-enhanced return forecasts?

First, textual depth is decisive. Sentiment extracted from full-length financial news articles consistently yields stronger predictive and economic performance than headline-based sentiment. Article-level signals derived from generative models such as ChatGPT-o3 and DeepSeek-R1 outperform headline-based alternatives and transformer-based sentiment derived from FinBERT. This indicates that the richer narrative context, capturing causal reasoning and forward-looking tone, provides more informative signals than condensed headlines.

Second, modelling flexibility is critical. Linear classifiers, including logistic regression, exhibit limited responsiveness to the inclusion of technical indicators and sentiment features. In contrast, nonlinear ensemble learners, specifically Random

Forest and XGBoost, consistently outperform linear benchmarks across predictive accuracy, trading metrics, and risk-adjusted evaluation. Sentiment and technical indicators generate meaningful gains primarily when nonlinear interactions can be represented flexibly.

Third, the strongest performance arises from the interaction of textual depth and nonlinear modelling. Article-level generative LLM sentiment integrated into Random Forest or XGBoost produces the most robust and economically meaningful outcomes, exceeding both headline-based sentiment and transformer-derived alternatives.

Thus, RQ4 is supported: predictive accuracy and economic robustness depend jointly on the richness of textual input and the flexibility of the modelling architecture.

RQ5: Does LLM-derived firm-specific narrative sentiment provide incremental predictive information and generate statistically and economically significant risk-adjusted performance in realistic out-of-sample trading settings?

Chapter V evaluates economic significance under a rigorous expanding-window out-of-sample design with risk adjustment using the Fama-French five-factor model.

Strategies based solely on lagged returns or implemented through logistic regression fail to generate economically meaningful abnormal performance. In contrast, nonlinear ensemble models, specifically Random Forest and XGBoost, that incorporate article-level generative LLM sentiment from ChatGPT-o3 and DeepSeek-R1 generate statistically significant risk-adjusted alpha at multi-day horizons.

Risk-adjusted performance metrics further confirm that the gains are not confined to classification improvements. Article-level generative sentiment yields stronger Sharpe ratios, lower drawdowns, and more stable trading outcomes relative to price-only, technical-only, or headline-based specifications. Importantly, these results are obtained under out-of-sample evaluation and established asset-pricing controls, mitigating concerns regarding overfitting or omitted factor exposures.

Accordingly, RQ5 is supported: LLM-extracted firm-specific narrative sentiment provides incremental predictive information and generates statistically and economically significant risk-adjusted performance in realistic trading settings.

Collectively, the answers to RQ1-RQ5 demonstrate that advances in firm-level sentiment extraction, particularly through context-sensitive generative LLMs such as ChatGPT-o1, ChatGPT-o3, and DeepSeek-R1, materially alter both predictive and economic conclusions in short-horizon asset pricing. The financial relevance of narrative sentiment depends on measurement alignment, prompting design, textual depth, and modelling flexibility. When these elements are combined appropriately, LLM-derived sentiment contributes both statistically and economically meaningful improvements in short-horizon return forecasting.

6.2. Policy Implications

The findings of this thesis carry important implications for financial regulation, market supervision, and the governance of artificial intelligence systems in capital markets.

First, the results indicate that firm-specific narrative sentiment constitutes a measurable component of the informational environment rather than merely qualitative “soft” commentary. Entity-level sentiment extracted from financial news exhibits economically meaningful short-horizon associations with price dynamics. Regulatory surveillance systems that rely primarily on keyword detection or coarse document-level tone measures may therefore underestimate the role of firm-specific narratives in shaping transient volatility and attention-driven price pressure. The evidence supports the development of entity-aligned, context-aware monitoring tools capable of identifying abrupt shifts in firm-level narrative sentiment, particularly during periods of heightened uncertainty.

Second, the findings underscore the importance of narrative disclosures in expectation formation. Forward-looking language, risk framing, and interpretive emphasis in financial communications can influence not only market reactions but also the speed with which information is incorporated into prices. Although the thesis does not

suggest persistent inefficiency, it demonstrates that narrative framing forms part of the economically relevant information set. This strengthens the case for clarity, comparability, and consistency in corporate disclosures and supports regulatory vigilance regarding strategically framed or potentially misleading narratives.

Third, the growing integration of LLMs into sentiment extraction, compliance systems, and risk monitoring introduces governance challenges. LLM-based analytical infrastructures are subject to model drift, opacity, and version instability, particularly when proprietary systems are updated over time. These features create risks for reproducibility, auditability, and supervisory oversight in regulated financial environments. The findings therefore support policy frameworks that emphasize documentation standards, version control, benchmarking transparency, and formal model risk management protocols for AI-based financial systems.

Finally, the documented horizon structure, where narrative effects strengthen over short multi-day intervals, suggests that narrative-driven attention shocks may contribute to short-term volatility clustering and correlated investor repositioning. Monitoring firm-specific narrative sentiment could therefore complement traditional systemic risk indicators by identifying early signals of attention-driven stress, especially around earnings announcements, regulatory developments, or macroeconomic events that generate complex interpretive narratives.

6.3. Limitations and Future Research Directions

Despite its contributions, the thesis operates within clearly defined empirical boundaries.

First, the analysis focuses on a limited cross-section of large U.S. technology firms over a relatively short sample period. While this design reduces microstructure noise and ensures dense textual coverage, it constrains generalizability. Narrative effects may differ across smaller firms, less-covered equities, or sectors characterized by greater informational frictions and limits to arbitrage.

Second, although trading strategies are evaluated using expanding-window out-of-sample procedures and risk-adjusted performance metrics, transaction costs, slippage, and market impact are not explicitly modelled. High-turnover strategies may exhibit lower implementable profitability once realistic execution frictions are incorporated. A full assessment of practical viability would therefore require detailed cost modelling and liquidity constraints.

Third, the use of proprietary LLMs raises concerns regarding reproducibility and stability. Model providers periodically update weights and internal system behaviour, potentially altering output distributions under identical prompts. While benchmarking mitigates this concern at a given point in time, long-term deployment would require systematic monitoring, version control, and performance revalidation.

Fourth, interpretability remains limited. Although LLM-based extraction improves attribution fidelity relative to document-level proxies, the internal reasoning processes of generative models are not fully transparent. In regulated financial settings, explainability frameworks may be necessary to ensure auditability and supervisory compliance.

Fifth, the thesis does not establish structural causality. Sentiment signals may be endogenous to prior market movements, and news production itself may respond to price dynamics. The results demonstrate predictive and economic relevance under disciplined evaluation but do not claim causal identification. Establishing causal mechanisms would require alternative research designs, including natural experiments, instrumental variables, or exogenous variation in media exposure.

A further limitation concerns textual scope. The empirical analysis relies exclusively on professionally produced financial news. While this ensures high-quality narratives with clear attribution, it excludes other influential information channels such as earnings call transcripts, regulatory filings, analyst reports, and social media discourse. Moreover, the study measures sentiment conditional on publication and does not model the endogenous process governing which narratives are reported or how they

are framed. As a result, the analysis captures the pricing of published narratives but does not disentangle potential media selection effects.

An additional limitation concerns language coverage. The analysis relies exclusively on English-language financial news, despite the inherently multilingual nature of global capital markets. As a result, the generalizability of the findings to non-English or multilingual information environments remains an open question.

These limitations motivate several avenues for future research.

A first direction concerns external validity. Extending firm-specific sentiment extraction to broader equity universes, international markets, and smaller capitalization firms would clarify whether narrative effects strengthen under higher informational frictions.

Second, linguistic generalizability warrants further investigation. Future research should examine whether the predictive content of narrative sentiment persists in non-English and multilingual settings. Developing cross-lingual or multilingual sentiment extraction frameworks would enable a systematic assessment of whether language structure, translation effects, and regional information environments influence the strength and timing of sentiment-based return predictability.

Third, regime dependence warrants systematic investigation. Narrative predictability may vary across macroeconomic conditions, crisis periods, or high-uncertainty environments characterized by heightened attention and belief dispersion.

Fourth, implementation-focused research should incorporate explicit transaction cost modelling, liquidity constraints, and portfolio construction considerations to assess real-world profitability more precisely.

Fifth, future work may expand beyond single-source textual analysis toward multisource narrative integration. Combining financial news with earnings calls, regulatory filings, analyst commentary, or other disclosure channels could provide a

more comprehensive representation of narrative formation and diffusion. Comparing timing and persistence across sources would clarify how different information channels contribute to price discovery.

Sixth, more refined temporal modelling may illuminate the speed and structure of narrative incorporation. Intraday data, flexible lag specifications, and diffusion modelling could deepen understanding of how interpretive information propagates through markets.

Finally, governance-focused research represents an important methodological frontier. Developing explainable sentiment extraction frameworks, drift-detection mechanisms, and standardized benchmarking protocols for evolving LLM architectures will be essential as AI-based systems become embedded within financial decision processes.

In conclusion, this dissertation demonstrates that carefully designed, firm-specific narrative sentiment measures derived from LLMs substantially improve short-horizon return predictability and generate economically and statistically significant risk-adjusted abnormal returns under stringent asset pricing tests. More fundamentally, it establishes that in narrative-driven financial markets, advances in predictive performance must be grounded in disciplined measurement, transparent modelling choices, and rigorous economic validation. By integrating modern language models with asset pricing frameworks, the thesis contributes a coherent methodological blueprint for credible AI-driven research in empirical finance.

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AI USE STATEMENT

I take full responsibility for ensuring that the work submitted in fulfilment of the degree requirements is my own and original. In accordance with applicable institutional and European Union guidelines on the use of artificial intelligence (AI), I declare that AI tools were used solely for language support, including the identification of language errors and the improvement of clarity and expression.

APPENDICES

APPENDIX I: Data Availability Statement

Dataset 1: Target-Level Financial Sentiment in News Headlines

The manually annotated dataset of firm-specific financial news headlines developed in this dissertation is publicly available at:

https://github.com/iftikharm895/Target-Level_Financial_Sentiment_Analysis

The repository provides the URLs of the news headlines along with the corresponding manual sentiment annotations.

Dataset 2: Target-Based Financial Sentiment in Full-Length News Articles

The manually annotated dataset of firm-specific full-length financial news articles developed in this dissertation is publicly available at:

https://github.com/iftikharm895/Target-Based_Sentiment_Analysis_in_Financial_News

The repository includes the URLs of the news articles together with the associated manual sentiment annotations.

Due to Bloomberg copyright restrictions, the full text of news headlines and articles cannot be publicly distributed. Researchers with access to a Bloomberg Terminal can reconstruct the datasets using the provided URLs.